

When RAG Meets LFM: Towards Retrieval-Augmented Large Foundation Models

Website: <https://advanced-rag.github.io/RAG-Meets-LFMs>

Survey: <https://arxiv.org/pdf/2405.06211>

Xu Yuan¹, Yajuan Ding¹, Chengliang Liu¹, Rui An¹, Chun-Hin Chan¹,

Yiqi Wang², Wenqi Fan¹, and Qing Li¹

¹The Hong Kong Polytechnic University

²National University of Defense Technology

June 9th (Day 1)

PAKDD 2026, Hong Kong, China



Tutorial Outline

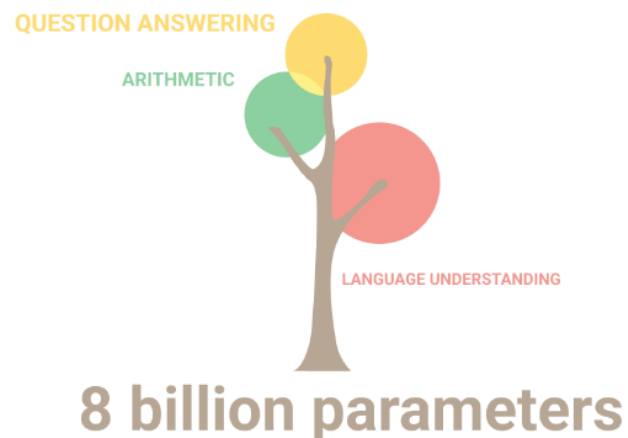


- ⊙ **Part 1: Introduction of Retrieval Augmented Large Foundation Models (RA-LFMs) (Dr. Yujuan Ding)**
- **Part 2: Architecture of RA-LFMs and Main Modules** (Xu Yuan)
- **Part 3: Learning Approach of RA-LFMs** (Chengliang Liu)
- **Part 4: Applications of RA-LFMs** (Chun-Hin Chan)
- **Part 5: Challenges and Future Directions of RA-LFMs** (Dr. Yujuan Ding)
- **Part 6: Q&A**

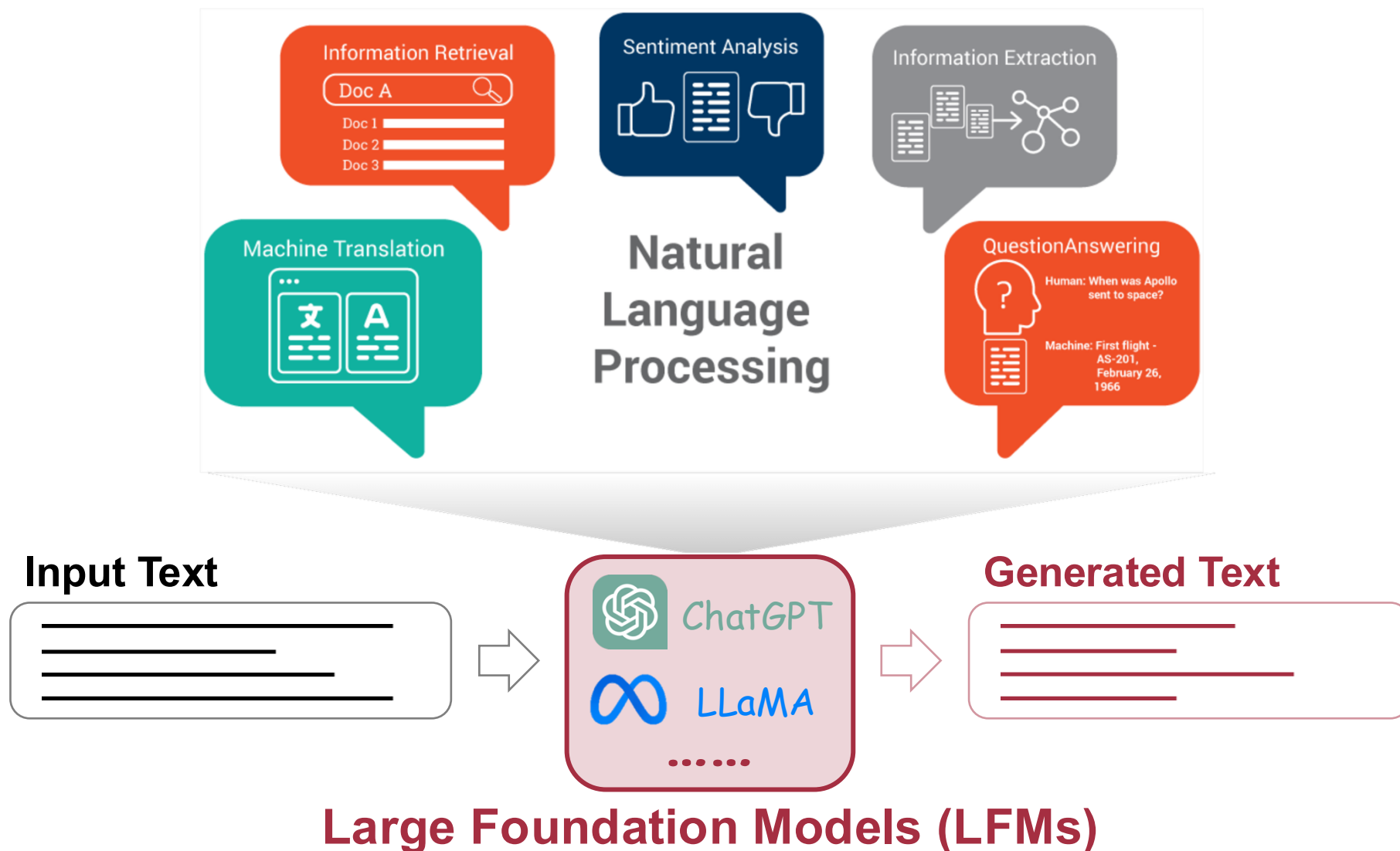
Website of this tutorial
Check out the slides and more information!



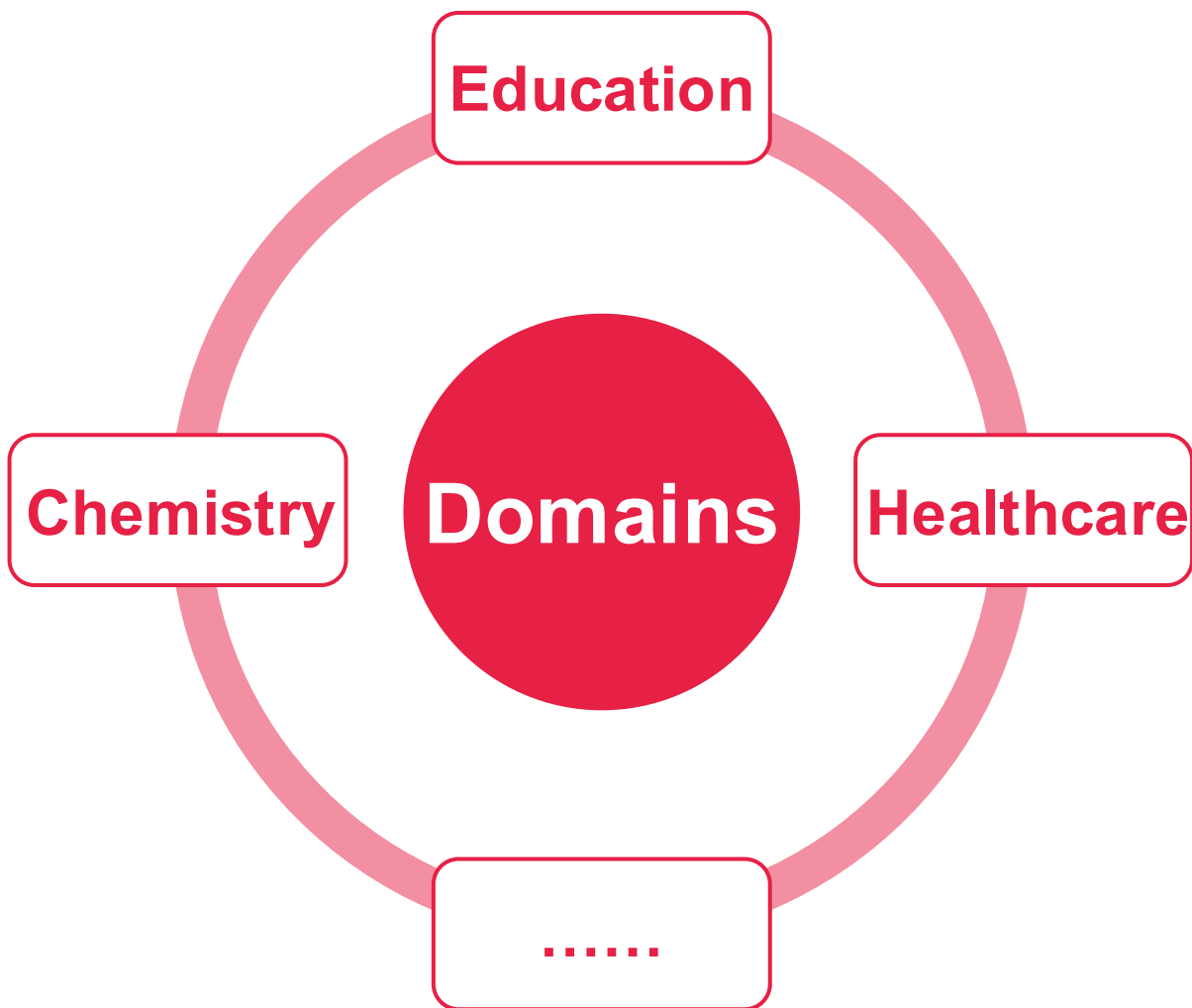
Large Foundation Models (LFMs)



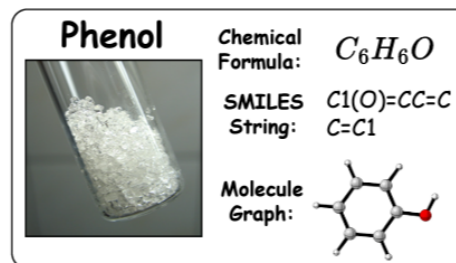
Large Foundation Models (LFMs)



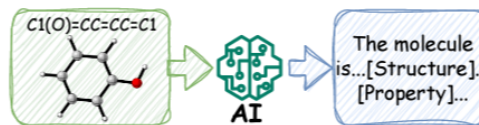
LFMs in Downstream Domains



❑ Molecule discovery, etc.



(a) Molecule Representations.



(b) Molecule Captioning.



ChatGPT

(a) Molecule Captioning

Please show me a description of this molecule: "C1=CC=C(C(=C1)OC2=CC=CC=C2"

The molecule is an aromatic ether in which the oxygen is attached to two phenyl substituents. It has been found in muscat grapes and vanilla. It has a role as a plant metabolite.

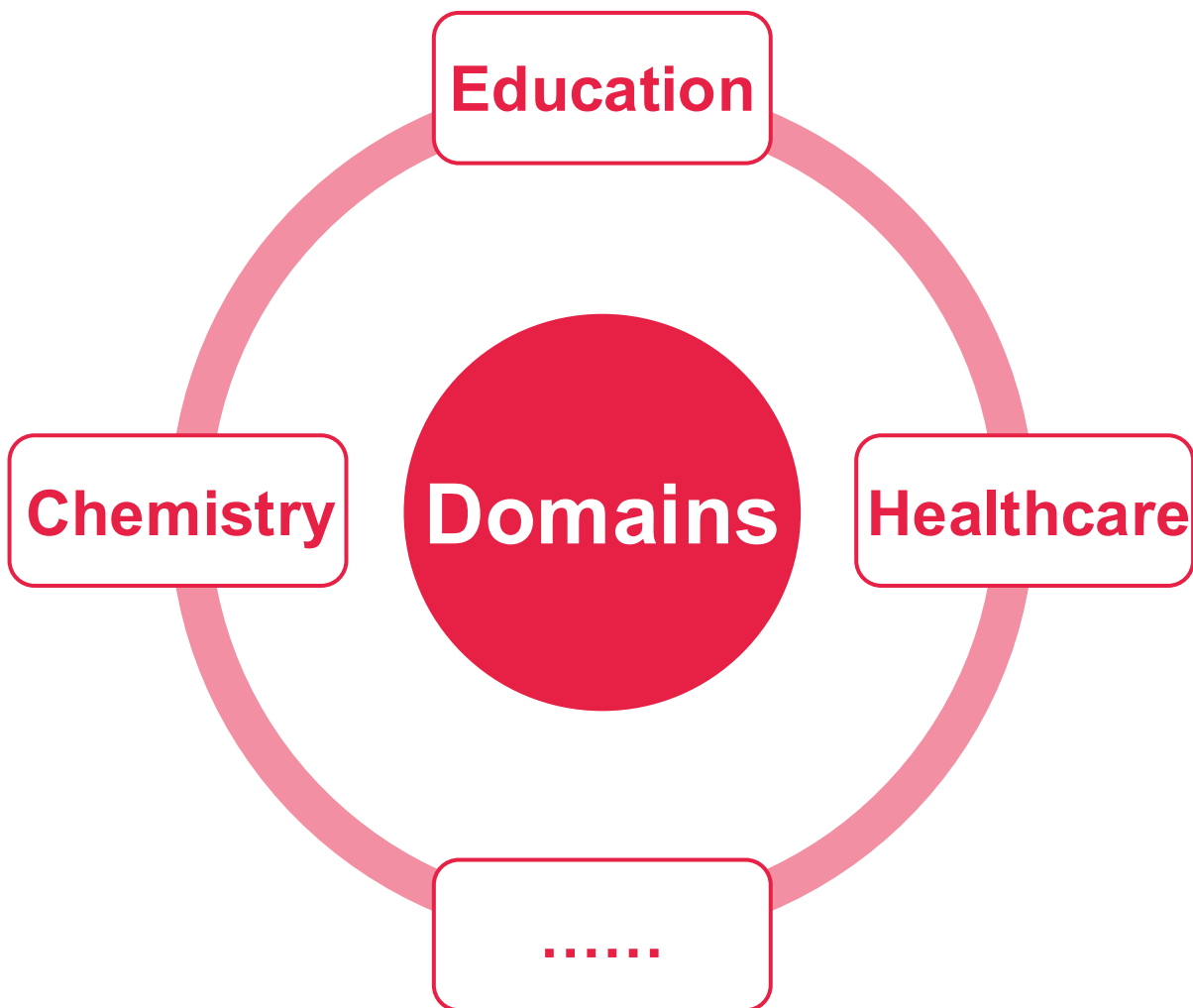
(b) Text-based Molecule Generation

Help me generate a molecule based on the given description:

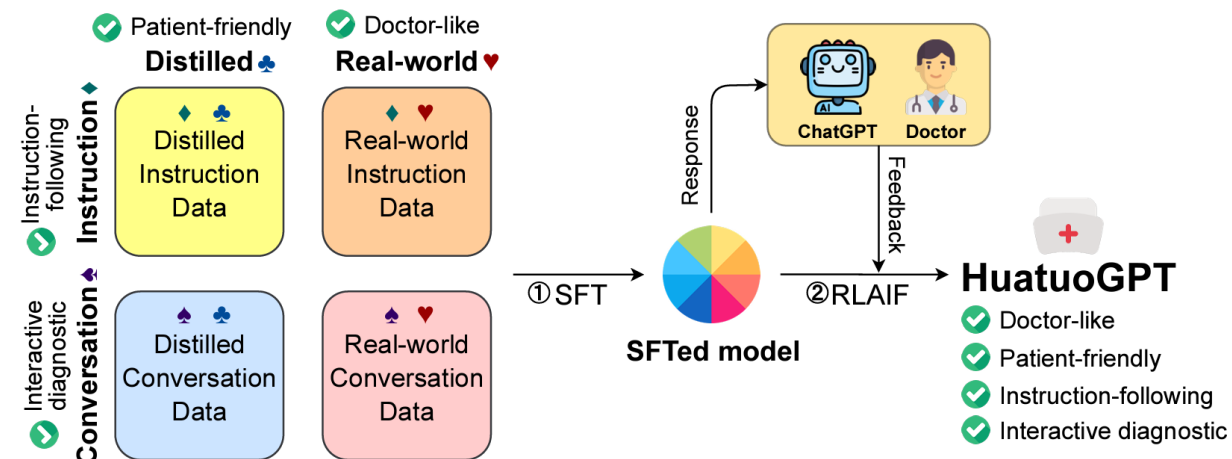
"The molecule is a quinolinemonocarboxylate that is the conjugate base of xanthurenic acid, obtained by deprotonation of the carboxy group. It has a role as an animal metabolite. It is a conjugate base of a xanthurenic acid."

C1=CC2=C(C(=C1)[O-])NC(=CC2=O)C(=O)O

LFMs in Downstream Domains



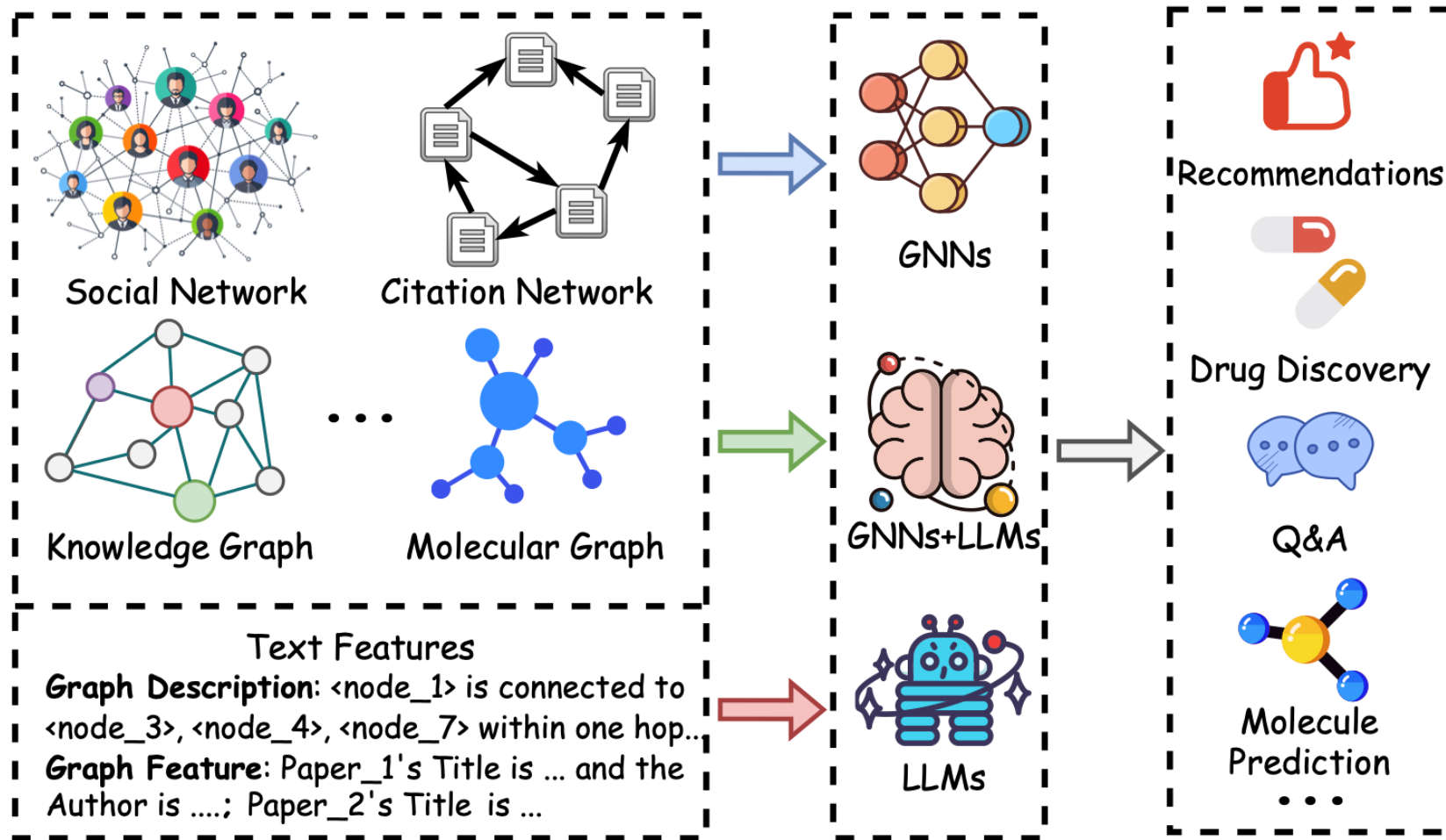
❑ Medical consultation, etc.



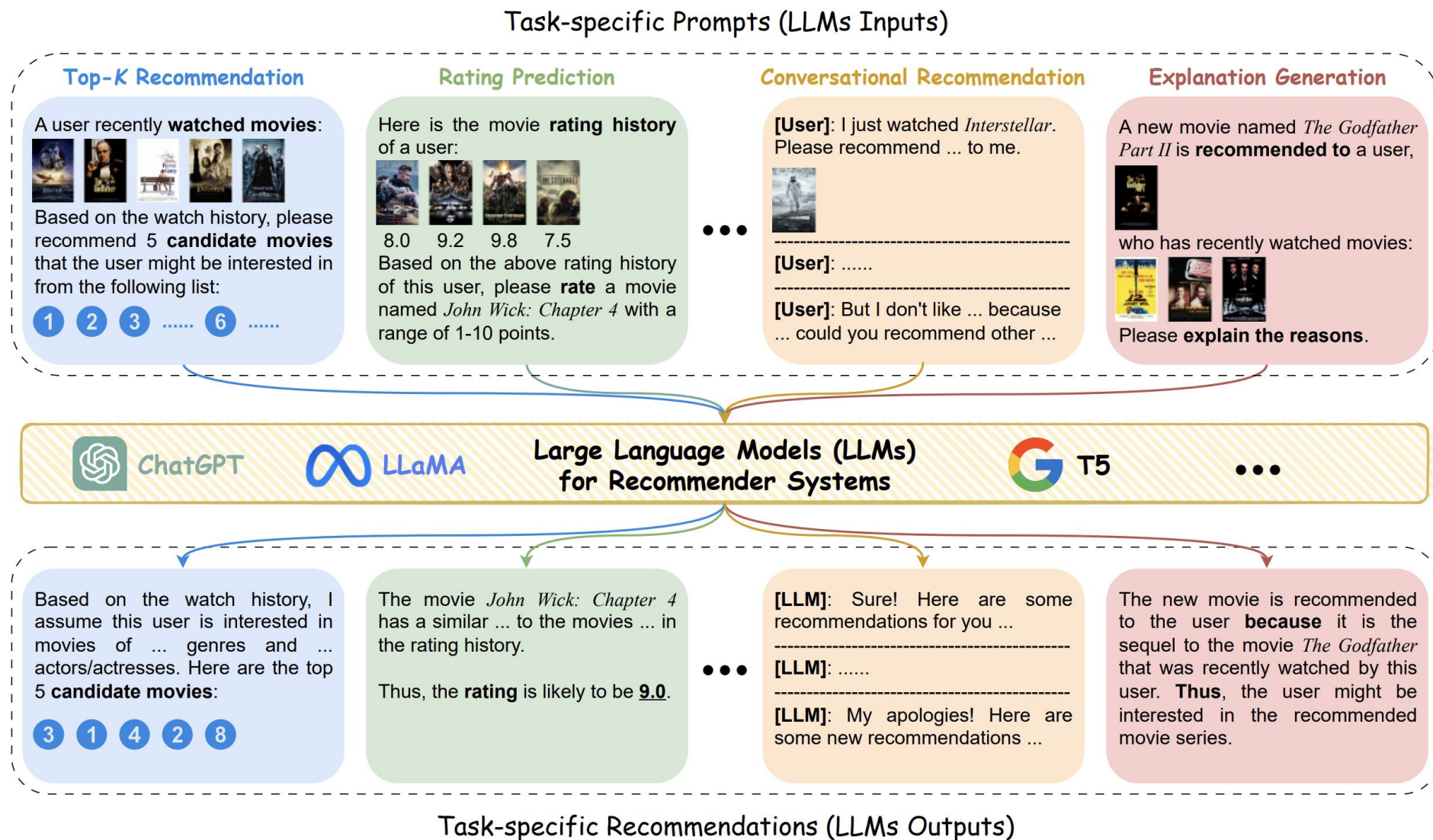
❑ Curriculum & Teaching, etc.



LFMs on Graph-structured Data



LFMs in Recommender Systems



Challenges and Risks of LFM

❑ Hallucination

The generation of inaccurate, nonsensical, or detached text, posing potential risks and challenges for organizations utilizing these models.



❑ Domain-specific knowledge & expertise

LFMs might not perform well in many domain-specific fields like medicine, law, finance, and more, because of the lack of domain-specific knowledge and expertise.



❑ Privacy

Various risks to data privacy and security exist at different stages of LFM, which becomes particularly acute in light of incidents where sensitive internal data was exposed to LFM.



❑ Inconsistency

Sometimes they nail the answer to questions, other times they regurgitate random facts from their training data.

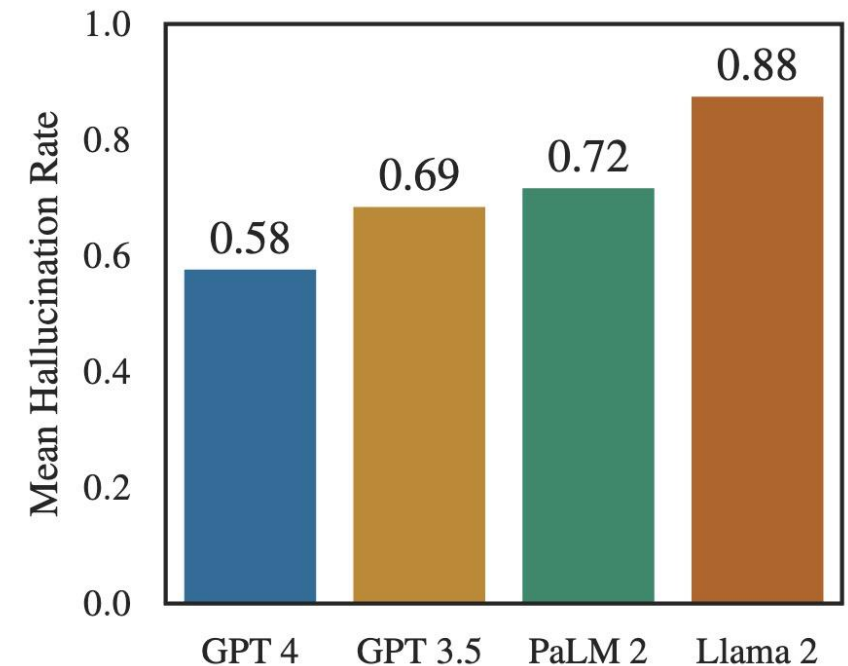
LFMs' Challenges in Vertical Domains

❑ Domain of Law



*In a new study by **Stanford RegLab** and **Institute for Human-Centered AI** researchers, it is demonstrated that legal hallucinations are pervasive and disturbing: **hallucination rates range from 69% to 88% in response to specific legal queries** for state-of-the-art language models.*

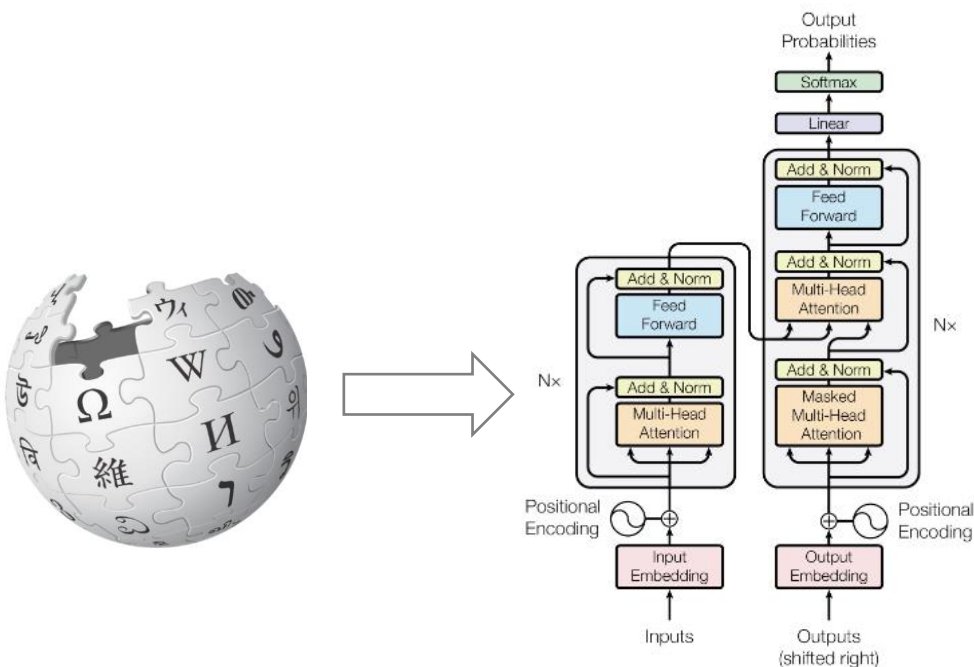
Hallucinations are common across all LFMs when they are asked a direct, verifiable question about a federal court case



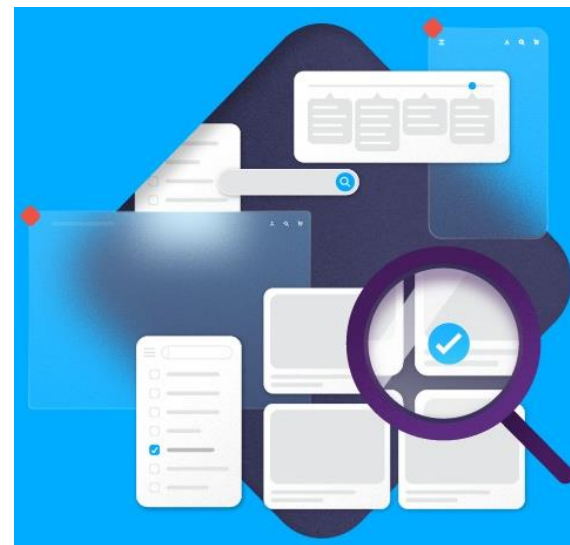
Why Large Foundation Models Work Well?

- ❑ Big Model + Big Training Data

Storing knowledge in the
parametric model !



Storing knowledge in the non-
parametric model?



Information Retrieval (IR)

Retrieval-Augmented Large Foundation Models (RA-LFMs)

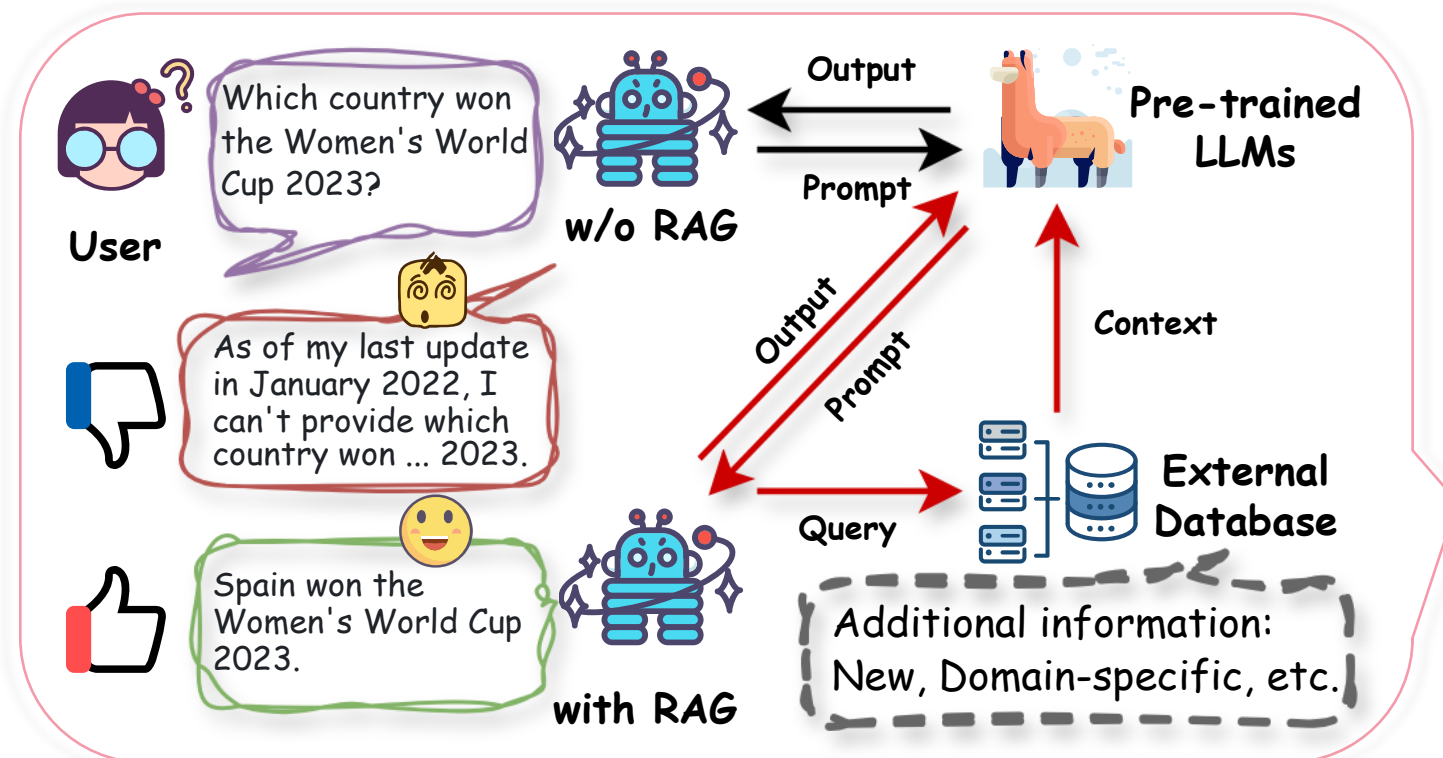
- ❑ LFM **cannot memorize all** (particularly long-tail) knowledge in their parameters
- ❑ Lack of **domain-specific knowledge, updated information**, etc



Hallucination & Unable to answer



Re-training / Finetuning ?



Costly & Heavy Work

Retrieval-Augmented Generation (RAG) for LFM:
RA-LFMs

Integrating Information Retrieval in Generation: **RA-LFM**

Data for Training LFM

- Low quality
- General
- Fixed
- Hard to update

External Knowledge Base

- High-quality knowledge
- Specialized knowledge
- Scalable
- Easy-updated

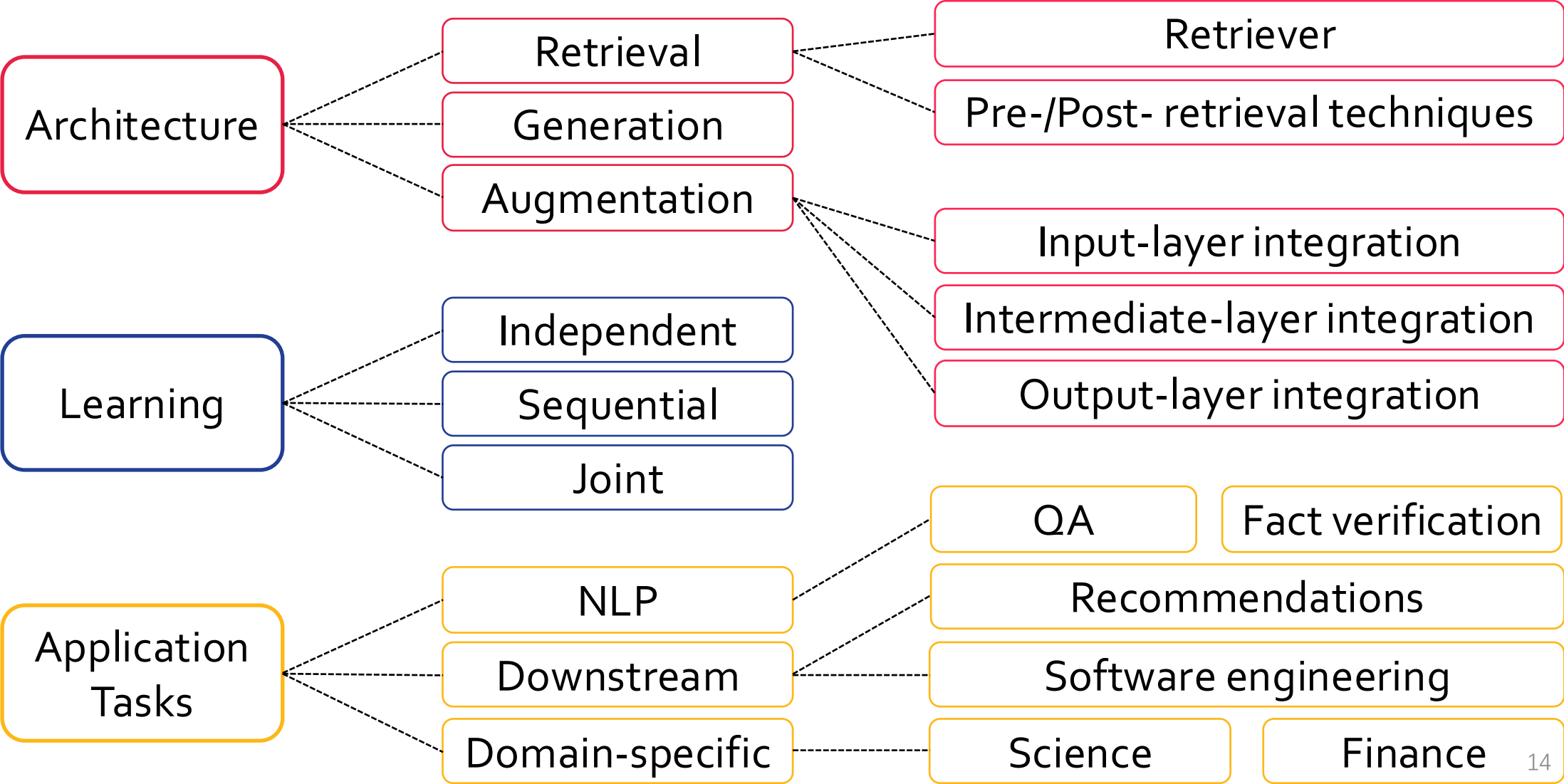


Content generation
Close-book exam
(Hard mode, have to
remember everything)

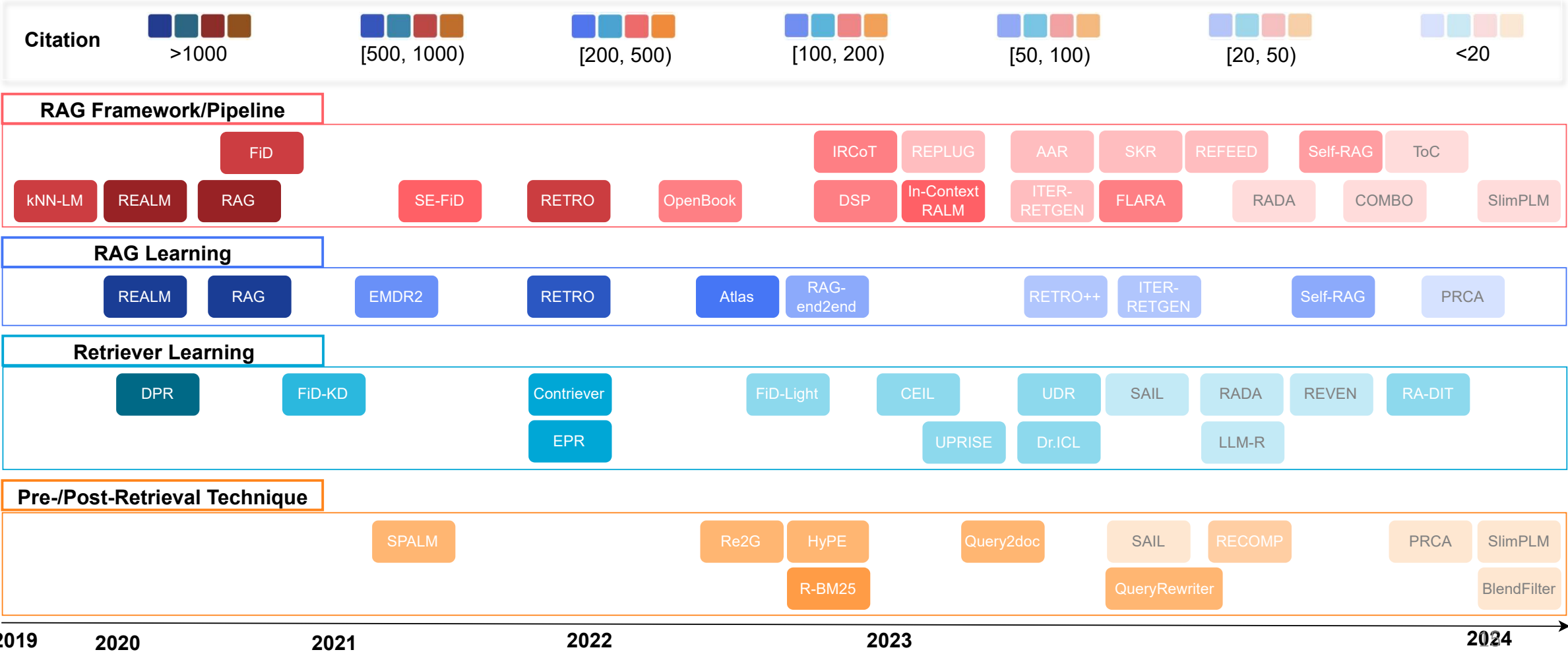
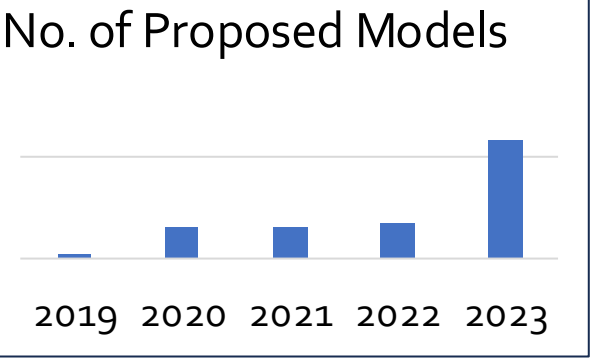
Information / Knowledge
retrieval

RA-LFMs
Open-book exam
(Easy mode, allow to search
in reference)

RA-LFM Research Taxonomy



RAG & RA-LFM Model Development



A Comprehensive Survey Paper

A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models

Wenqi Fan

wenqifan03@gmail.com
The Hong Kong Polytechnic
University, HK SAR

Yujuan Ding*

dingyujuan385@gmail.com
The Hong Kong Polytechnic
University, HK SAR

Liangbo Ning

BigLemon1123@gmail.com
The Hong Kong Polytechnic
University, HK SAR

Shijie Wang

shijie.wang@connect.polyu.hk
The Hong Kong Polytechnic
University, HK SAR

Hengyun Li

neilhengyun.li@polyu.edu.hk
The Hong Kong Polytechnic
University, HK SAR

Dawei Yin

yindawei@acm.org
Baidu Inc, China

Tat-Seng Chua

dcscts@nus.edu.sg
National University of Singapore,
Singapore

Qing Li

csqli@comp.polyu.edu.hk
The Hong Kong Polytechnic
University, HK SAR

Accepted by KDD'24

<https://arxiv.org/pdf/2405.06211>

Website of this tutorial
Check out the slides and more information! ➡



Recruitment

❑ Our research group (Prof. Qing LI & Dr. Wenqi FAN) is actively recruiting self-motivated **postdocs, Ph.D. students, research assistants**, etc. **Visiting scholars, interns, and self-funded students** are also welcome. Send us an email if you are interested.

❖ Research areas: machine learning (ML), data mining (DM), artificial intelligence (AI), deep learning (DNNs), large language models (LLMs), graph neural networks (GNNs), computer vision (CV), natural language processing (NLP), etc.

❖ Position details:

<https://wenqifano3.github.io/openings.html>



Tutorial Outline

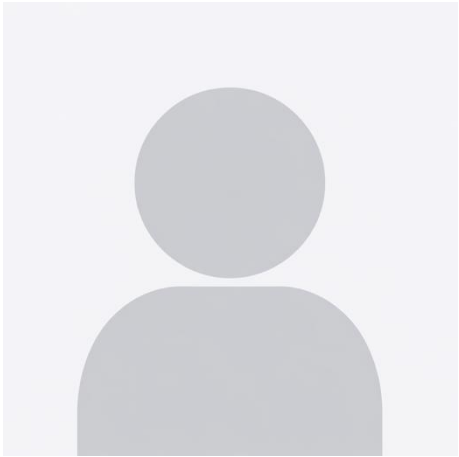


- **Part 1: Introduction** of Retrieval Augmented Large Foundation Models (RA-LFMs) (Dr. Yujuan Ding)
- ⦿ **Part 2: Architecture of RA-LFMs and Main Modules** (Xu Yuan)
- **Part 3: Learning** Approach of RA-LFMs (Chengliang Liu)
- **Part 4: Applications** of RA-LFMs (Chun-Hin Chan)
- **Part 5: Challenges and Future Directions** of RA-LFMs (Dr. Yujuan Ding)
- **Part 6: Q&A**

Website of this tutorial
Check out the slides and more information!



PART 2: Architecture of RA-LFMs and Main Modules

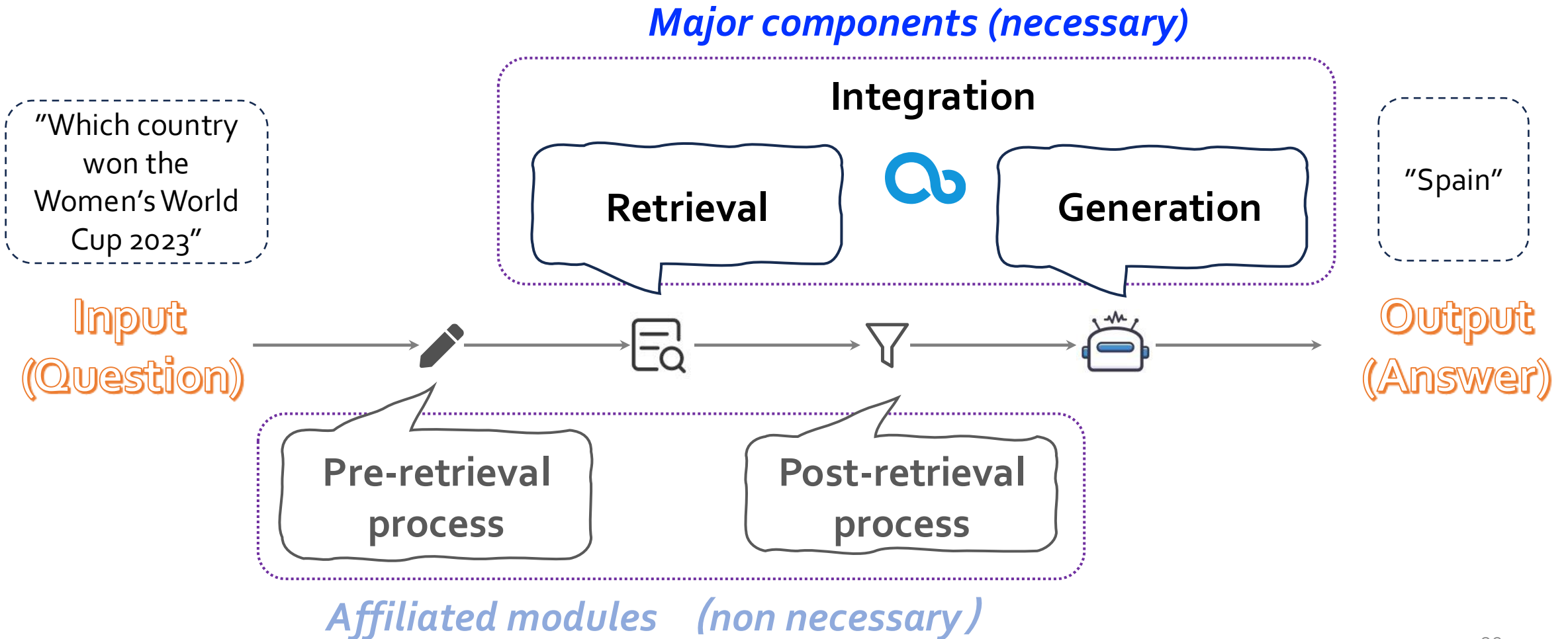


Presenter
Xu Yuan
HK PolyU

- **RA-LFM architecture overview**
- **Retriever in RA-LFMs**
- **Retrieval results integration**
- **Pre/Post-retrieval techniques**
- **Special RA-LFM paradigms**

RA-LFM Architecture: Standard Pipeline

- ❑ Technical component illustration in a RA-LFM for the Q&A task



A Simple Retrieval-Augmented Generation Model

❑ RAG

Integration: concatenating each retrieved passage with the question

Retrieval: DPR + MIPS

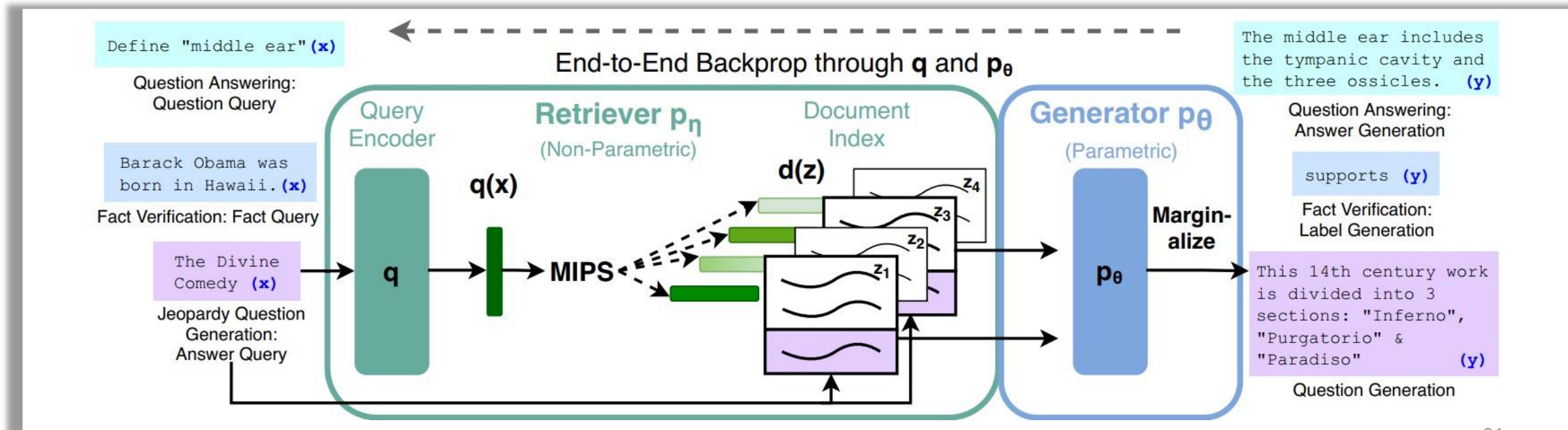


Generation: BART

Input

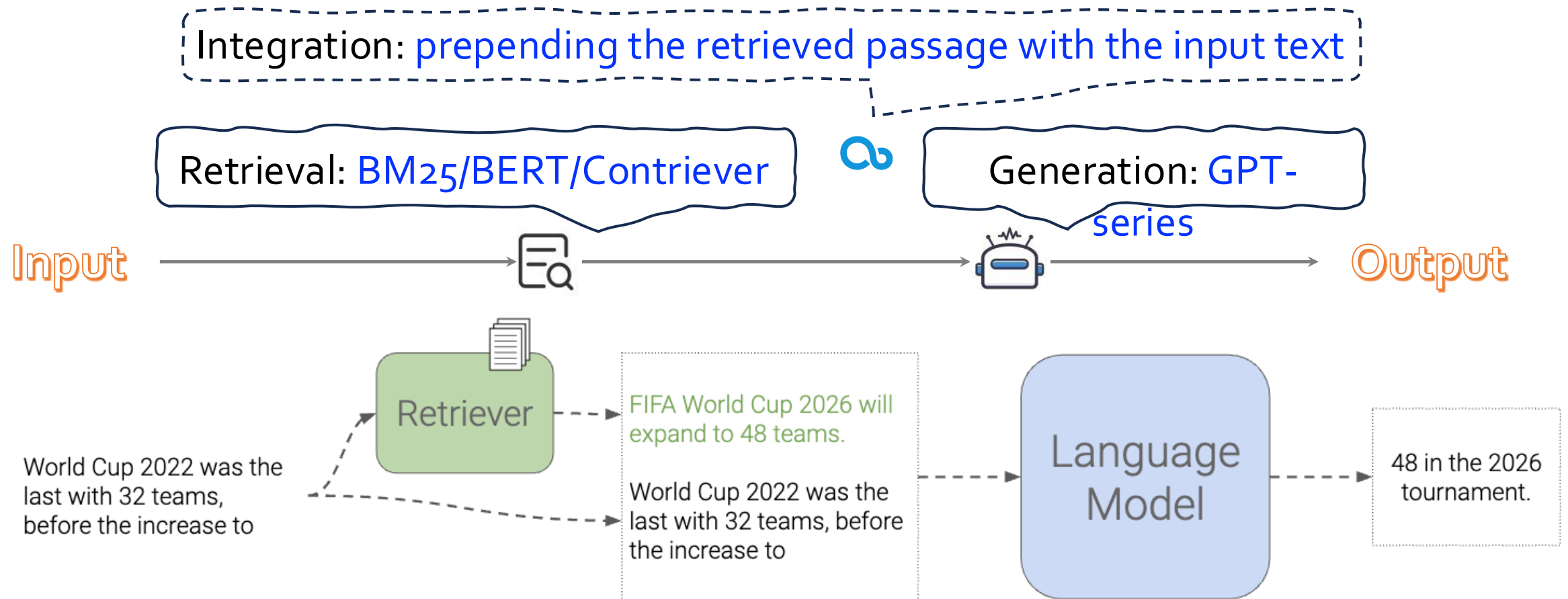


Output



A Simple Retrieval-Augmented Generation Model

□ In-Context RALM



PART 2: Architecture of RA-LFMs and Main Modules



Slides

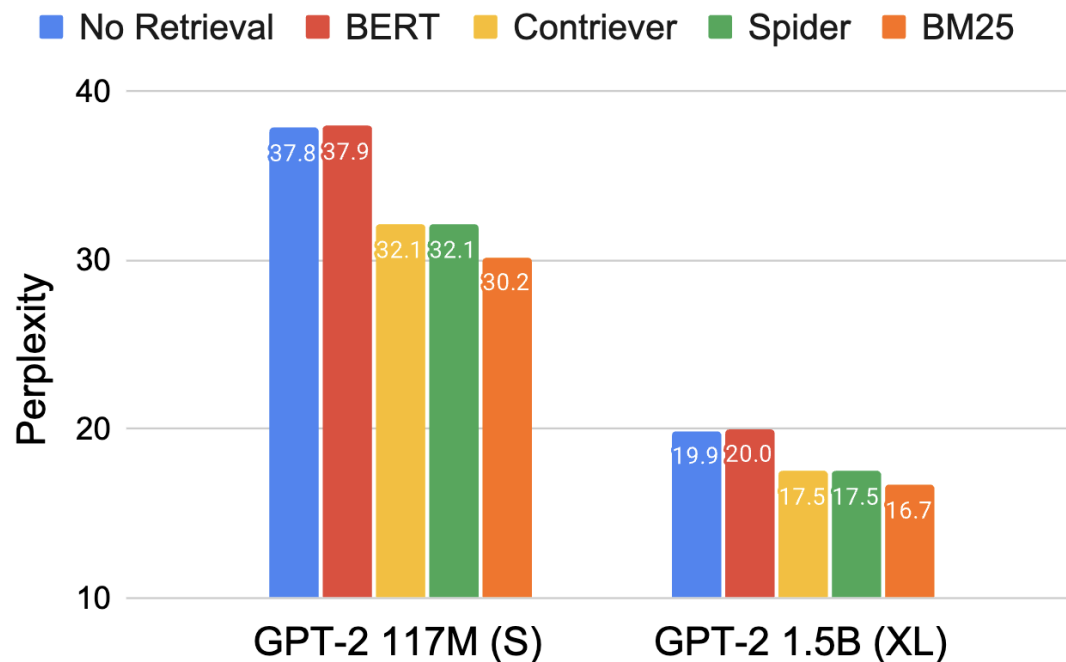


Website of this tutorial

- RA-LFM architecture overview
- **Retriever in RA-LFMs**
- Retrieval results integration
- Pre/Post-retrieval techniques
- Special RA-LFM paradigms

RA-LFM Architecture: Retriever Types

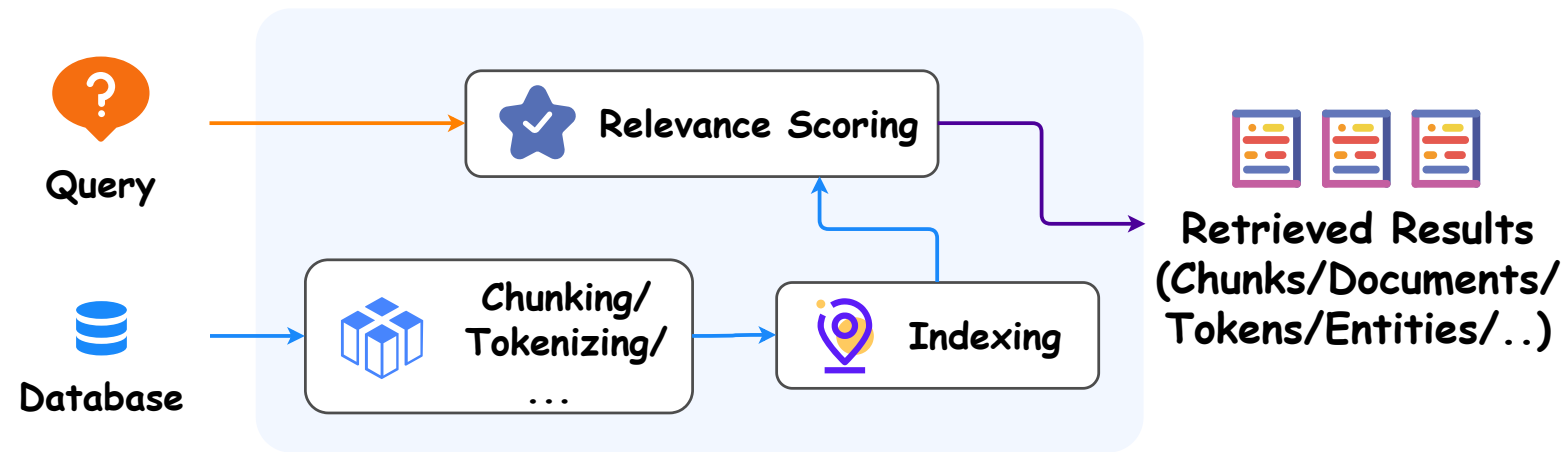
- Different types of retriever deliver different generation performance



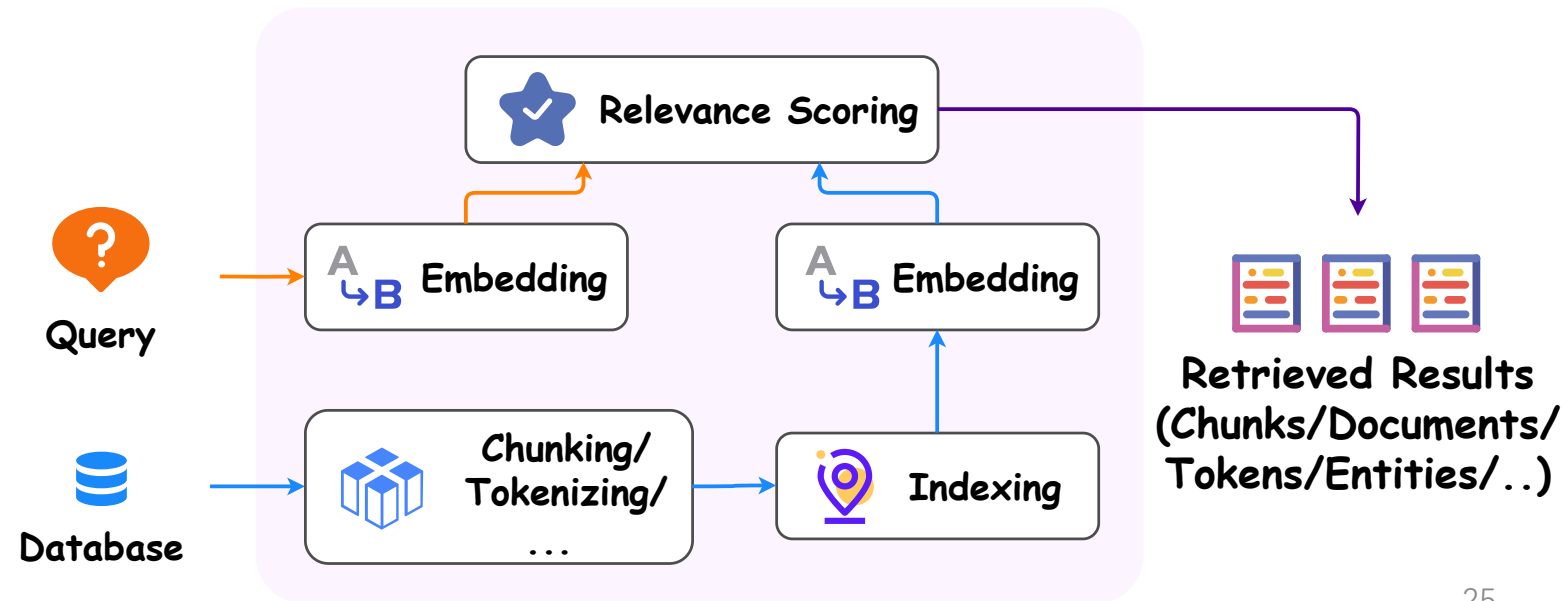
Relevance measurement	Retriever learning
Sparse	Task-specific pre-trained
Dense	General-purpose pre-trained

Dense v.s. Sparse Retrievers

Sparse Retrievers (SR)



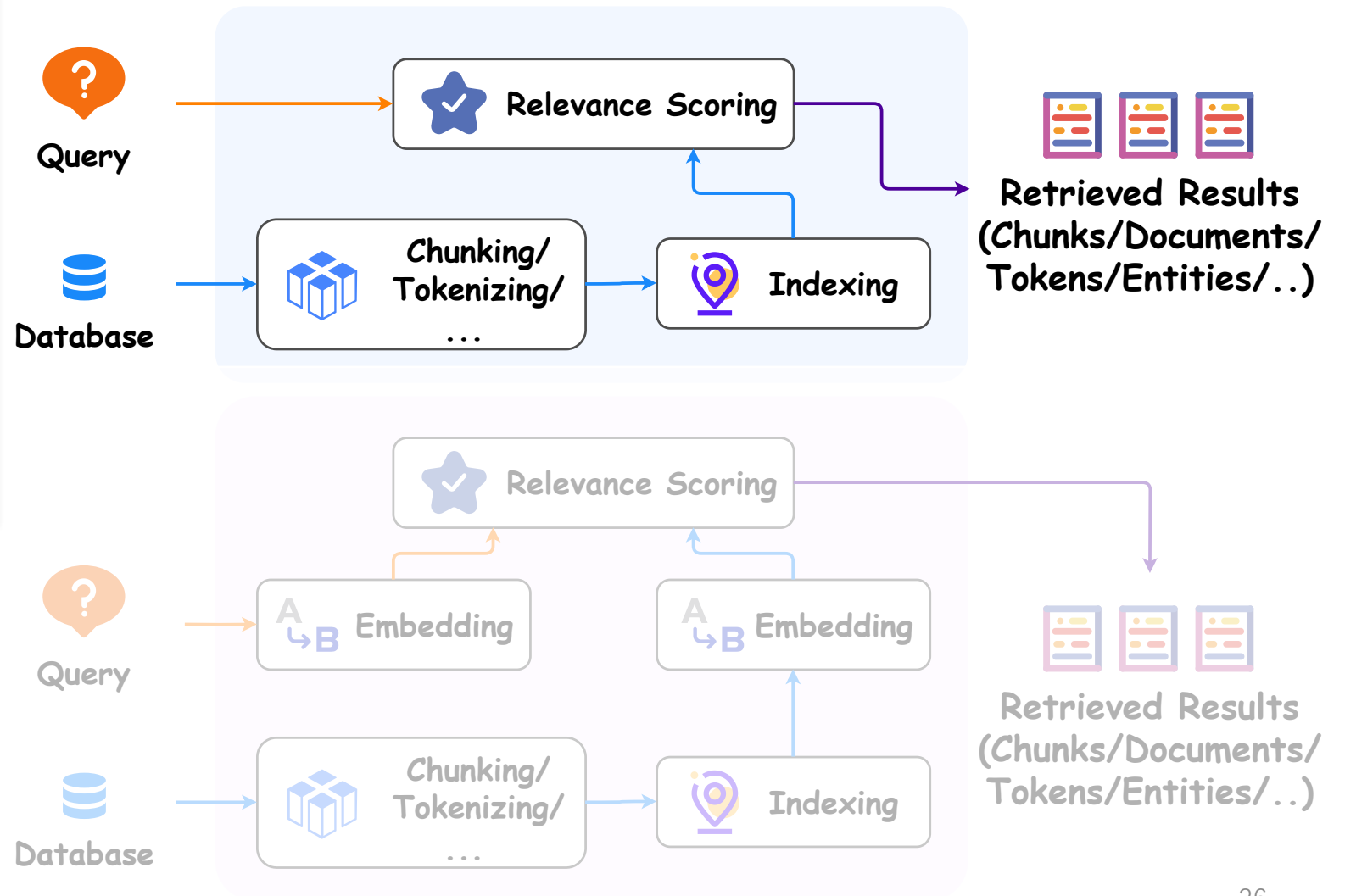
Dense Retrievers (DR)



Dense v.s. Sparse Retrievers

Sparse Retrievers (SR)

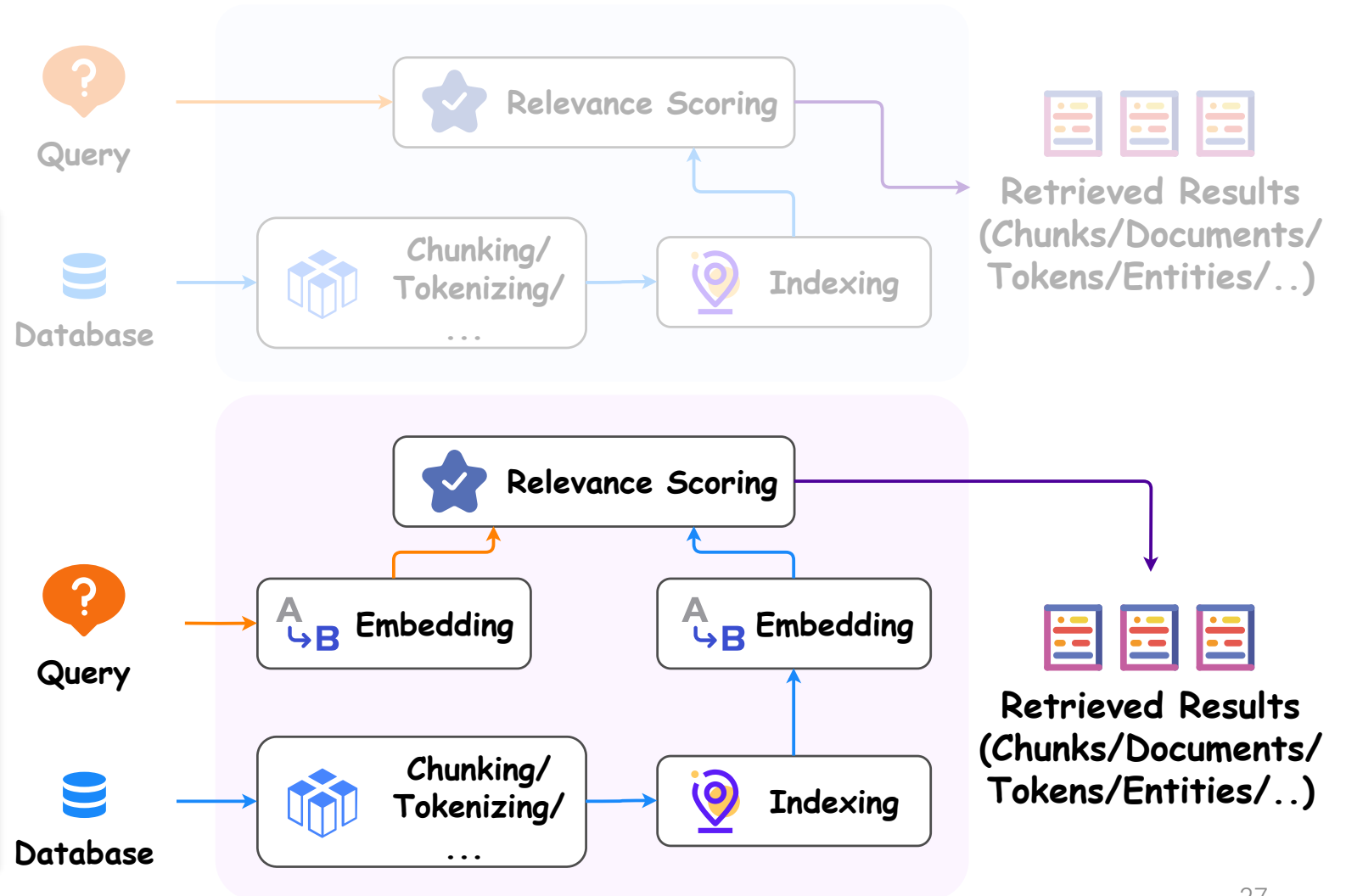
- Feasible to apply
- High efficiency
- Fine performance
- Example: TF-IDF, BM25



Dense v.s. Sparse Retrievers

Dense Retrievers (DR)

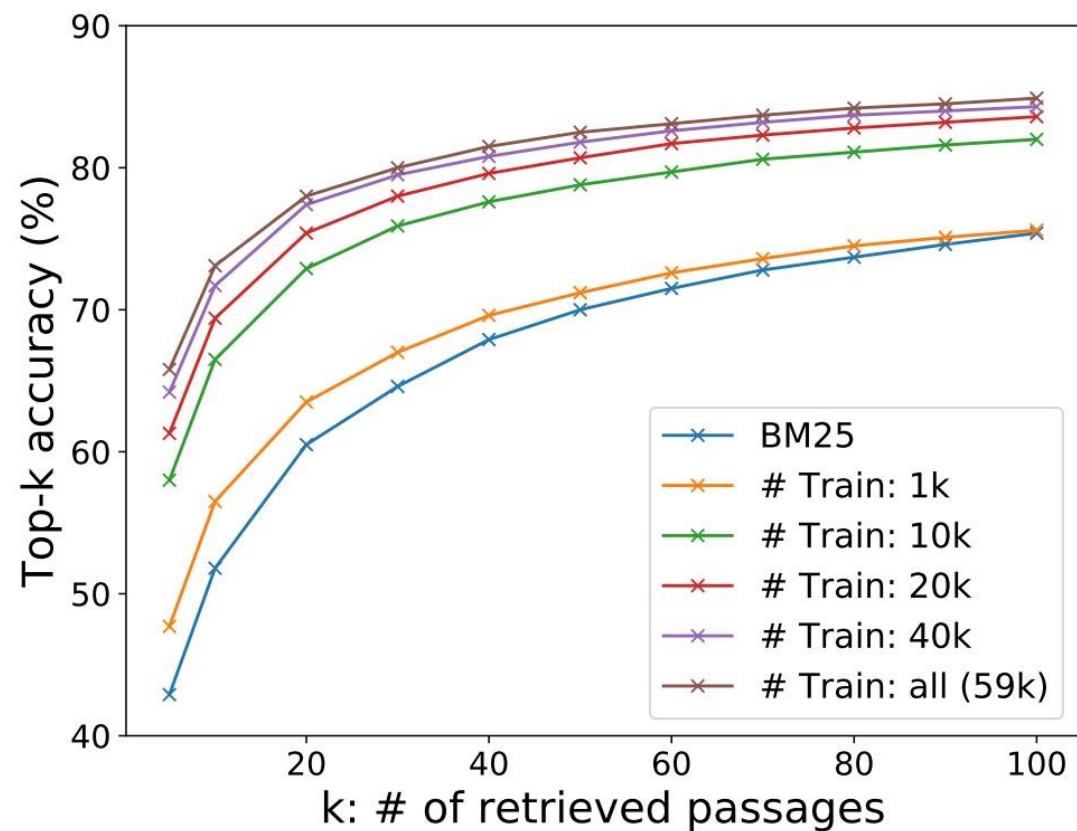
- Allowing fine-tuning
- Better adaptation
- Customizable for more retrieval goals
- Example: DPR, Contriever



Task-Specific Pre-trained Retriever (Supervised)

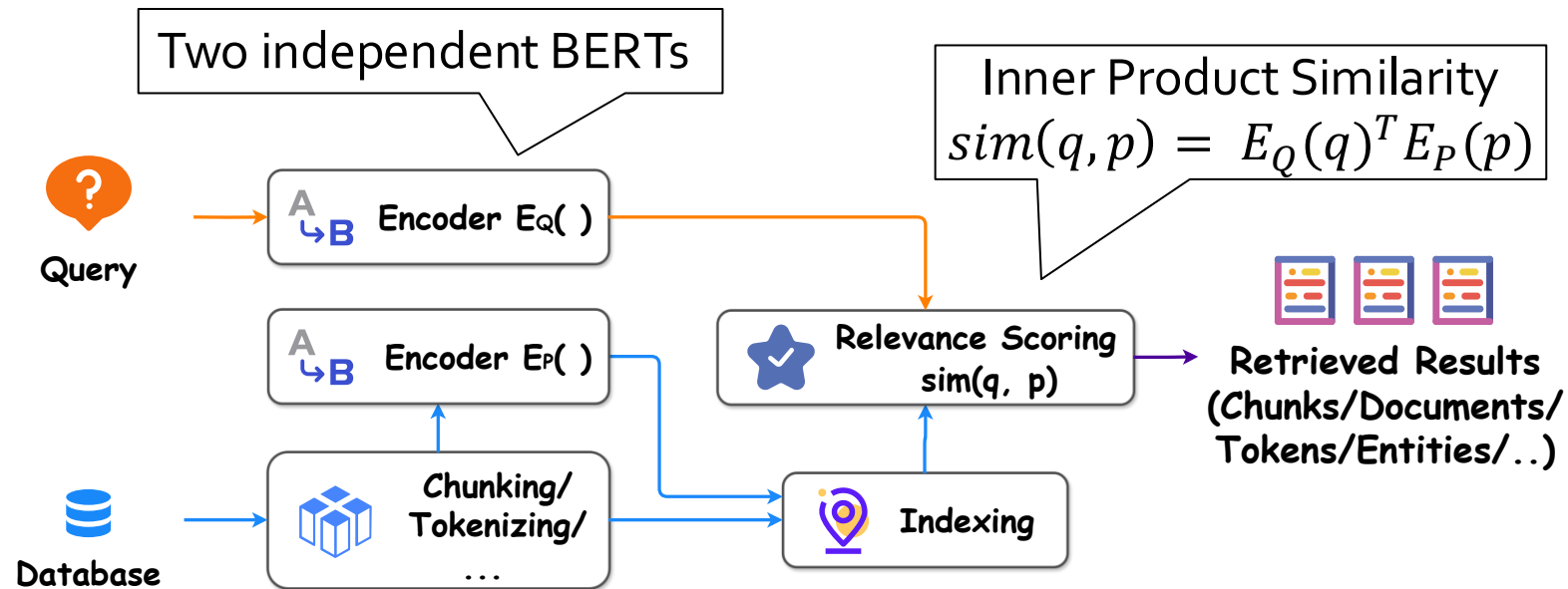
❑ **Dense Passage Retriever (DPR):** Pretrained for Question Answering (QA)

- ❖ It is generally believed that learning a good dense vector representation needs a large number of labeled pairs of question and contexts.
- ❖ Dense retrieval methods have never been shown to outperform TF-IDF/BM25 for open-domain QA before ORQA (Lee et al., 2019), which is unfortunately computationally intensive for pre-training and not specifically trained on Q&A data



Task-Specific Pre-trained Retriever (Supervised)

❑ Dense Passage Retriever (DPR): Pretrained for Question Answering (QA)



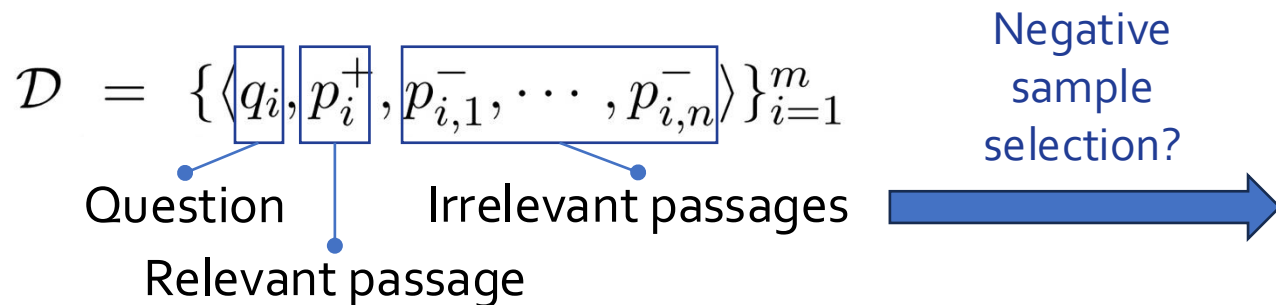
Task-Specific Pre-trained Retriever (Supervised)

❑ Dense Passage Retriever (DPR): Pretrained for Question Answering (QA)

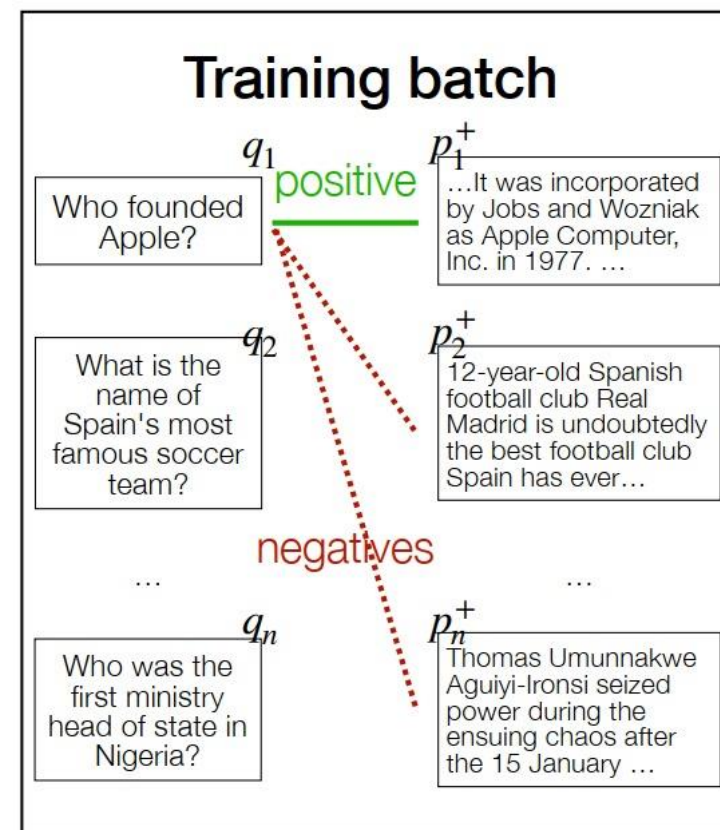
- Learning Objective

$$L(q_i, p_i^+, p_{i,1}^-, \dots, p_{i,n}^-)$$
$$= -\log \frac{e^{\text{sim}(q_i, p_i^+)}}{e^{\text{sim}(q_i, p_i^+)} + \sum_{j=1}^n e^{\text{sim}(q_i, p_{i,j}^-)}}$$

- Training data: Question-Passage Sets



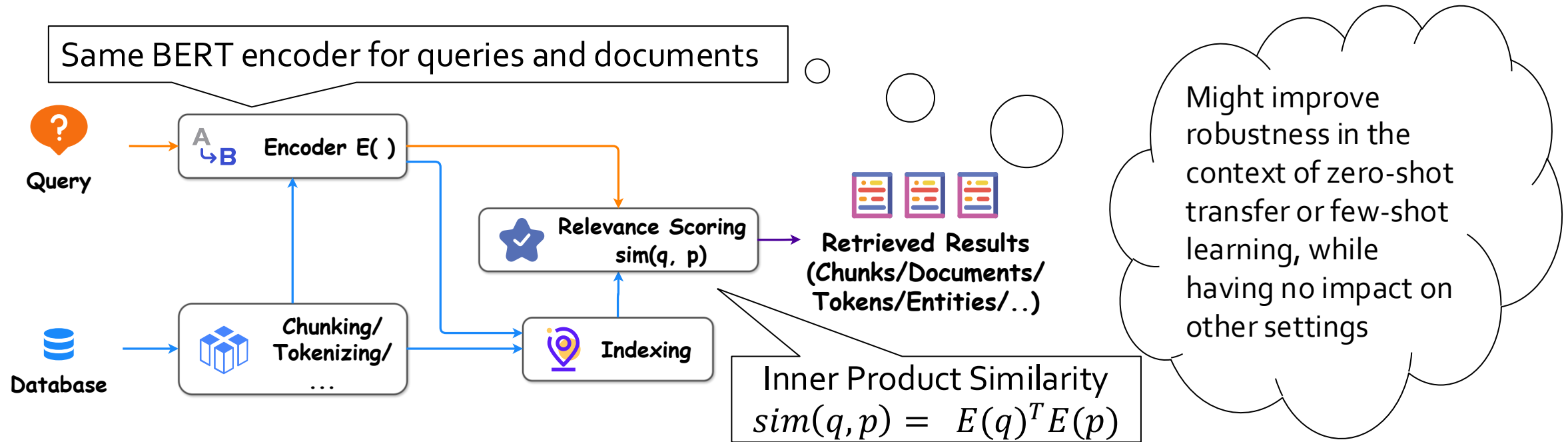
- Training with in-batch negatives



General-Purpose Pre-trained Retriever (Unsupervised)

❑ **Contriever:** Pre-trained with unsupervised learning

- ❖ supervised pretrained methods do not transfer well to new applications with no training data



Contrastive learning with unaligned documents

$$\mathcal{L}(q, k_+) = - \frac{\exp(s(q, k_+)/\tau)}{\exp(s(q, k_+)/\tau) + \sum_{i=1}^K \exp(s(q, k_i)/\tau)}$$

k_+ : positive pairs from a single document
 $(k_i)_{1,2,\dots,K}$: negative pairs with in-batch negative sampling and MoCo

DPR & Contriever Performance on OpenQA Tasks

End-to-end QA (Exact Match) Accuracy

Training	Model	NQ	TriviaQA	WQ	TREC	SQuAD
Single	BM25+BERT (Lee et al., 2019)	26.5	47.1	17.7	21.3	33.2
Single	ORQA (Lee et al., 2019)	33.3	45.0	36.4	30.1	20.2
Single	HardEM (Min et al., 2019a)	28.1	50.9	-	-	-
Single	GraphRetriever (Min et al., 2019b)	34.5	56.0	36.4	-	-
Single	PathRetriever (Asai et al., 2020)	32.6	-	-	-	56.5
Single	REALM _{Wiki} (Guu et al., 2020)	39.2	-	40.2	46.8	-
Single	REALM _{News} (Guu et al., 2020)	40.4	-	40.7	42.9	-
Single	BM25	32.6	52.4	29.9	24.9	38.1
	DPR	41.5	56.8	34.6	25.9	29.8
	BM25+DPR	39.0	57.0	35.2	28.0	36.7
Multi	DPR	41.5	56.8	42.4	49.4	24.1
	BM25+DPR	38.8	57.9	41.1	50.6	35.8

Both widely applied in
RAG and RA-LFMs

DPR in
RAG, FiD, RETRO,
EPR, UDR, ...

Contriever in
Self-RAG, Atlas,
RAVEN, ...

Inverse Cloze Task (Sachan et al., 2021)

Masked salient spans (Sachan et al., 2021)

BM25 (Ma et al., 2021)

Contriever

supervised model: DPR (Karpukhin et al., 2020)

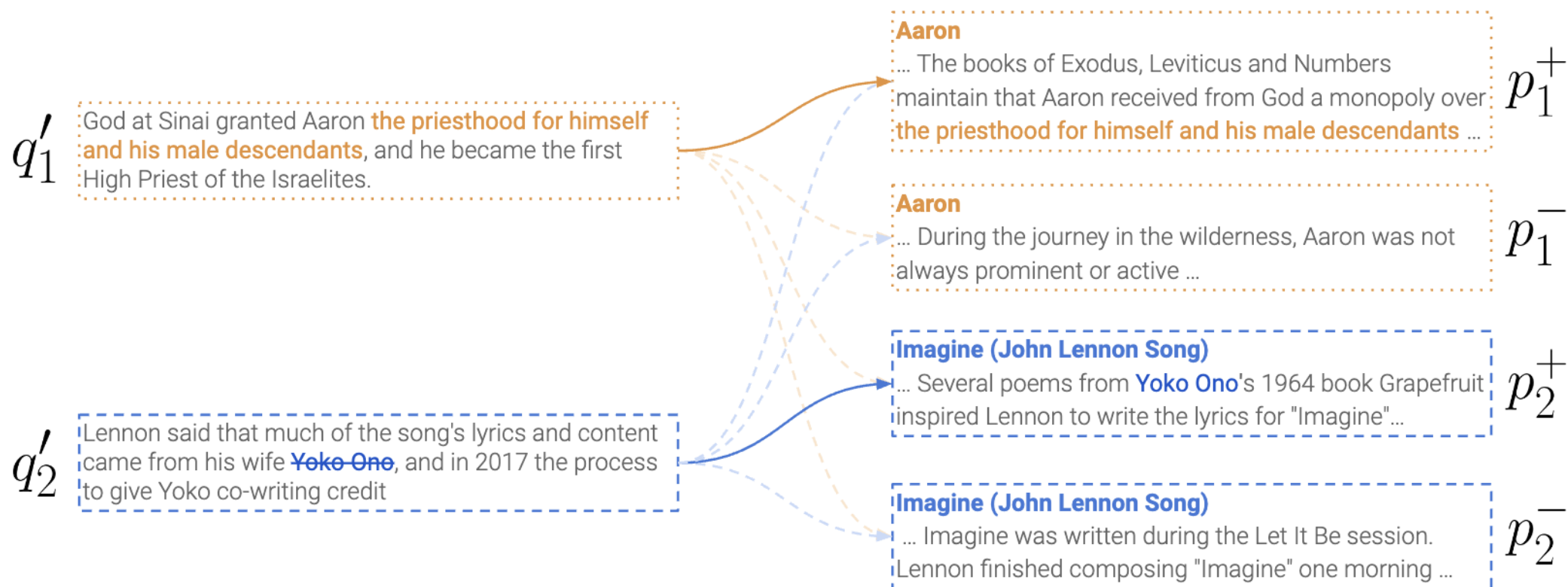
NaturalQuestions			TriviaQA		
R@5	R@20	R@100	R@5	R@20	R@100
32.3	50.9	66.8	40.2	57.5	73.6
41.7	59.8	74.9	53.3	68.2	79.4
-	62.9	78.3	-	76.4	83.2
47.8	67.8	82.1	59.4	74.2	83.2
-	78.4	85.4	-	79.4	85.0

Both better than
the sparse retriever!

Task-Specific Pre-trained Retriever (Unsupervised)

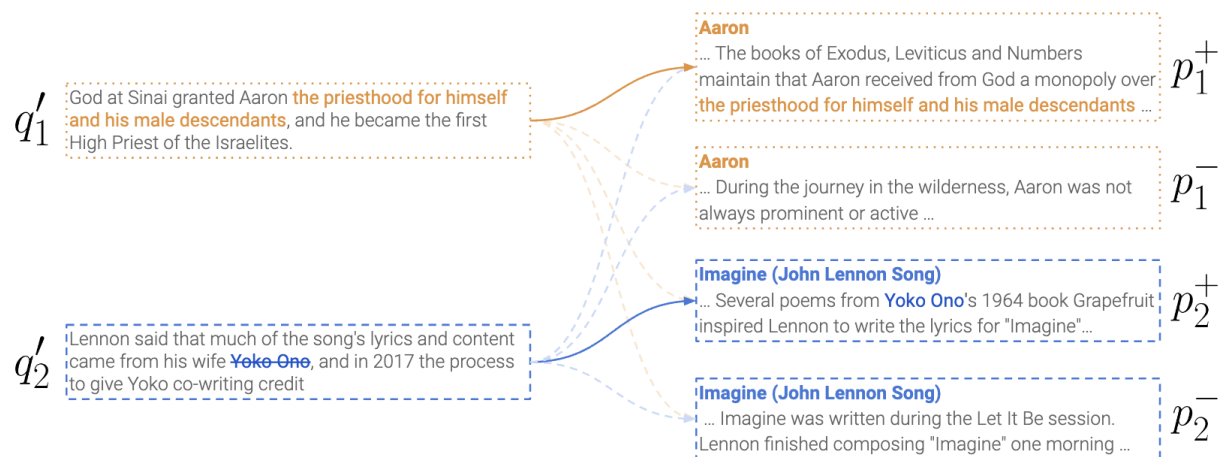
❑ Spider (Span-based unsupervised dense retriever)

- ❖ **Recurring Span Retrieval:** It is based on the notion of recurring spans within a document: given two paragraphs with the same recurring span, we construct a query from one of the paragraphs, while the other is taken as the target for retrieval

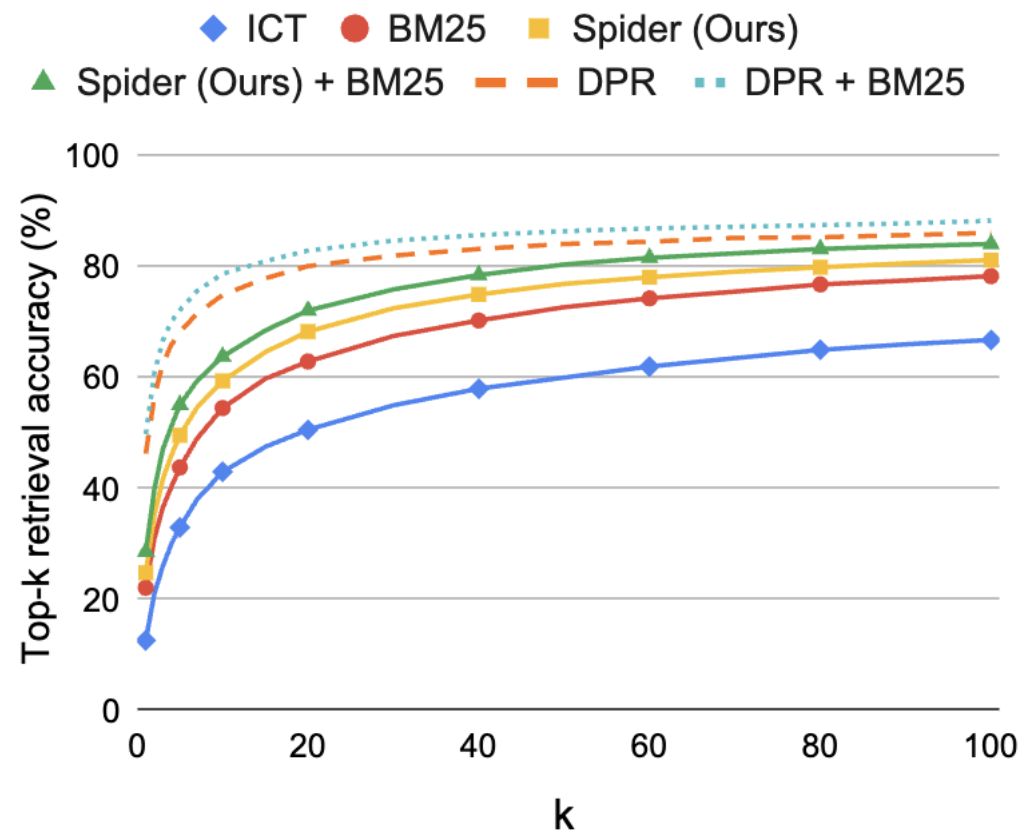


Task-Specific Pre-trained Retriever (Unsupervised)

❑ Learning and results of Spider

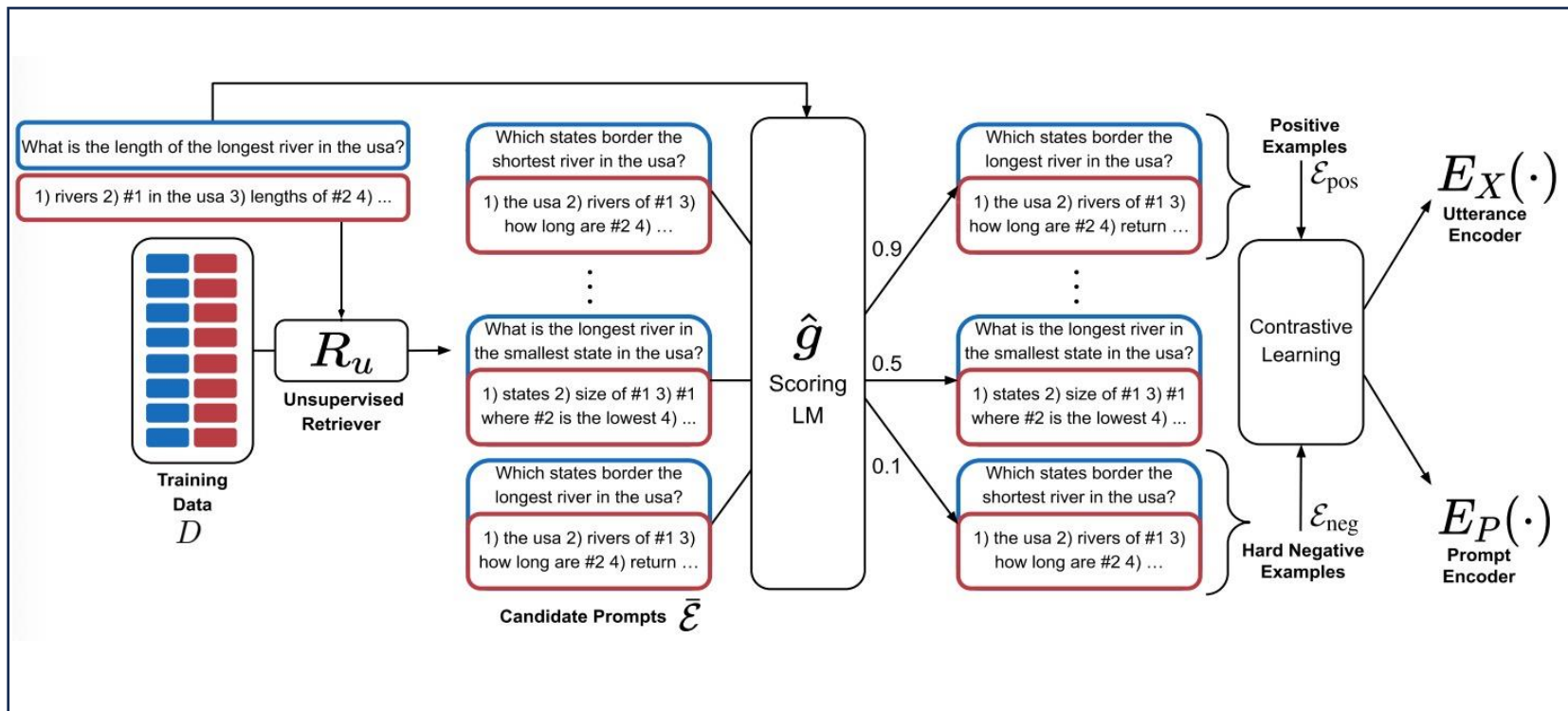


$$-\log \frac{\exp(s(q'_i, p_i^+))}{\sum_{j=1}^m (\exp(s(q'_i, p_j^+)) + \exp(s(q'_i, p_j^-)))}$$

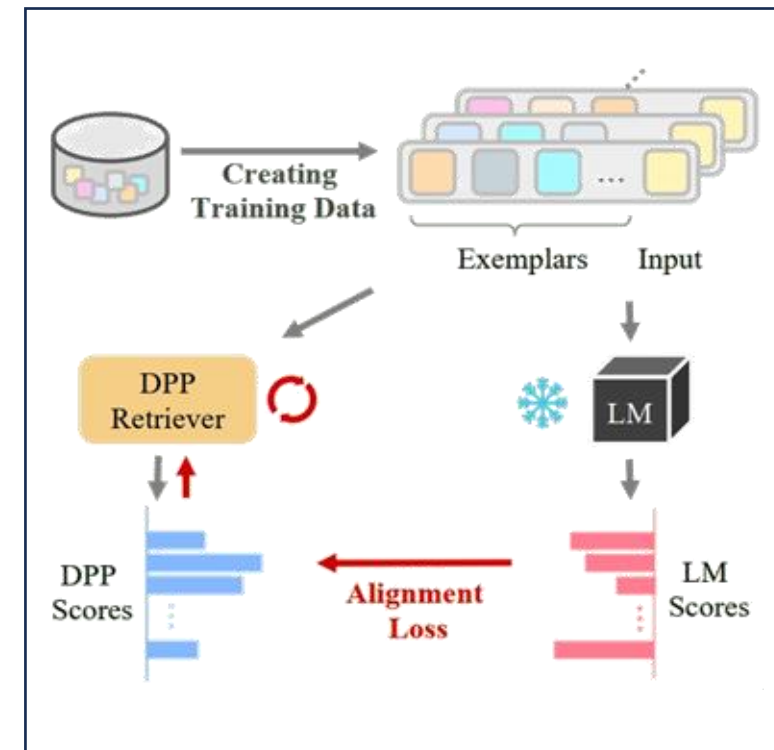


Retrievers for In-Context Learning of LFM

□ Prompt Retriever



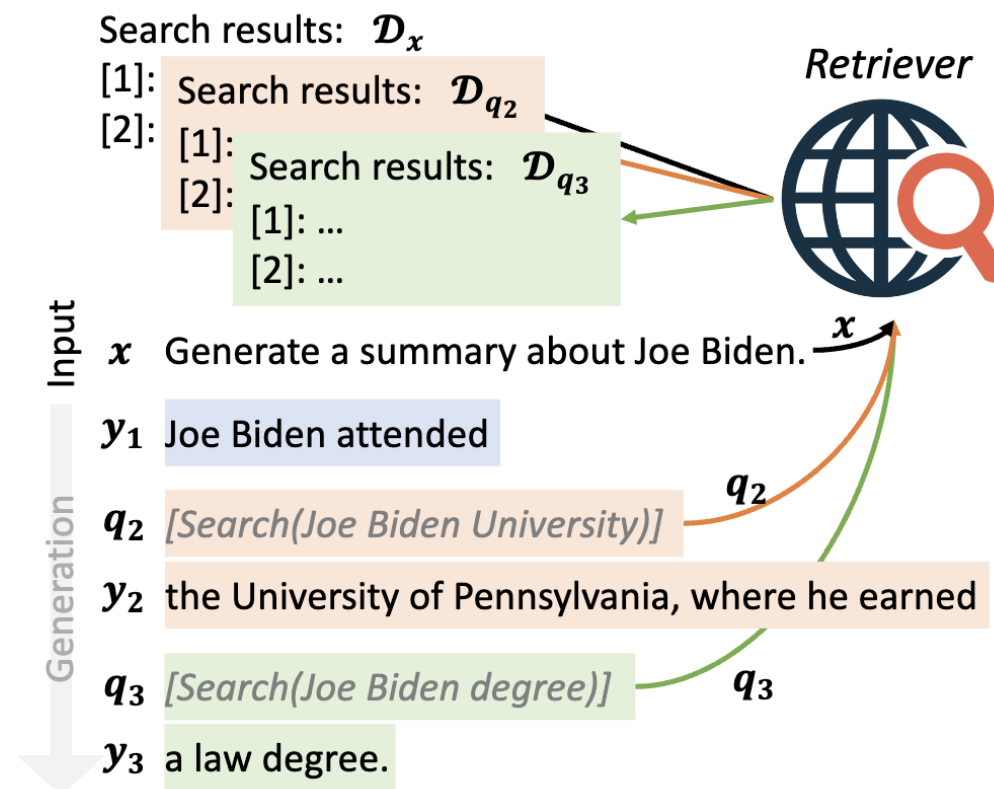
□ Exemplar Retriever (CEIL)



Search Engine as Retrievers

❑ Traditional retrieval methods

- ❖ May be difficult to update to real-time web documents
- ❖ May be a limit to the number of documents storable in the pre-defined database
- ❖ Will not take advantage of the high quality ranking that has been finely tuned in Internet Search engines over decades of use



PART 2: Architecture of RA-LFMs and Main Modules



Slides

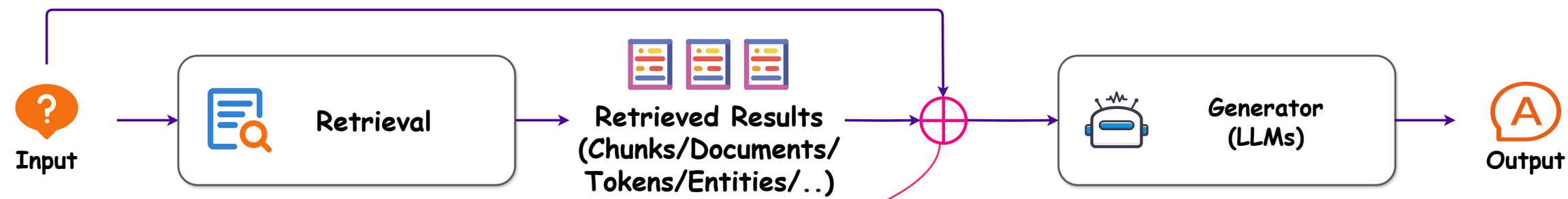


Website of this tutorial

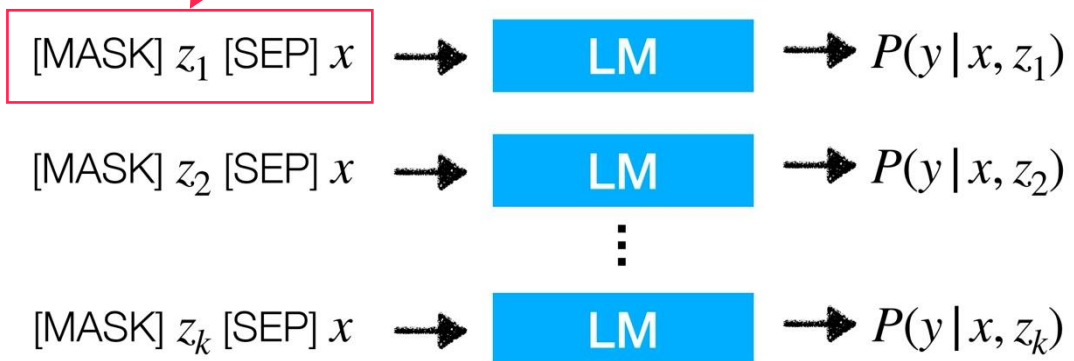
- RA-LFM architecture overview
- Retriever in RA-LFMs
- Retrieval results integration
- Pre/Post-retrieval techniques
- Special RA-LFM paradigms

Retrieved Results Integration: Input-layer Integration

❑ REALM



Integrating the retrieved passage $\mathbf{z}$ and $\mathbf{x}$ the original input

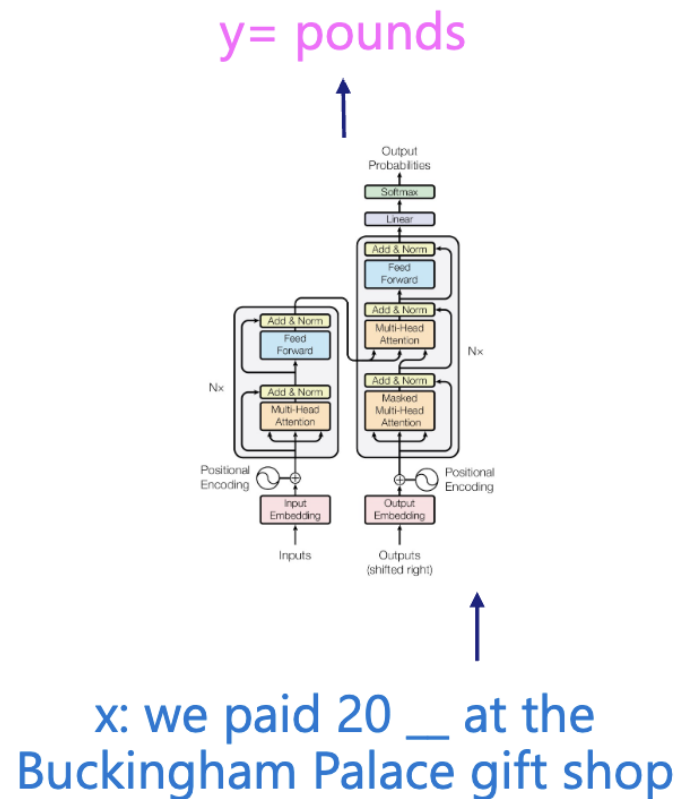


Weighted aggregating the prediction results based on all retrieved passages

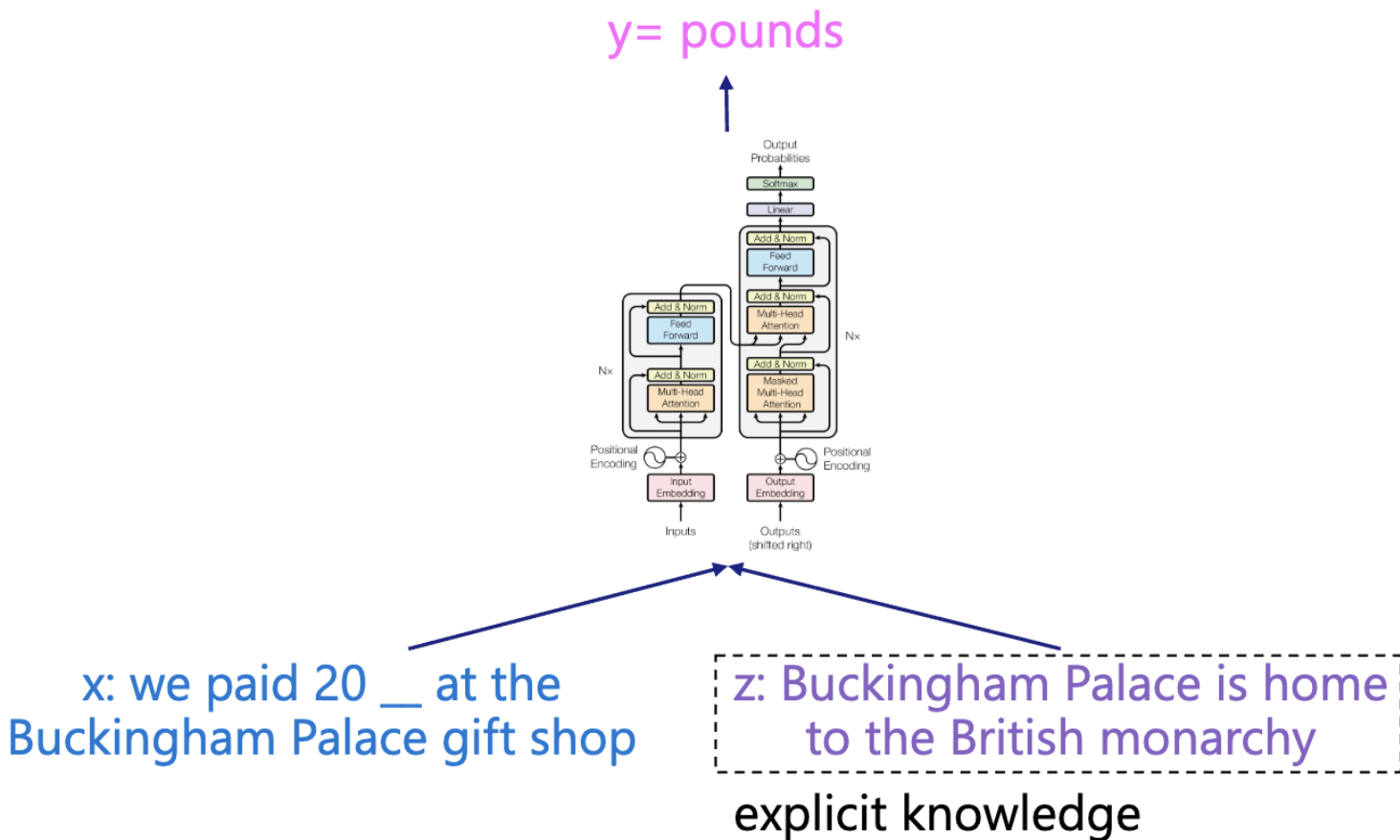
$$\sum_{z \in \mathcal{D}} P(z | x) P(y | x, z)$$

Retrieval-Augmented Generator

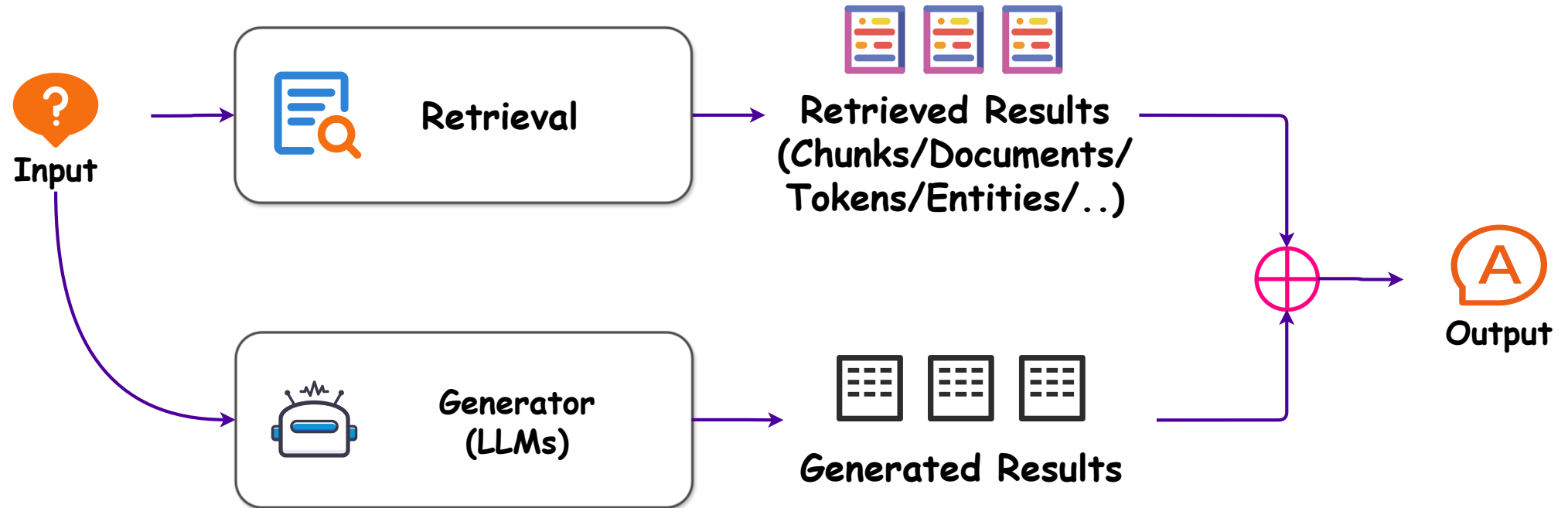
Typical encoder: $p(y|x)$



Knowledge-augmented encoder: $p(y|x, z)$

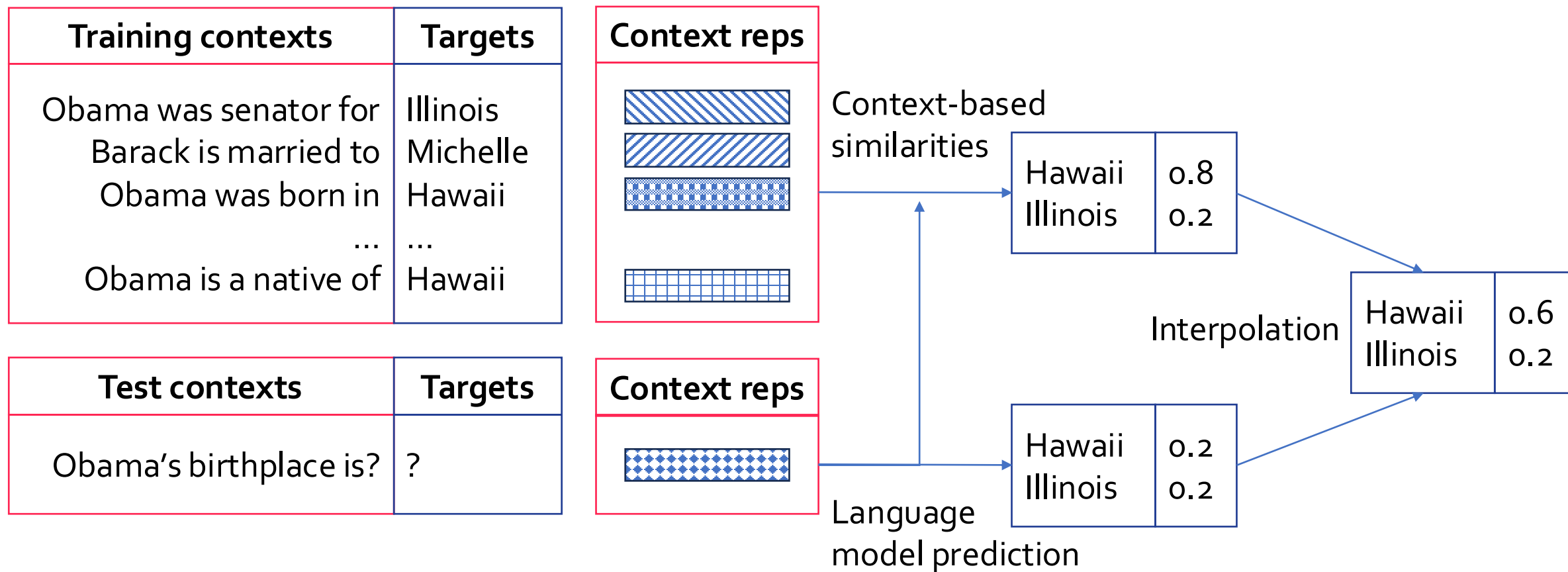


Retrieved Results Integration: Output-layer integration

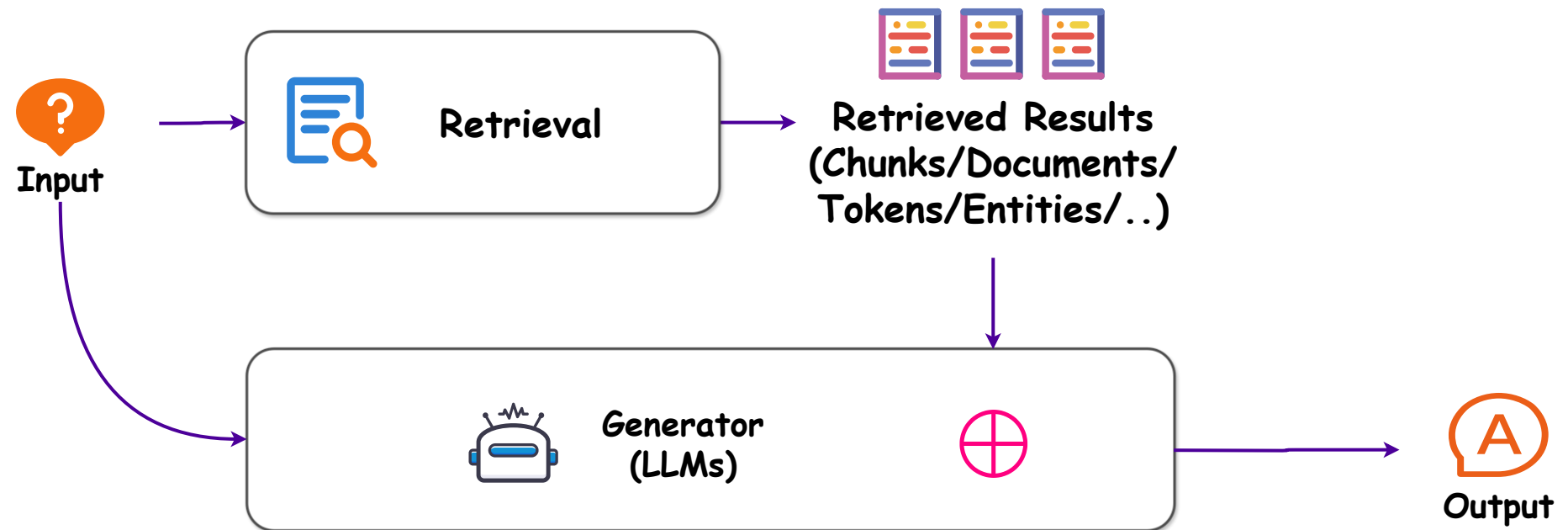


RA-LFM Architecture: Output-layer Integration

- ❑ **kNN-LM**: Combining retrieved probabilities and predicted ones in generation

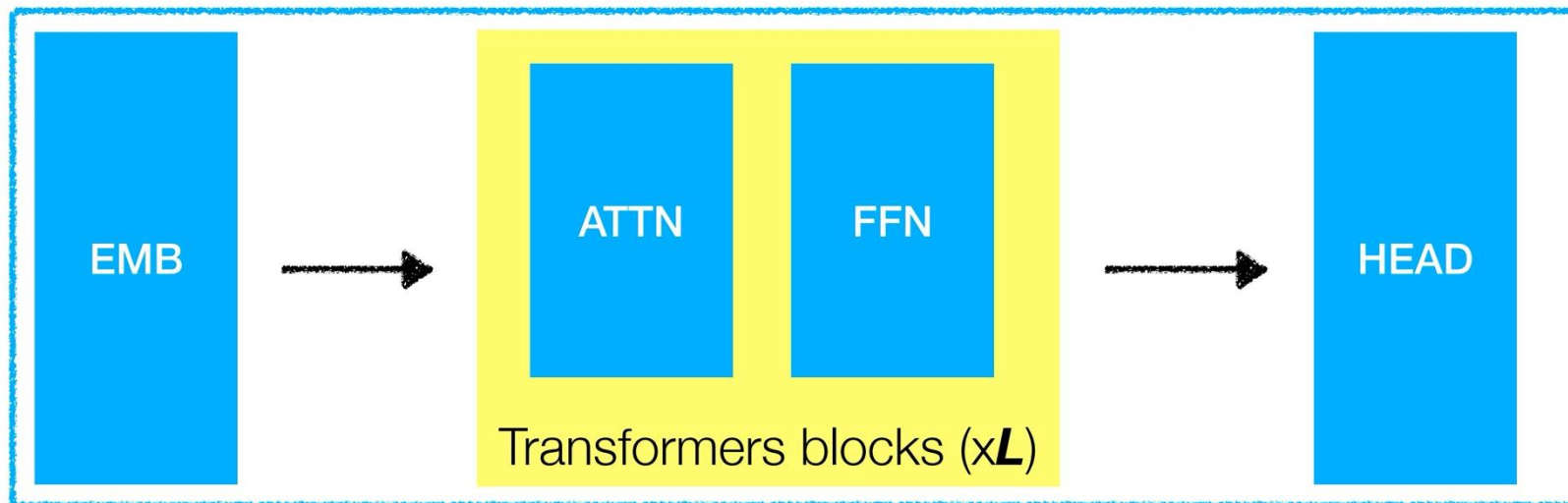


Retrieved Results Integration: Intermediate-layer Integration



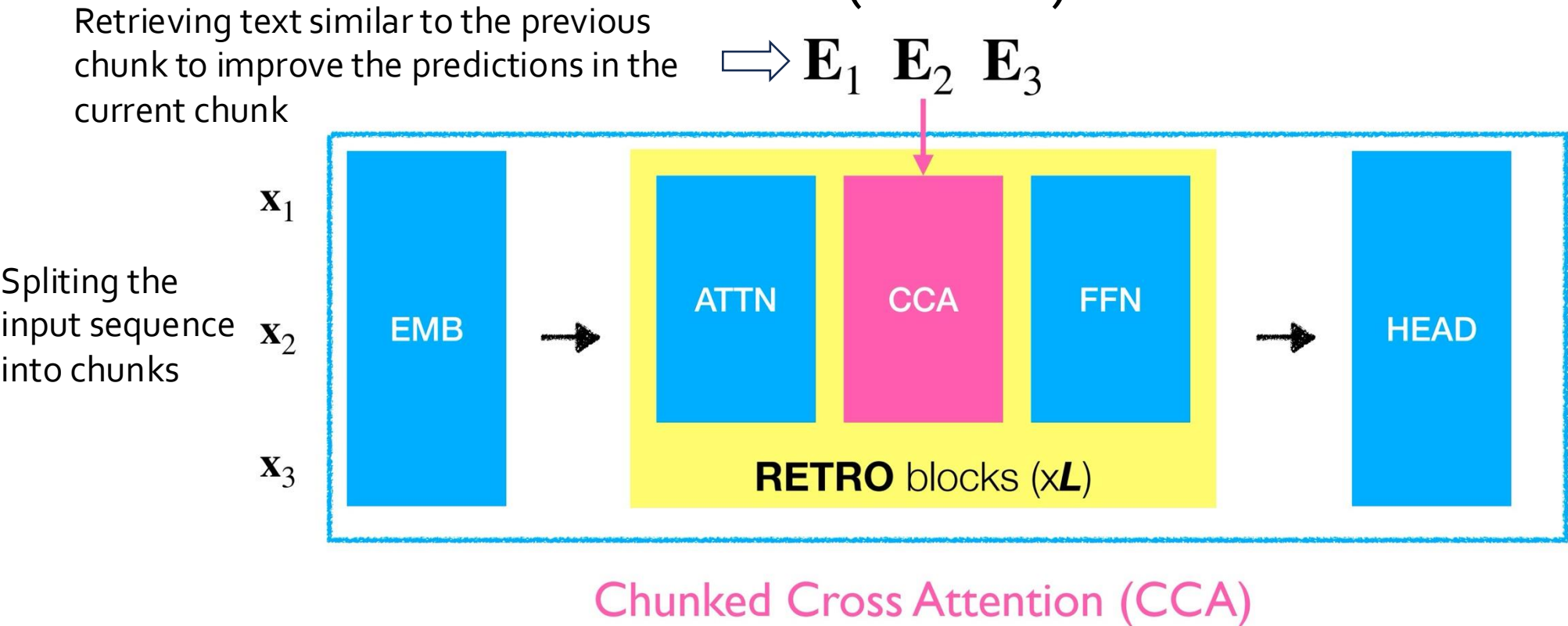
Retrieved Results Integration: Intermediate-layer Integration

Regular Decoder

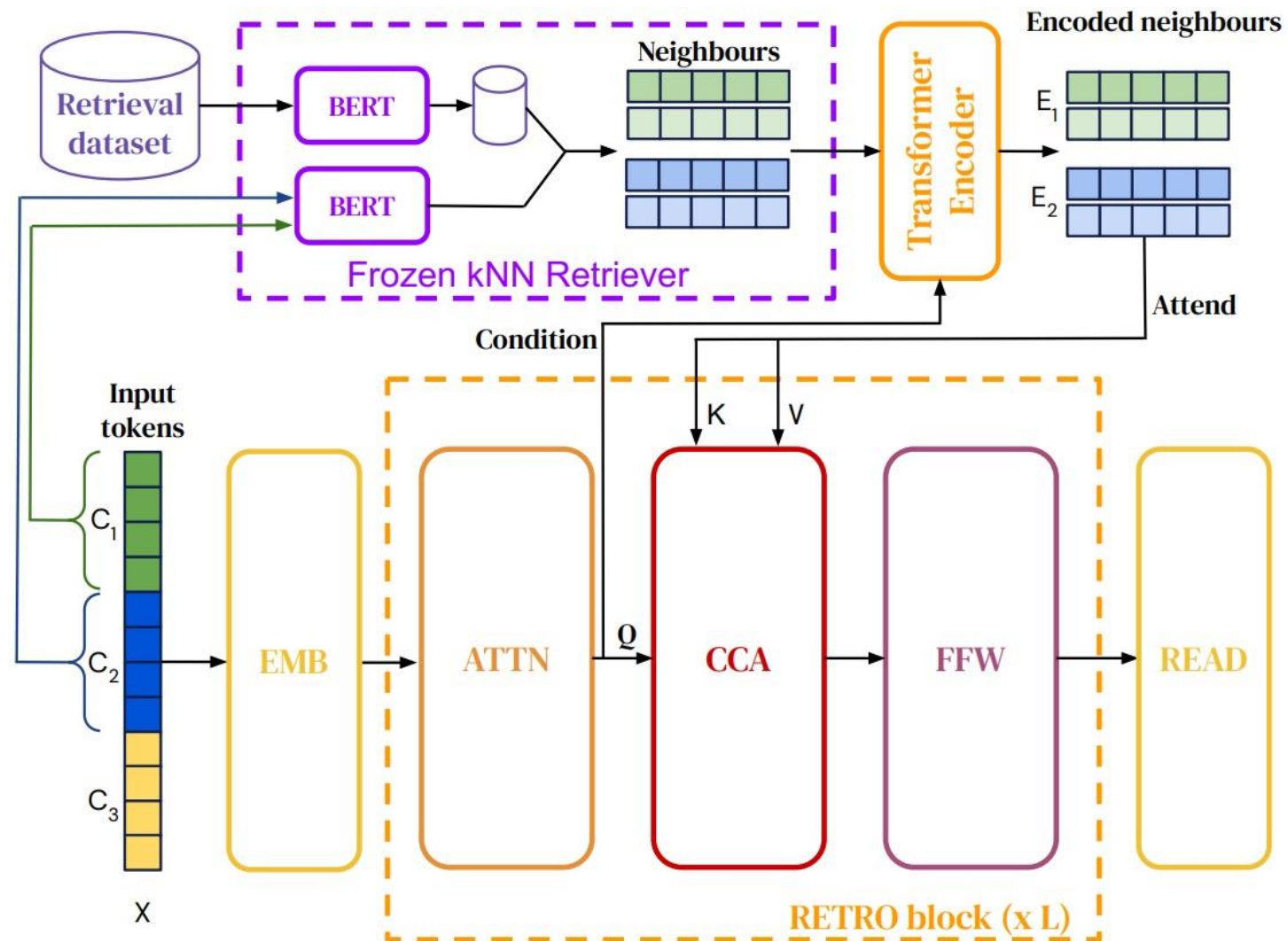


Retrieved Results Integration: Intermediate-layer Integration

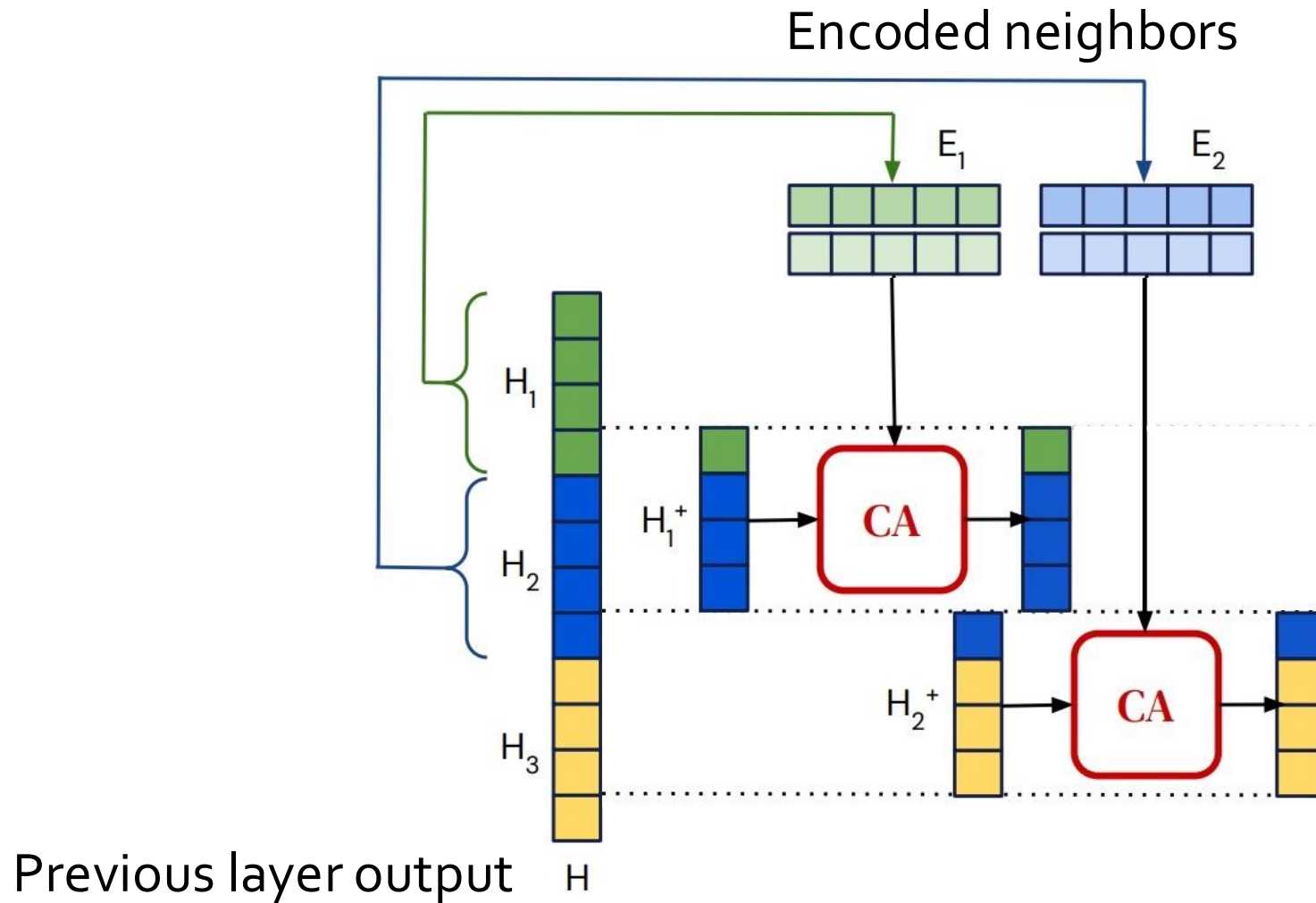
Decoder to incorporate retrieved results (RETRO)



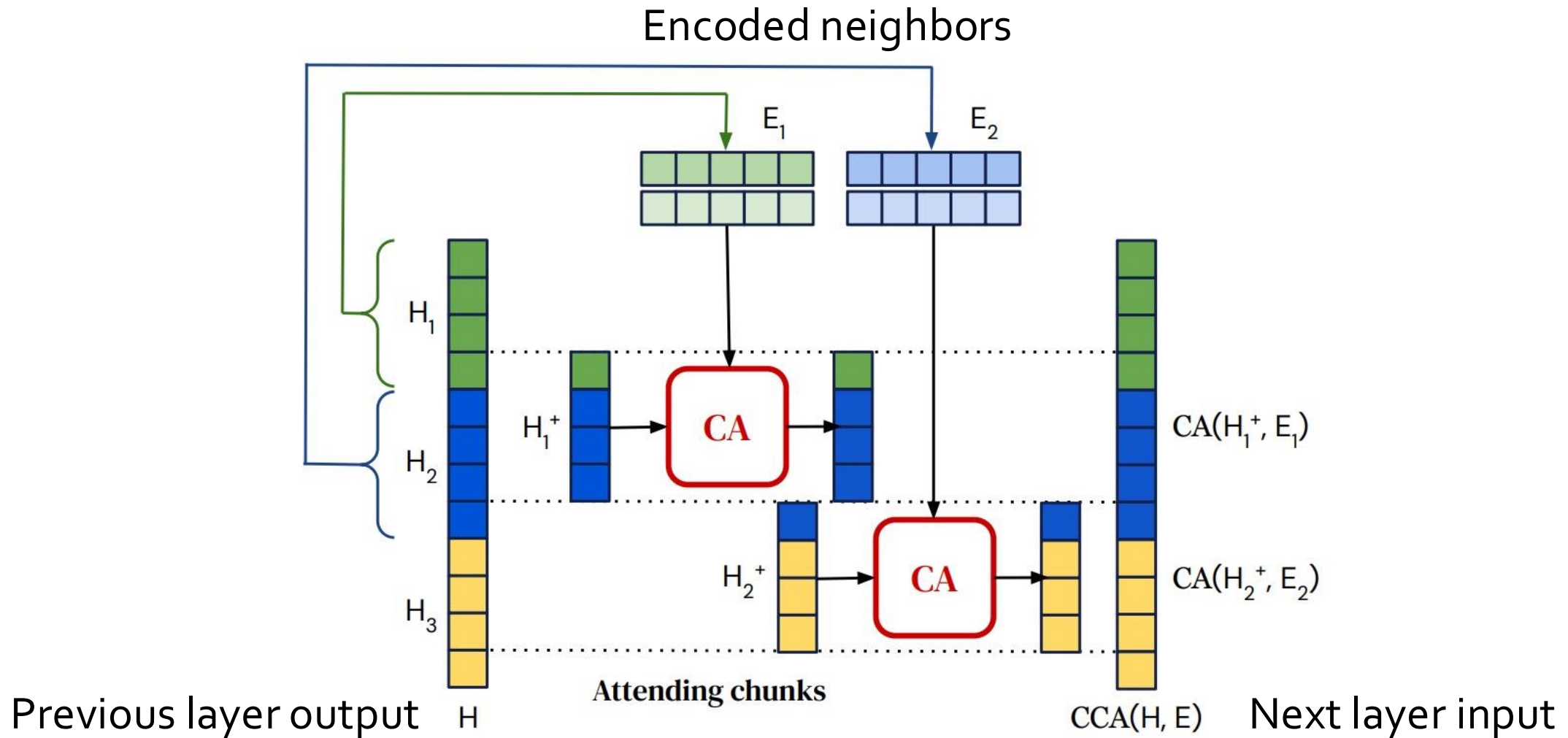
Retrieved Results Integration: Intermediate-layer Integration



Retrieved Results Integration: Intermediate-layer Integration



Retrieved Results Integration: Intermediate-layer Integration



PART 2: Architecture of RA-LFMs and Main Modules



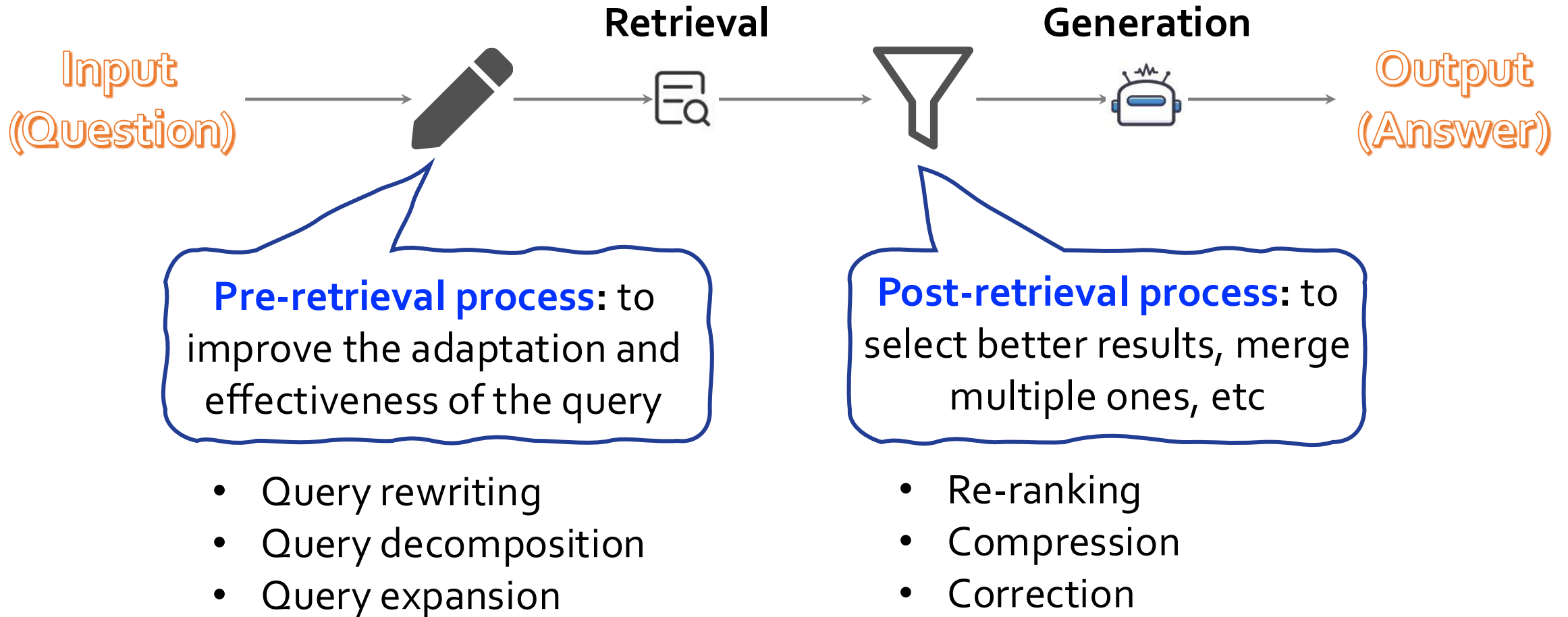
Slides



Website of this tutorial

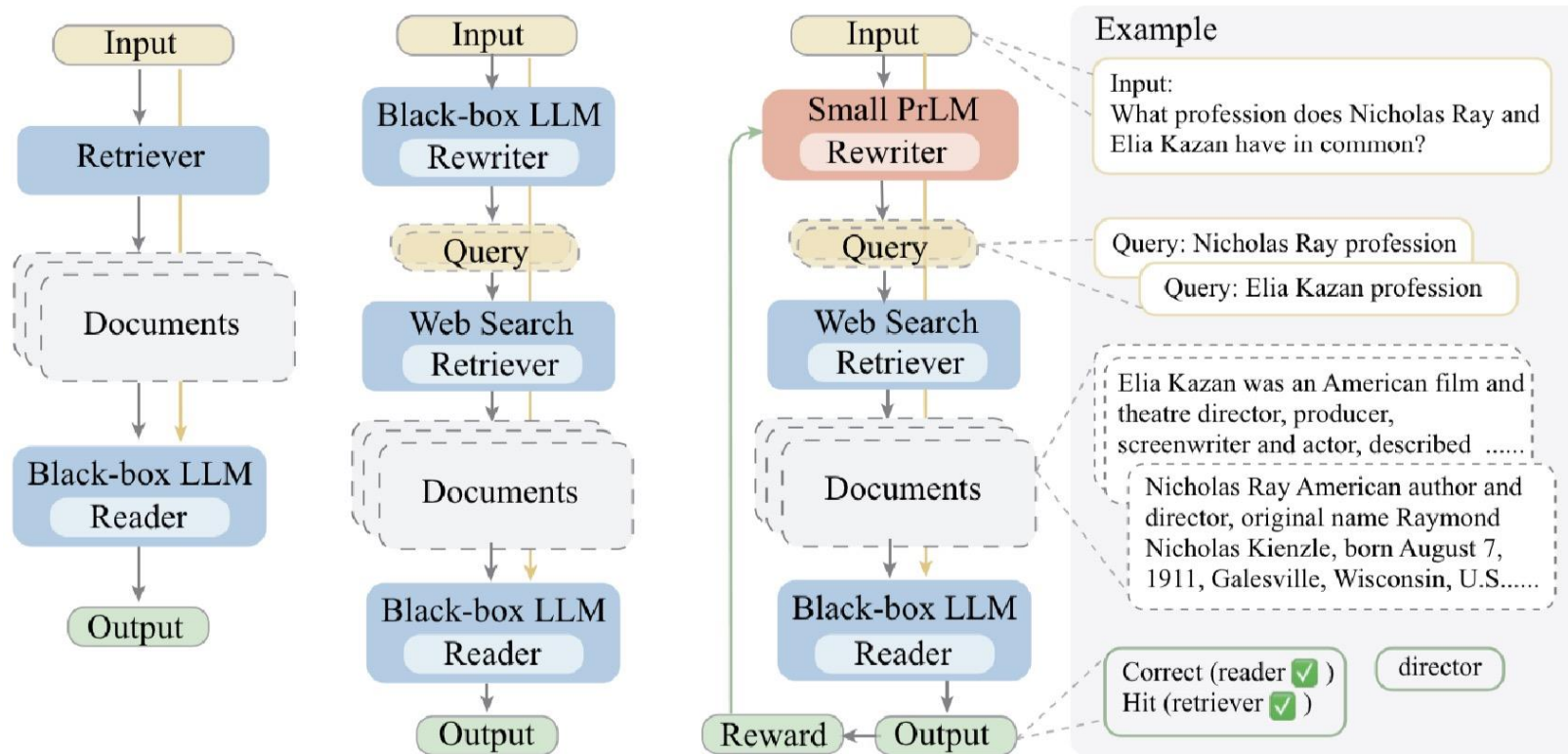
- RA-LFM architecture overview
- Retriever in RA-LFMs
- Retrieval results integration
- Pre/Post-retrieval techniques
- Special RA-LFM paradigms

Pre/Post-Retrieval Techniques



Pre-Retrieval Techniques

❑ **Query Rewriting:** to improve the adaptation of the query

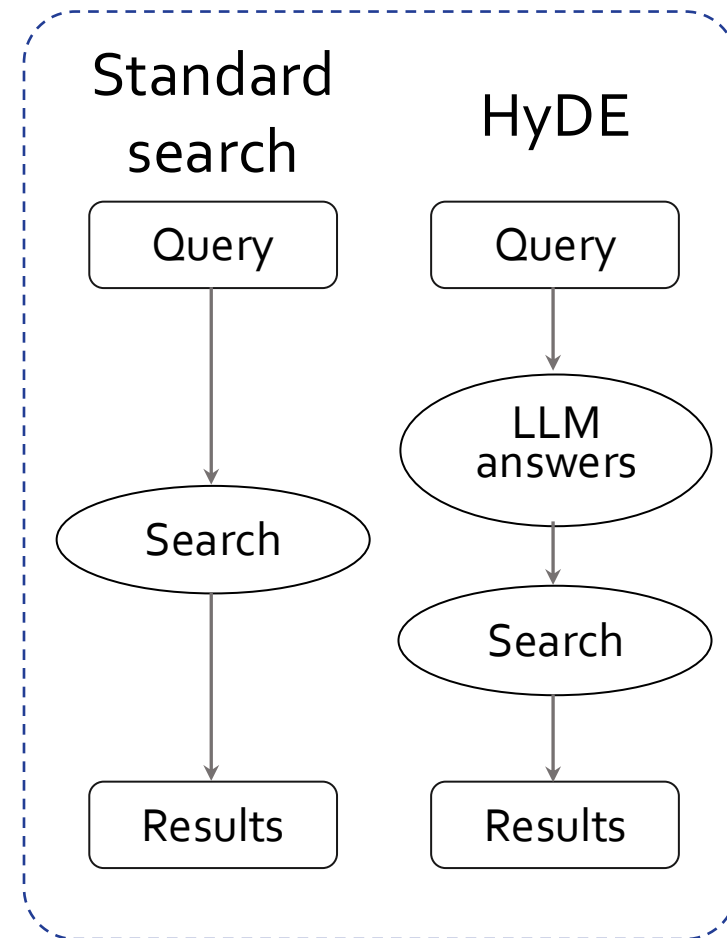
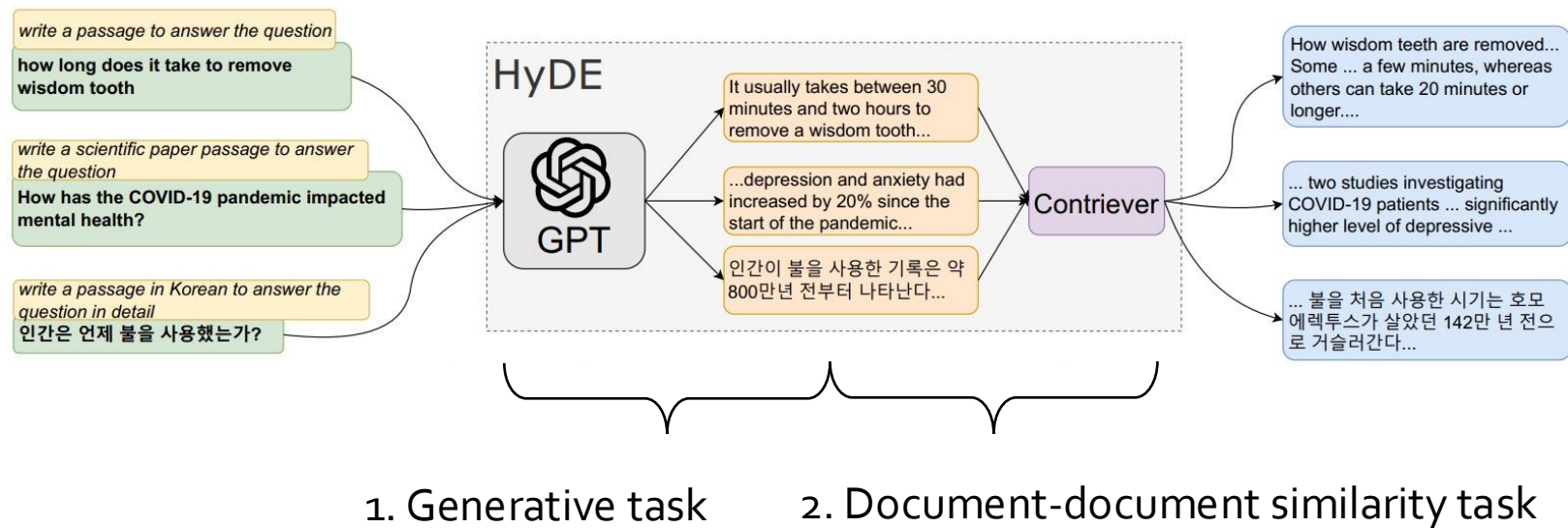


Model	EM	F ₁
<i>HotpotQA</i>		
Direct	32.36	43.05
Retrieve-then-read	30.47	41.34
LLM rewriter	32.80	43.85
Trainable rewriter	34.38	45.97
<i>AmbigNQ</i>		
Direct	42.10	53.05
Retrieve-then-read	45.80	58.50
LLM rewriter	46.40	58.74
Trainable rewriter	47.80	60.71
<i>PopQA</i>		
Direct	41.94	44.61
Retrieve-then-read	43.20	47.53
LLM rewriter	46.00	49.74
Trainable rewriter	45.72	49.51

Works on different QA settings

Pre-Retrieval Techniques

❑ HyDE: Hypothetical Document Embeddings



Pre-Retrieval Techniques

❑ Query Expansion

LLM Prompts

Write a passage that answers the given query:

Query: what state is this zip code 85282

Passage: Welcome to TEMPE, AZ 85282.
85282 is a rural zip code in Tempe, Arizona.
The population is primarily white...

...

Query: when was pokemon green released

Passage:

Method	Fine-tuning	MS MARCO dev			TREC DL 19
		MRR@10	R@50	R@1k	nDCG@10
Sparse retrieval					
BM25	✗	18.4	58.5	85.7	51.2*
+ query2doc	✗	21.4 ^{+3.0}	65.3 ^{+6.8}	91.8 ^{+6.1}	66.2^{+15.0}
BM25 + RM3	✗	15.8	56.7	86.4	52.2
docT5query (Nogueira and Lin)	✓	27.7	75.6	94.7	64.2
Dense retrieval w/o distillation					
ANCE (Xiong et al., 2021)	✓	33.0	-	95.9	64.5
HyDE (Gao et al., 2022)	✗	-	-	-	61.3
DPR _{bert-base} (our impl.)	✓	33.7	80.5	95.9	64.7
+ query2doc	✓	35.1^{+1.4}	82.6^{+2.1}	97.2^{+1.3}	68.7^{+4.0}

New query = original query + generated documents

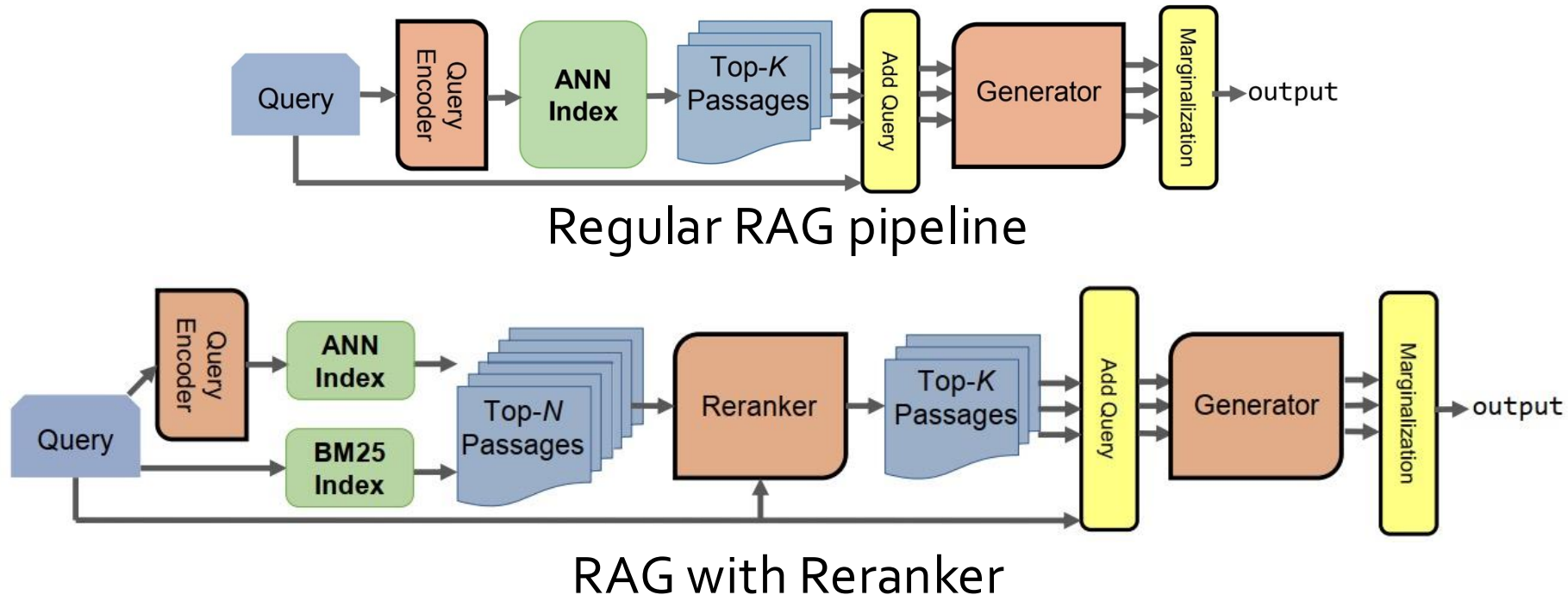
$$q^+ = \text{concat}(q, [\text{SEP}], d')$$

Works for both sparse and
dense retrievers

Post-Retrieval Techniques

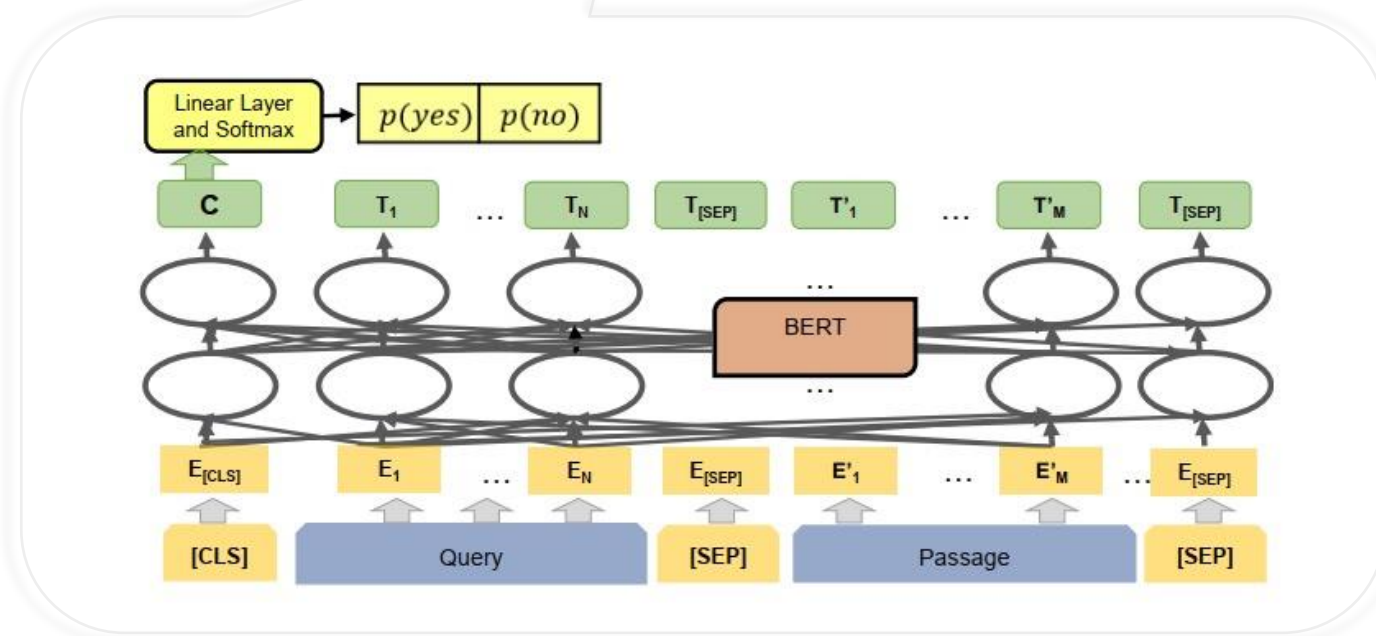
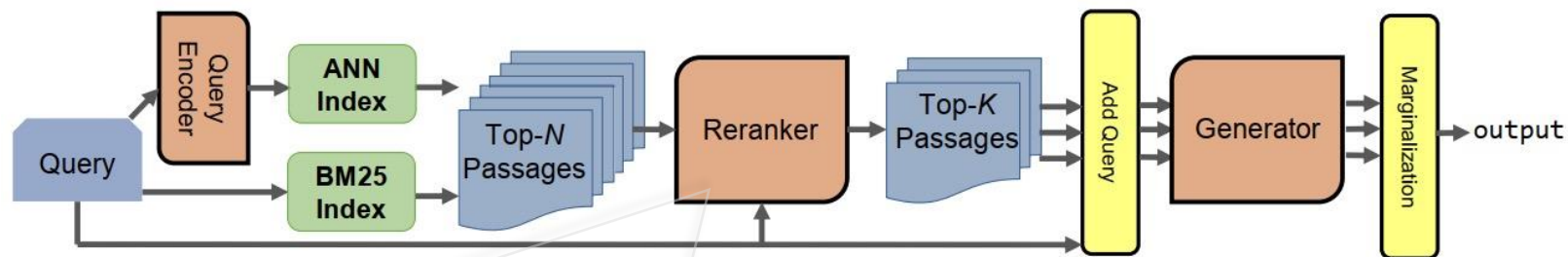
❑ Retrieved Result Rerank (Re2G)

- ❖ Results from initial retrieval can be greatly improved through the use of a reranker
- ❖ Reranker allows merging retrieval results from sources with incomparable scores, e.g., BM25 and neural initial retrieval



Retrieved Result Rerank (Re2G) Model

- ❑ Reranker: interaction model based on the sequence-pair classification



Retrieved Result Rerank (Re2G) Performance

	T-REx		NQ		TriviaQA		FEVER		WoW	
	R-Prec	R@5	R-Prec	R@5	R-Prec	R@5	R-Prec	R@5	R-Prec	R@5
BM25	46.88	69.59	24.99	42.57	26.48	45.57	42.73	70.48	27.44	45.74
DPR Stage 1	49.02	63.34	56.64	64.38	60.12	64.04	75.49	84.66	34.74	60.22
KGI ₀ DPR	65.02	75.52	64.65	69.60	60.55	63.65	80.34	86.53	48.04	71.02
Re ² G DPR	67.16	76.42	65.88	70.90	62.33	65.72	84.13	87.90	47.09	69.88
KGI ₀ DPR+BM25	60.48	80.06	36.91	66.94	40.81	64.79	65.95	90.34	35.63	68.47
Reranker Stage 1	81.22	87.00	70.78	73.05	71.80	71.98	87.71	92.43	55.50	74.98
Re ² G Reranker	81.24	88.58	70.92	74.79	60.37	70.61	90.06	92.91	57.89	74.62

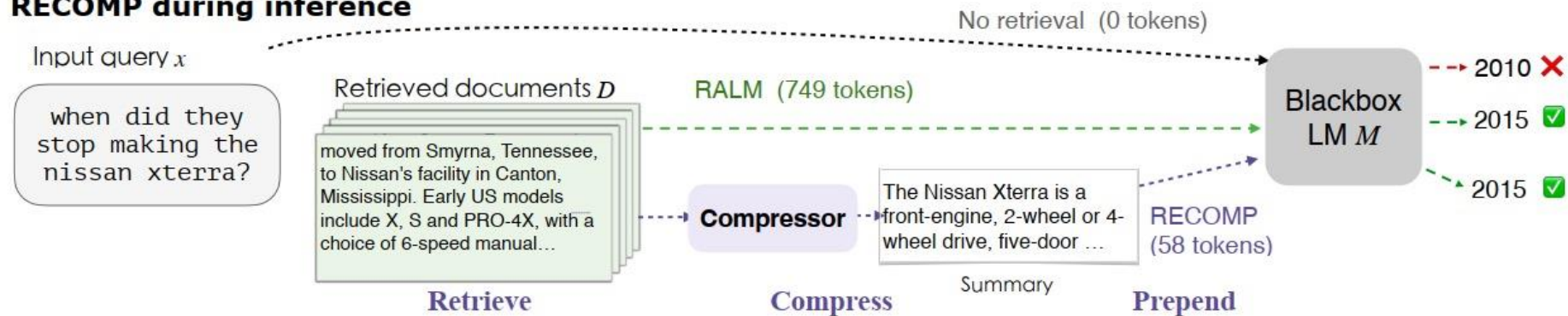
Significantly outperforms pipelines without the *Rerank* stage

Post-Retrieval Techniques

❑ Retrieved Result Compression

- ❖ To reduce the computational costs and also relieve the burden of LMs to identify relevant information in long retrieved documents.

RECOMP during inference



❑ Compressor Learning Objectives

- ❖ Concise
- ❖ Effective
- ❖ Faithful

Retrieved Result Compression Performance

❑ QA tasks

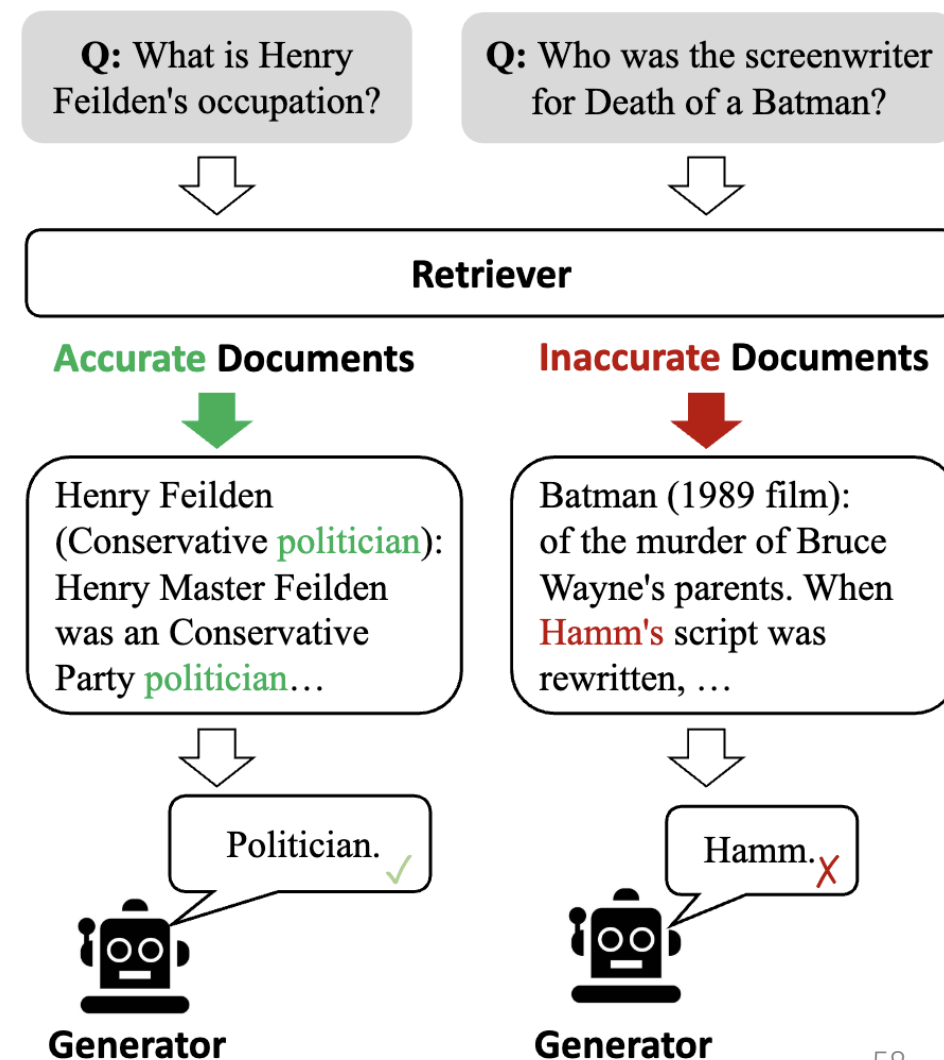
In-Context evidence	# tok	NQ EM	F1	# tok	TQA EM	F1	# tok	HotpotQA EM	F1
-	0	21.99	29.38	0	49.33	54.85	0	17.80	26.10
<i>RALM without compression</i>									
Top 1 documents	132	33.07	41.45	136	57.84	64.94	138	28.80	40.58
Top 5 documents	660	39.39	48.28	677	62.37	70.09	684	32.80	43.90
<i>Phrase/token level compression</i>									
Top 5 documents (NE)	338	23.60	31.02	128	54.96	61.19	157	22.20	31.89
Top 5 documents (BoW)	450	28.48	36.84	259	58.16	65.15	255	25.60	36.00
<i>Extractive compression of top 5 documents</i>									
Oracle	34	60.22	64.25	32	79.29	82.06	70	41.80	51.07
Random	32	23.27	31.09	31	50.18	56.24	61	21.00	29.86
BM25	36	25.82	33.63	37	54.67	61.19	74	26.80	38.02
DPR	39	34.32	43.38	41	56.58	62.96	78	27.40	38.15
Contriever	36	30.06	31.92	40	53.67	60.01	78	28.60	39.48
Ours	37	36.57	44.22	38	58.99	65.26	75	30.40	40.14

Outperforms representative sparse and dense retrievers

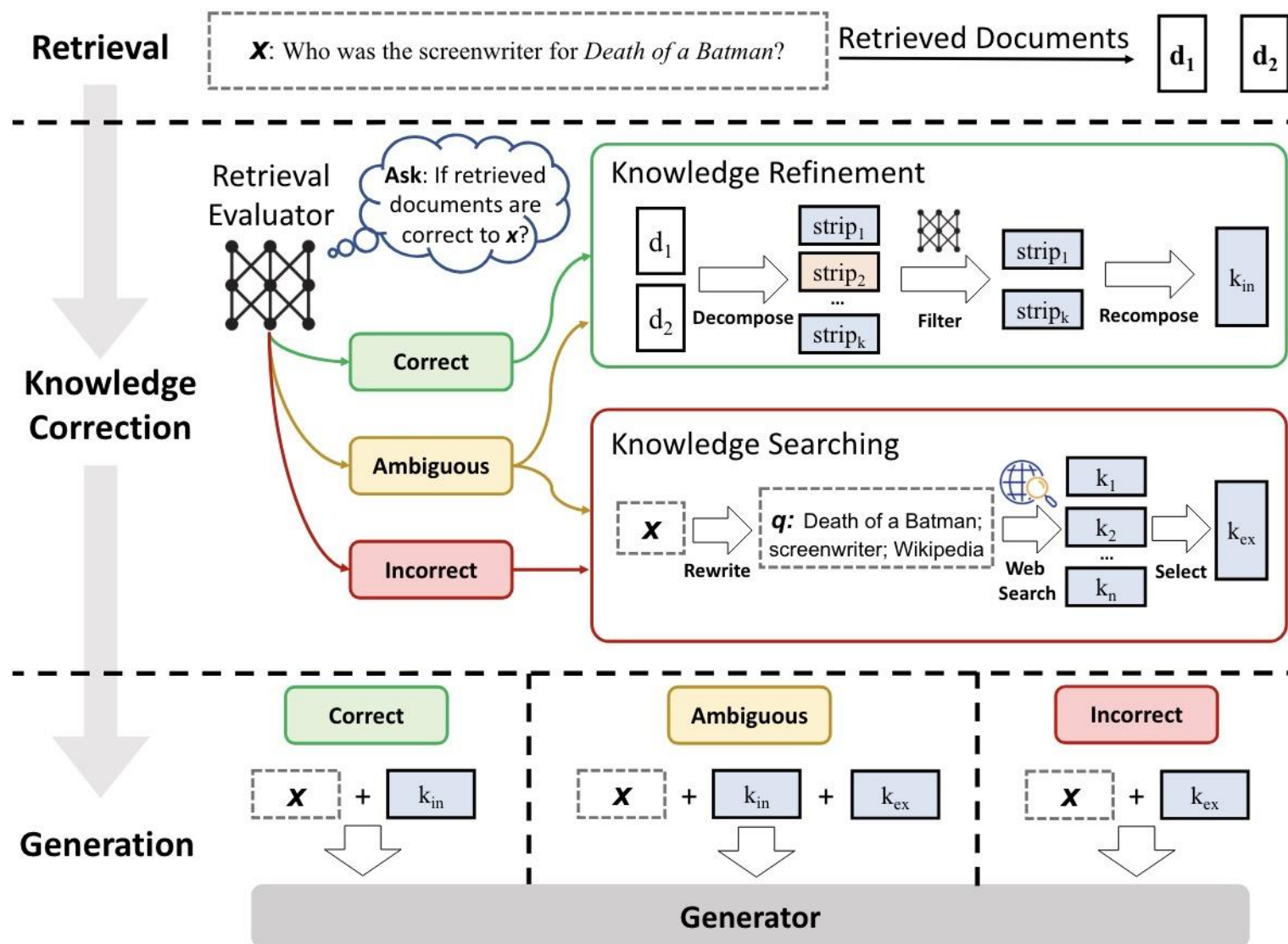
Post-Retrieval Techniques: Corrective RAG

❑ Corrective Retrieval Augmented Generation (CRAG)

- ❖ Although retrieval-augmented generation (RAG) is a practicable complement to LLMs, it relies heavily on the **relevance of retrieved documents**
- ❖ A lightweight **retrieval evaluator** is designed to assess the overall quality of retrieved documents for a query, returning a **confidence degree** based on which different knowledge retrieval actions can be triggered

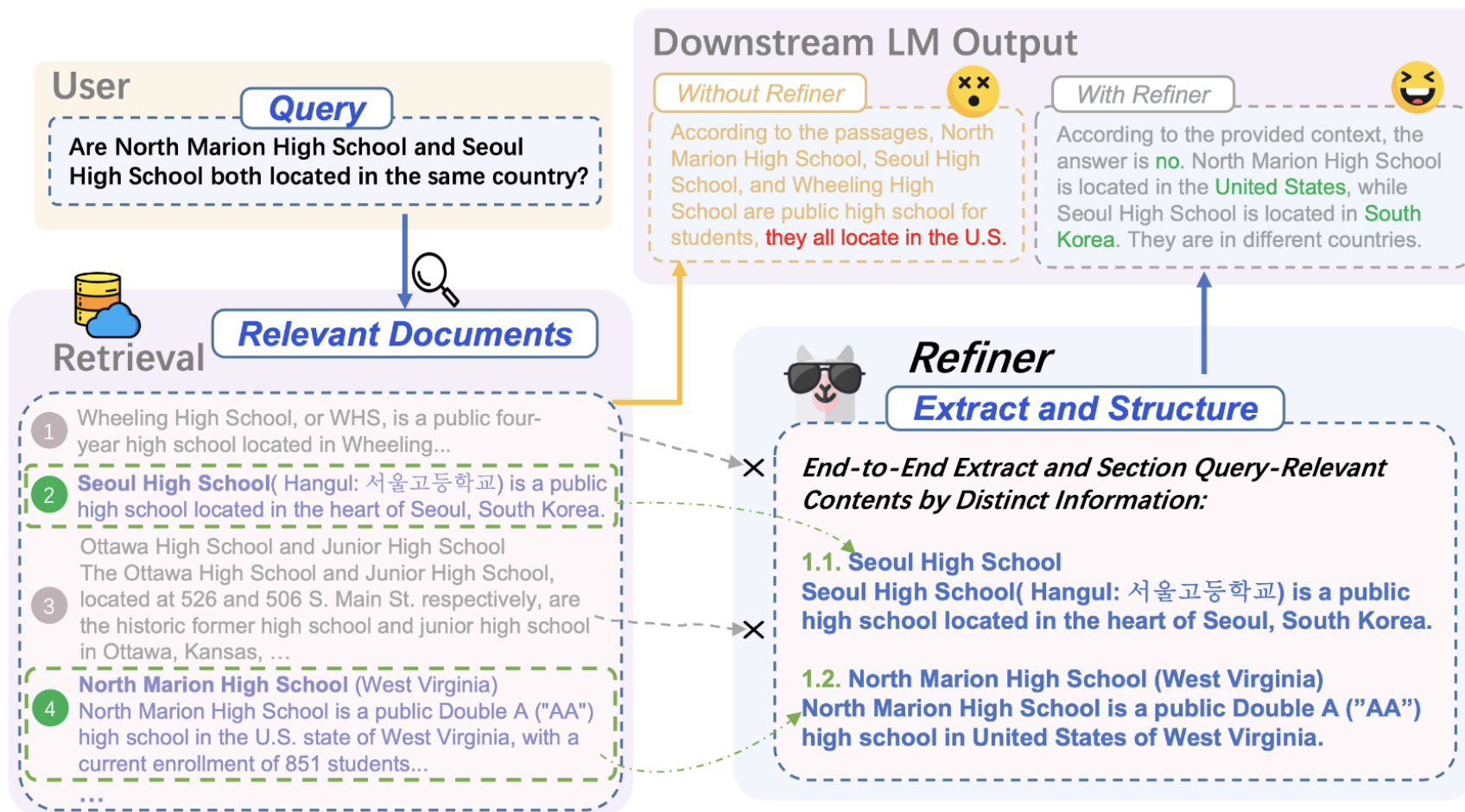


Post-Retrieval Techniques: Corrective RAG



Post-Retrieval Techniques: Refiner

- ❑ **Refiner**: leveraging a single decoder-only LLM to adaptively extract query relevant contents verbatim along with the necessary context



PART 2: Architecture of RA-LFMs and Main Modules



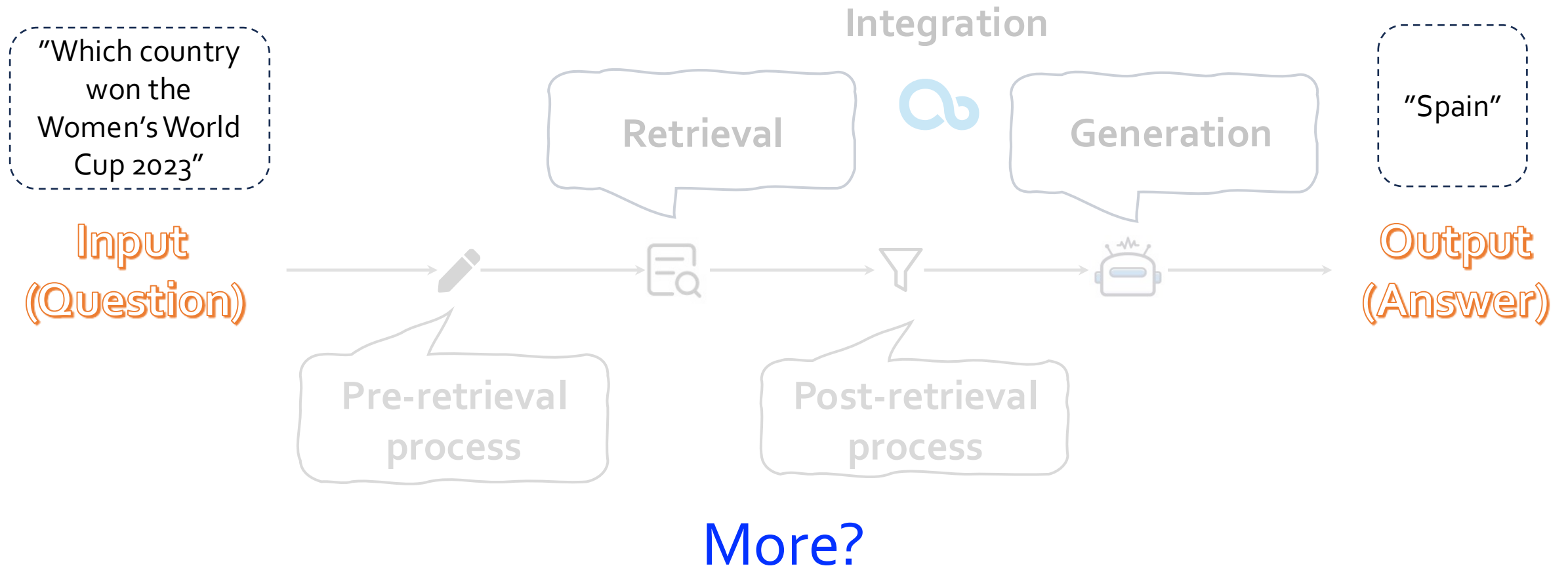
Slides



Website of this tutorial

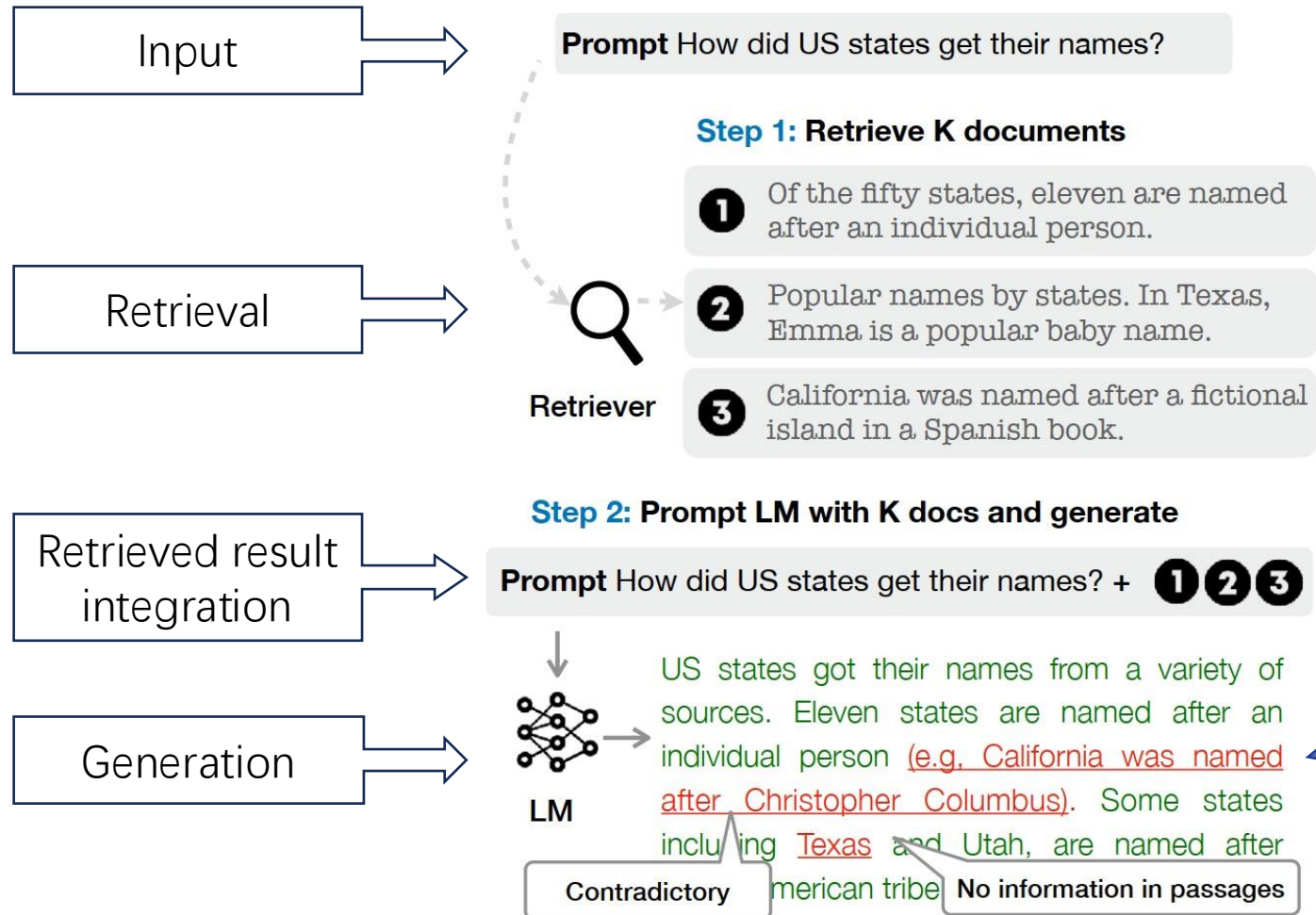
- **RA-LFM architecture overview**
- **Retriever in RA-LFMs**
- **Retrieval results integration**
- **Pre/Post-retrieval techniques**
- **Special RA-LFM paradigms**

Beyond Standard Pipelines and Components?



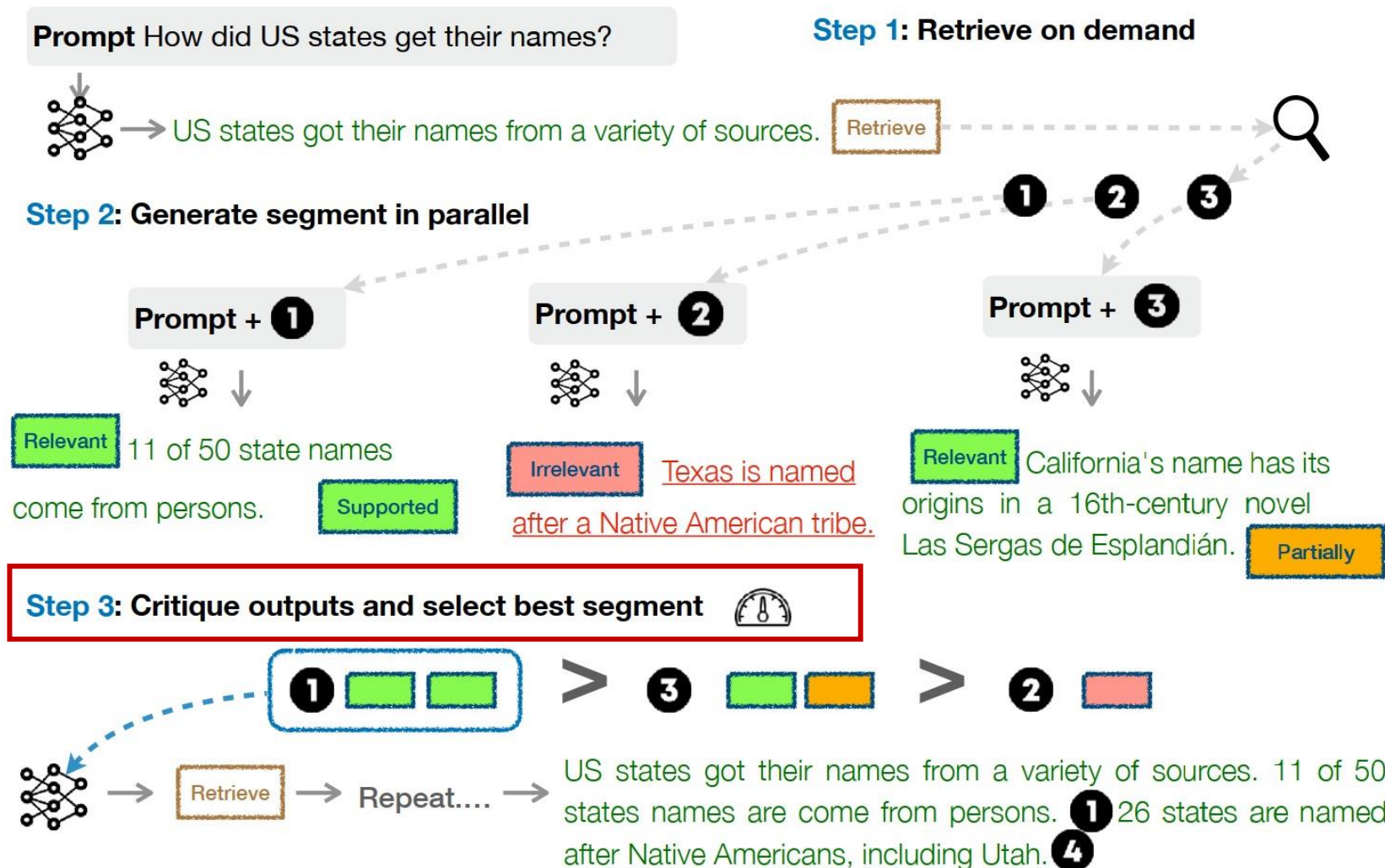
Special RAG Pipeline: Self-Reflective RAG (SELF-RAG)

❑ General Retrieval-Augmented Generation (RAG)



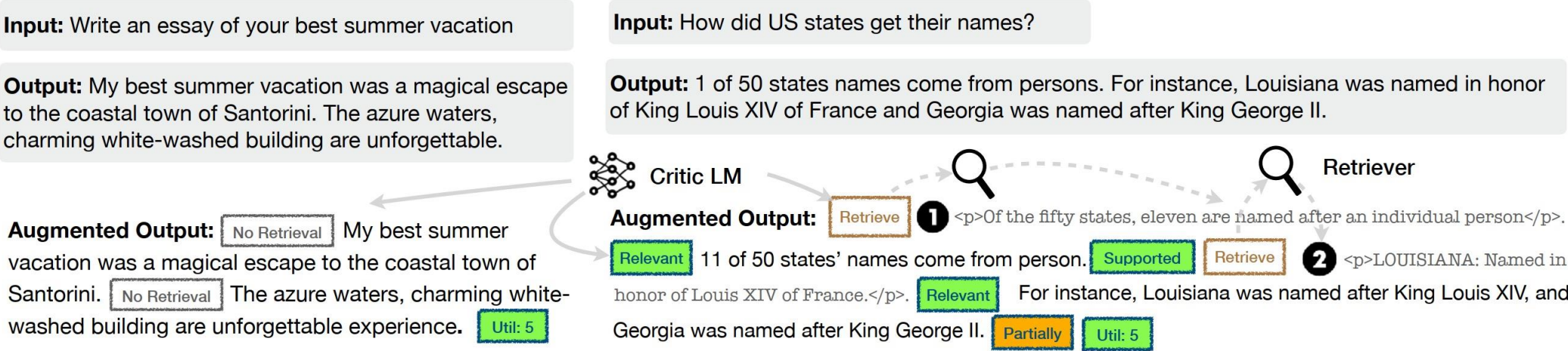
Retrieval results may not be necessary or helpful!

SELF-RAG Overview



Key Technical Design in SELF-RAG

❑ Critic Model Training



❑ Four types of reflection tokens used in SELF-RAG

Type	Input	Output	Definitions
Retrieve	$x / x, y$	{yes, no, continue}	Decides when to retrieve with $\mathcal{R}$
ISREL	x, d	{ relevant , irrelevant}	d provides useful information to solve x .
ISSUP	x, d, y	{ fully supported , partially supported, no support}	All of the verification-worthy statement in y is supported by d .
ISUSE	x, y	{ 5 , 4, 3, 2, 1}	y is a useful response to x .

SELF-RAG Algorithm

Algorithm 1 SELF-RAG Inference

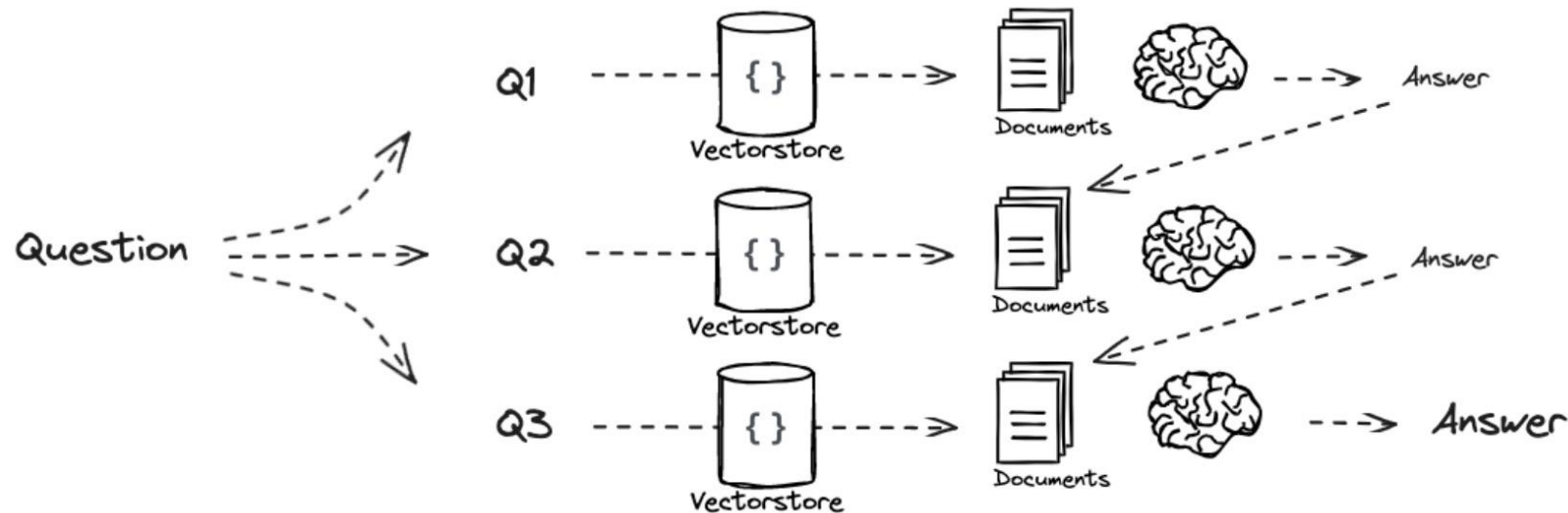
Require: Generator LM $\mathcal{M}$, Retriever $\mathcal{R}$, Large-scale passage collections $\{d_1, \dots, d_N\}$

- 1: **Input:** input prompt x and preceding generation $y_{<t}$, **Output:** next output segment y_t
 - 2: $\mathcal{M}$ predicts **Retrieve** given $(x, y_{<t})$
 - 3: **if** **Retrieve** == Yes **then**
 - 4: Retrieve relevant text passages $\mathbf{D}$ using $\mathcal{R}$ given (x, y_{t-1}) ▷ Retrieve
 - 5: $\mathcal{M}$ predicts **ISREL** given x, d and y_t given $x, d, y_{<t}$ for each $d \in \mathbf{D}$ ▷ Generate
 - 6: $\mathcal{M}$ predicts **ISSUP** and **ISUSE** given x, y_t, d for each $d \in \mathbf{D}$ ▷ Critique
 - 7: Rank y_t based on **ISREL**, **ISSUP**, **ISUSE** ▷ Detailed in Section 3.3
 - 8: **else if** **Retrieve** == No **then**
 - 9: $\mathcal{M}_{gen}$ predicts y_t given x ▷ Generate
 - 10: $\mathcal{M}_{gen}$ predicts **ISUSE** given x, y_t ▷ Critique
-

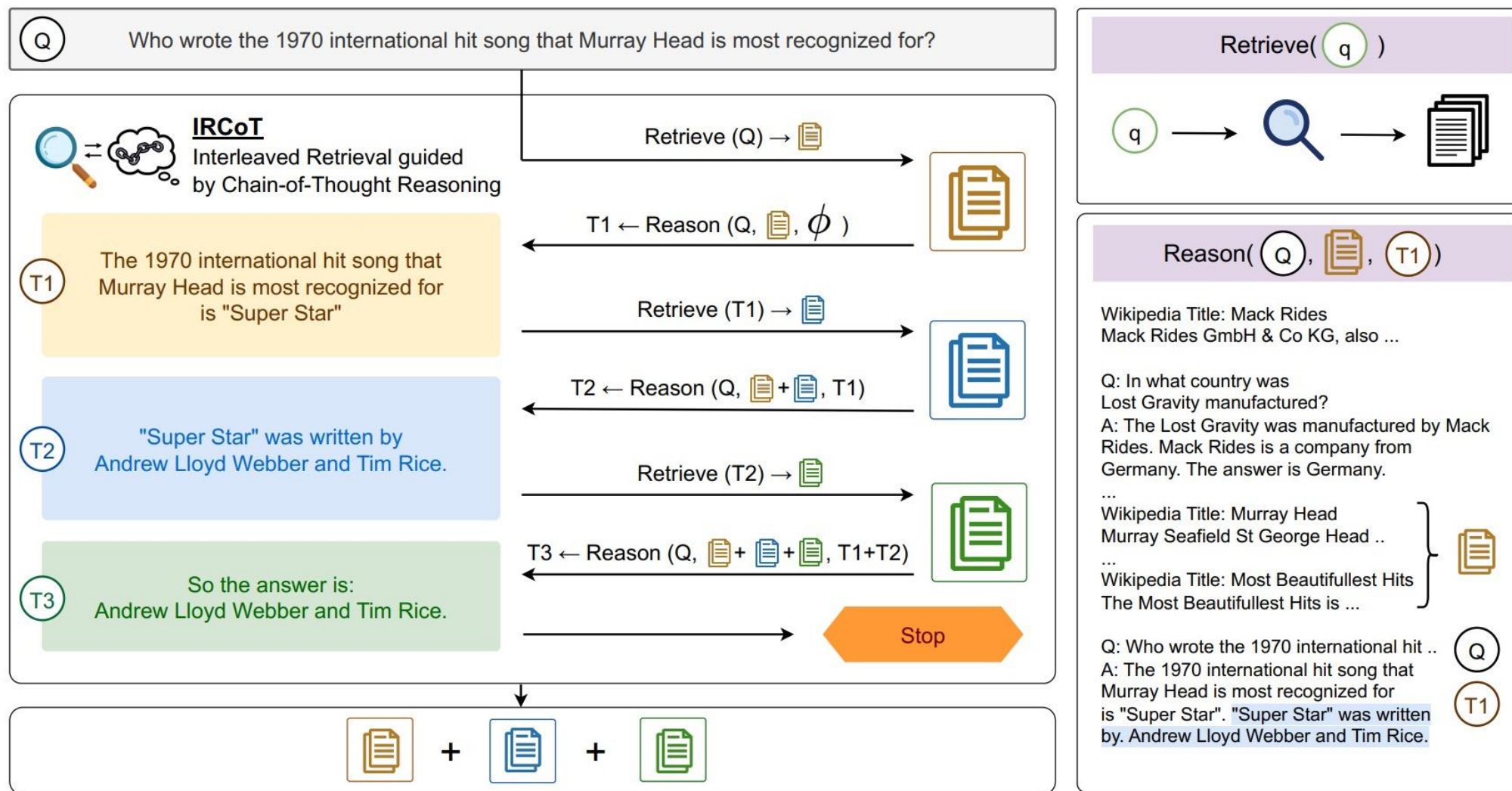
Special RAG Pipeline: Recursively Answer

❑ Chain-of-Thought + RAG

- ❖ One-step retrieve-and-read approach is insufficient for multi-step QA
- ❖ What to retrieve depends on what has already been derived, which in turn may depend on what was previously retrieved

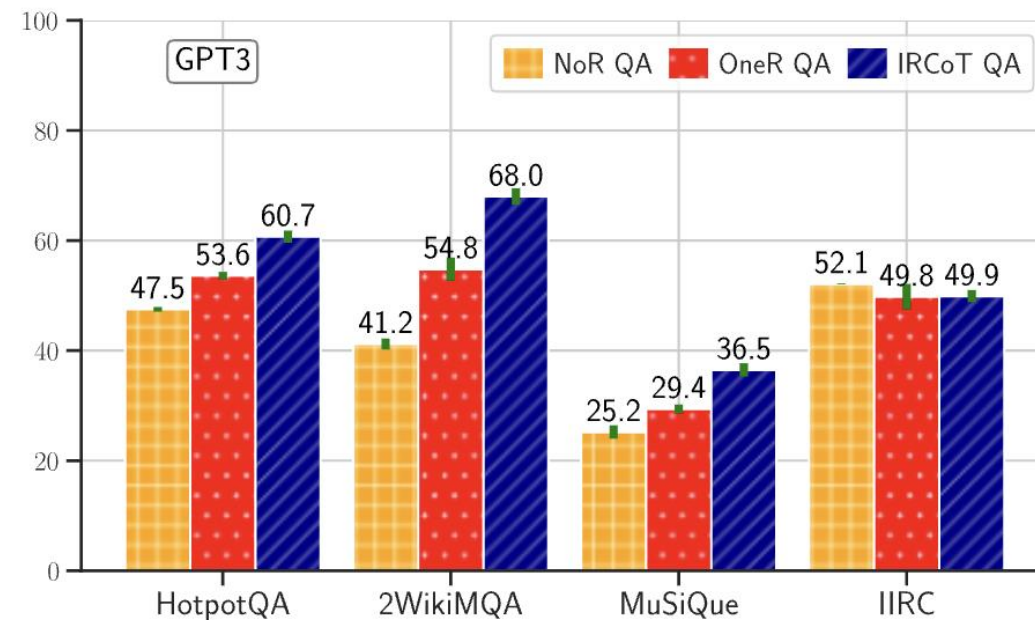
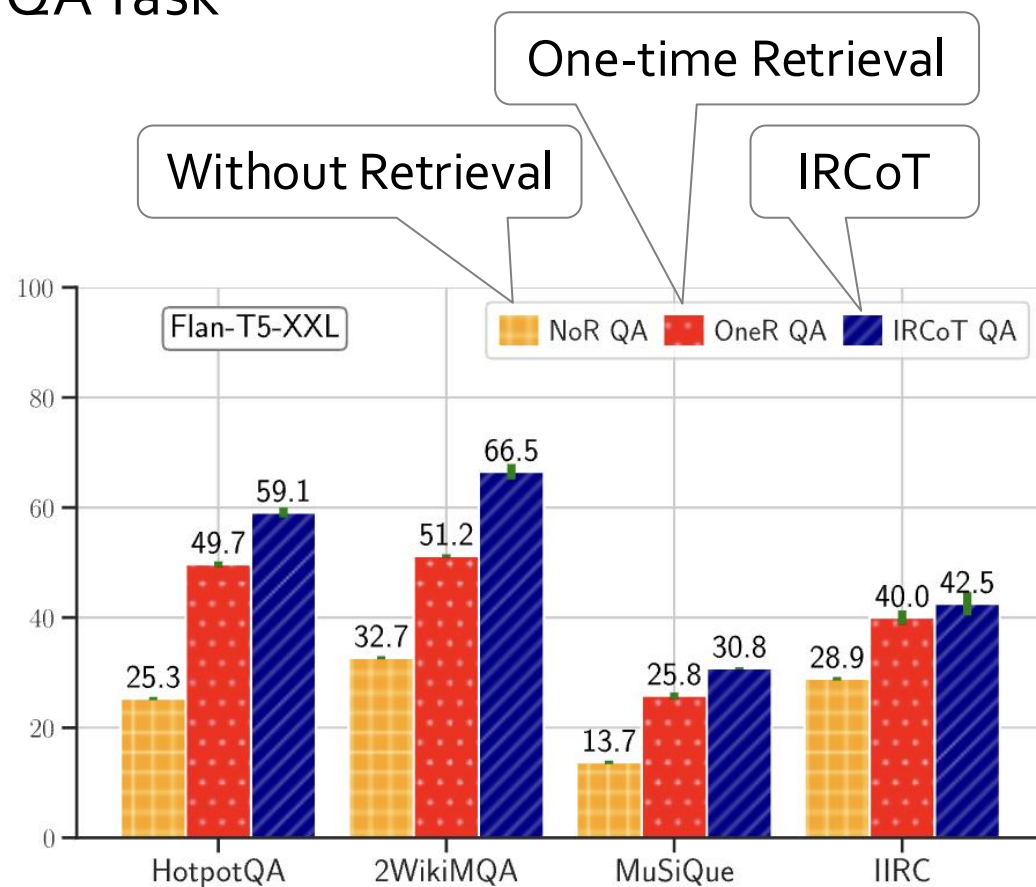


Interleaved Retrieval guided by Chain-of-Thought (IRCoT)



IRCoT Performance

QA Task



Tutorial Outline

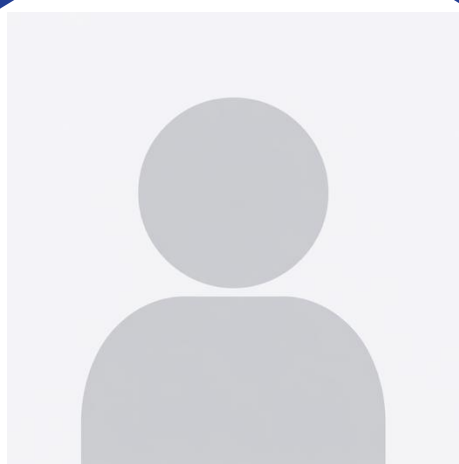


- **Part 1: Introduction** of Retrieval Augmented Large Foundation Models (RA-LFMs) (Dr. Yujuan Ding)
- **Part 2: Architecture** of RA-LFMs and **Main Modules** (Xu Yuan)
- ⦿ **Part 3: Learning Approach of RA-LFMs (Chengliang Liu)**
- **Part 4: Applications** of RA-LFMs (Chun-Hin Chan)
- **Part 5: Challenges and Future Directions** of RA-LFMs (Dr. Yujuan Ding)
- **Part 6: Q&A**

Website of this tutorial
Check out the slides and more information!



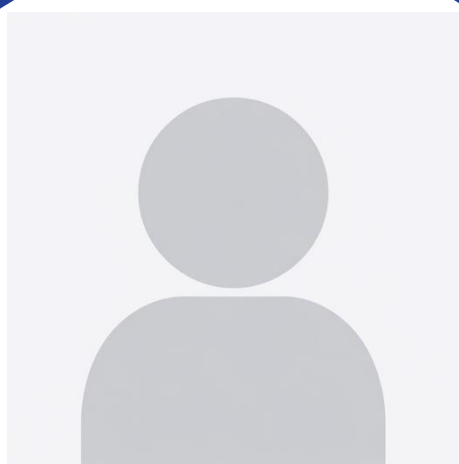
Part 3: RA-LFM Learning



Presenter
Chengliang Liu
HK PolyU

- **Training-free Methods**
- **Training-based Methods**
 - **Independent Learning**
 - **Sequential Learning**
 - **Joint Learning**

Part 3: RA-LFM Learning

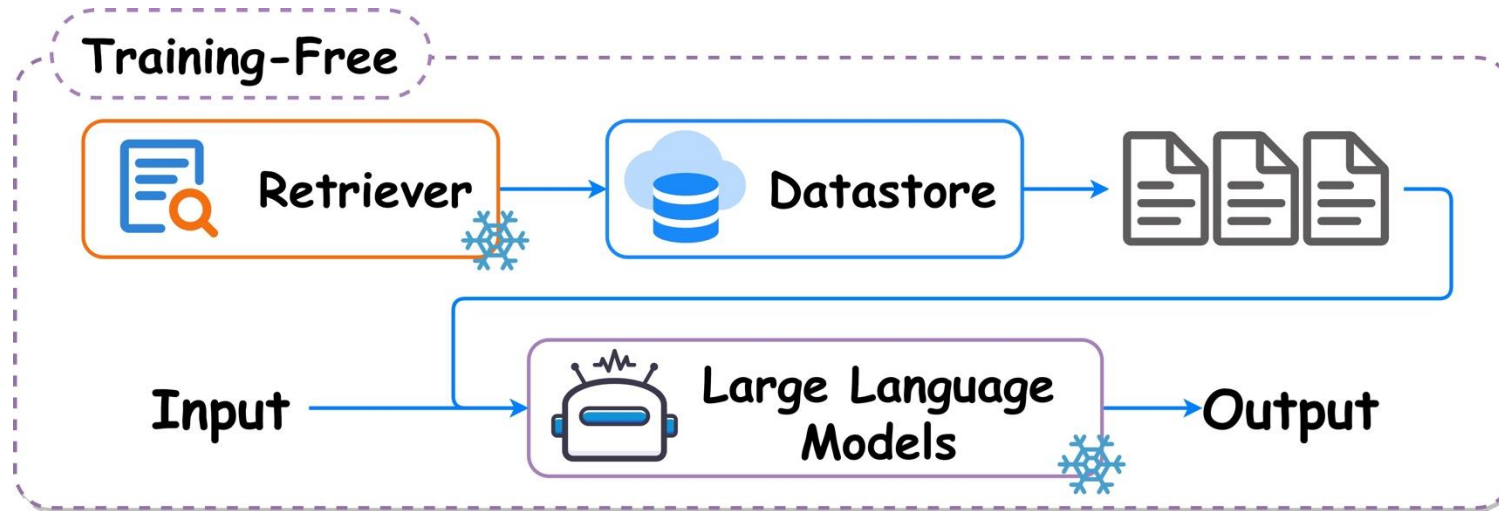


Presenter
Chengliang Liu
HK PolyU

- **Training-free Methods**
- Training-based Methods
 - Independent Learning
 - Sequential Learning
 - Joint Learning

RA-LFM Learning: Training-free

- **Retrieval models** and **language models** are both **frozen**.

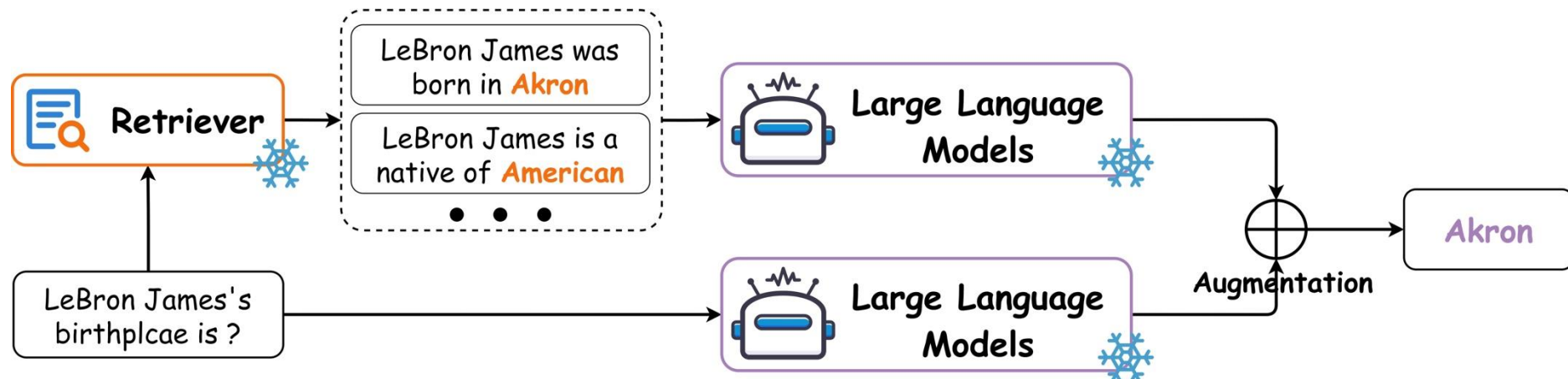


RA-LFM Learning: Training-free

- Prompt Engineering-based Methods

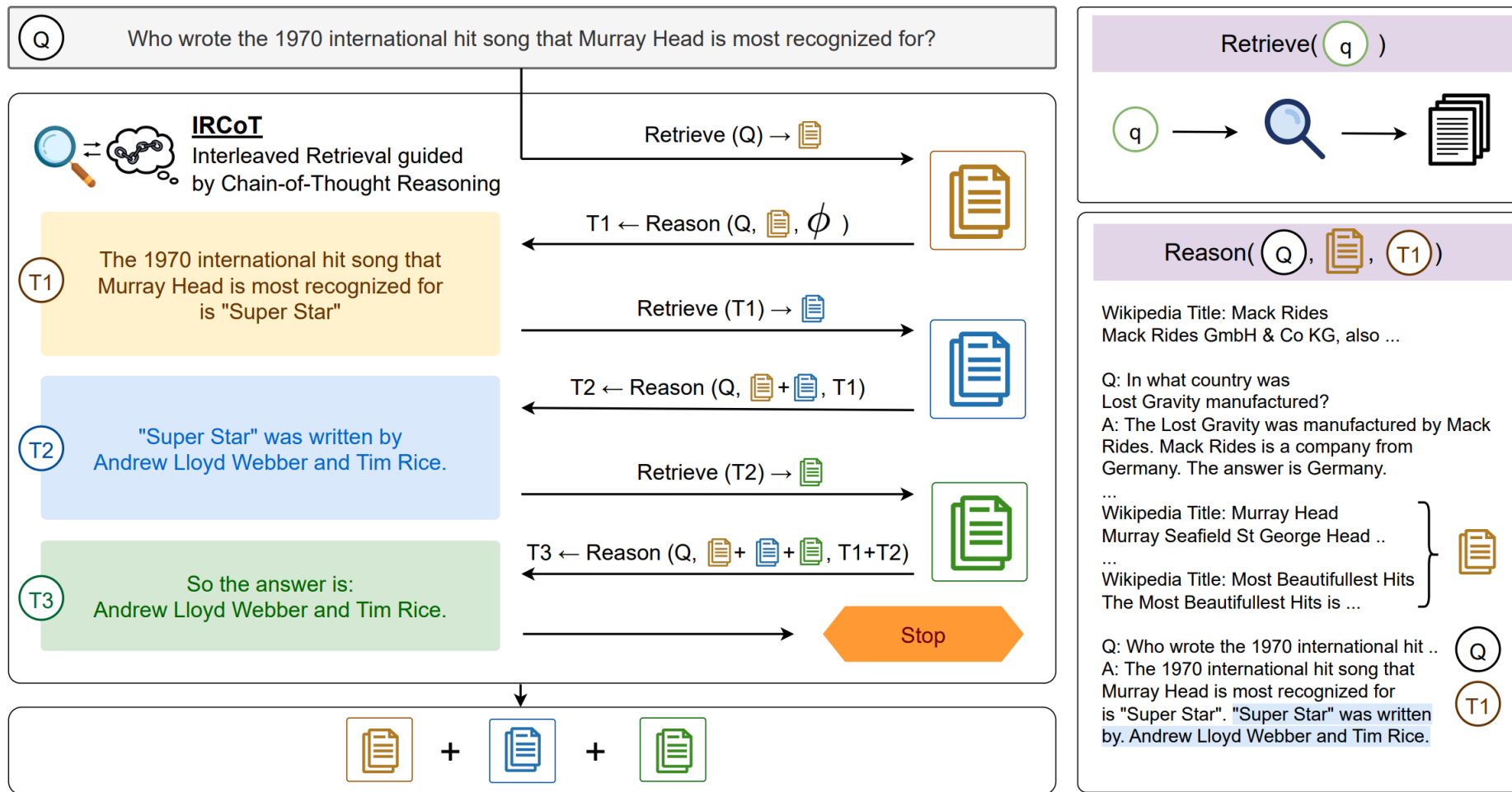


- Retrieval-Guided Token Generation Methods



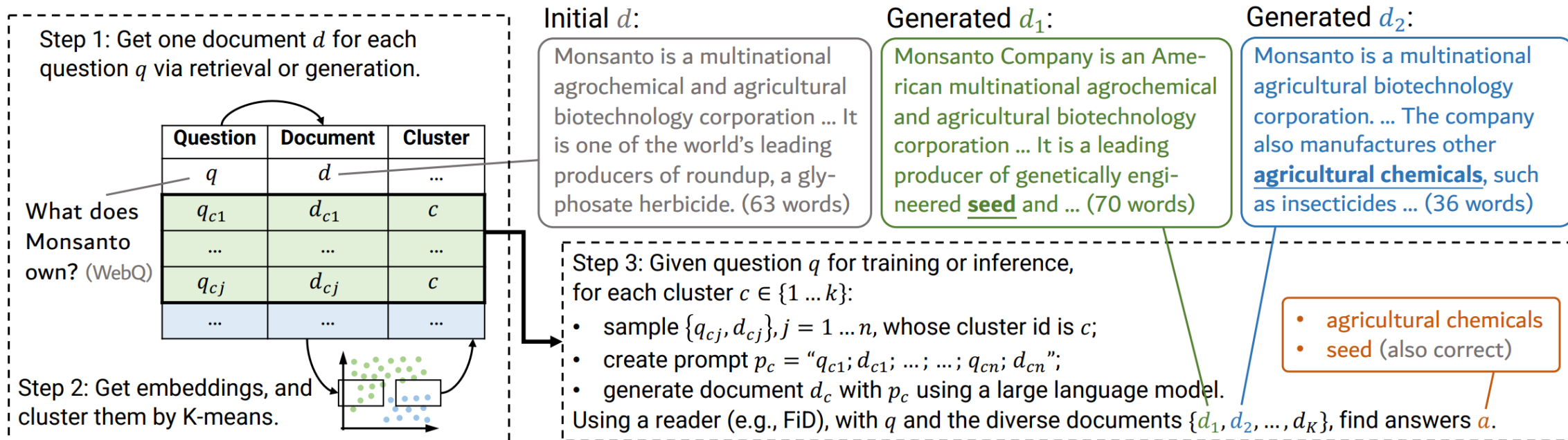
RA-LFM Learning: Training-free

- IRCoT



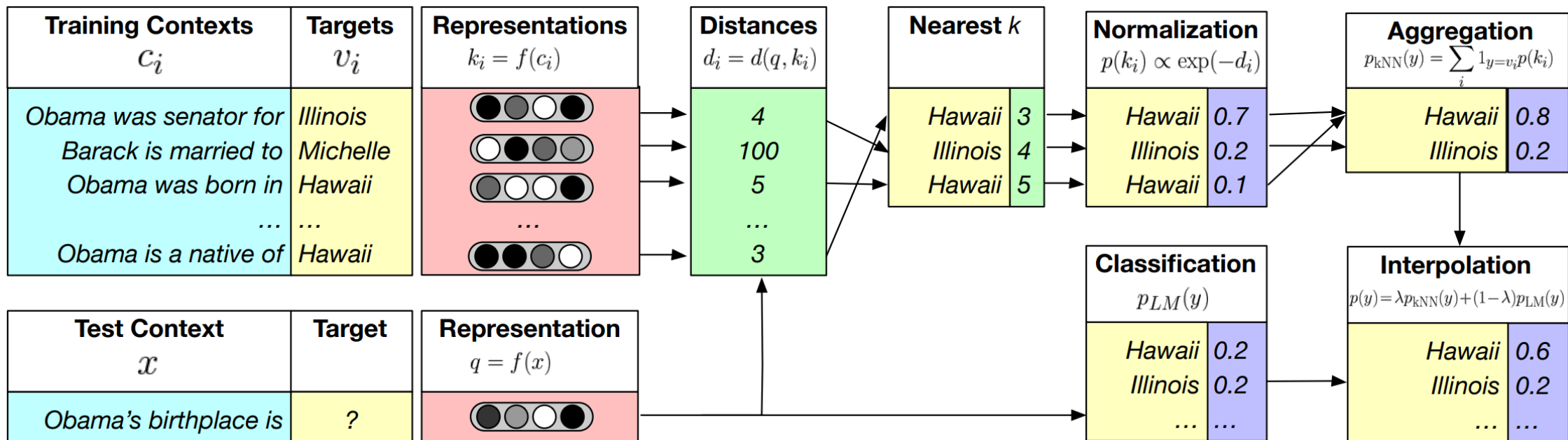
RA-LFM Learning: Training-free

- GENREAD



RA-LFM Learning: Training-free

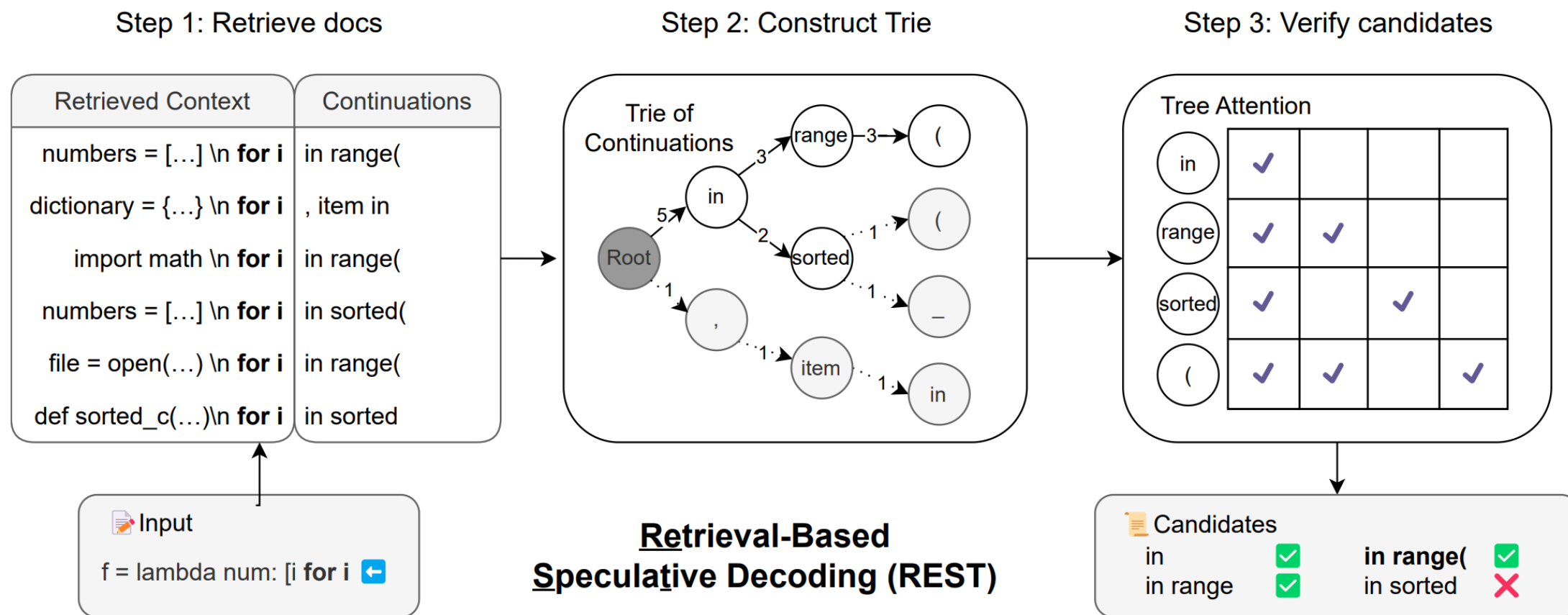
- k NN-LM**



$$p(y|x) = \lambda p_{kNN}(y|x) + (1 - \lambda) p_{LM}(y|x)$$

RA-LFM Learning: Training-free

- REST



RA-LFM Learning: Training-free

- ✓ Work with off-the-shelf models
- x All components are fixed and not trained
- x Might not achieve optimal learning result of the whole model

Part 3: RA-LFM Learning

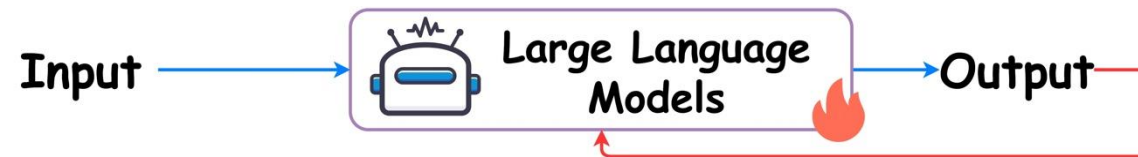


Website of this tutorial

- Training-free Methods
- Training-based Methods
 - **Independent Learning**
 - Sequential Learning
 - Joint Learning

RA-LFM Learning: Independent Training

- **Retrieval models** and **language models** are trained independently.
 - Independent training of large language models.



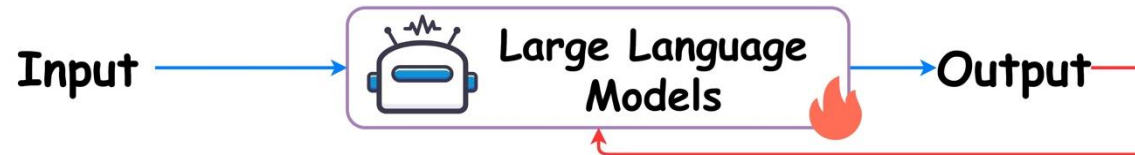
- Independent training of Retriever.



RA-LFM Learning: Independent Training

- **Retrieval models** and **language models** are trained **independently**.

- Independent training of large language models.

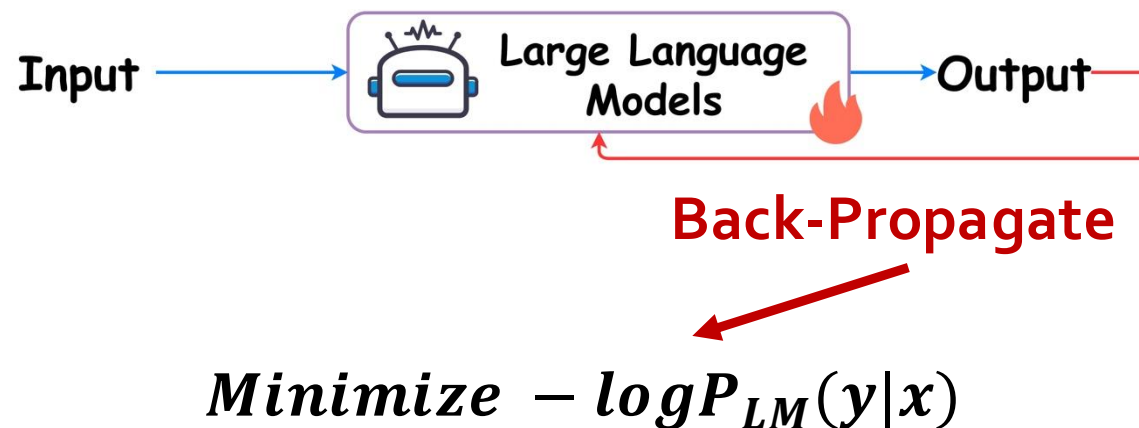


- Independent training of Retriever.



RA-LFM Learning: Independent Training

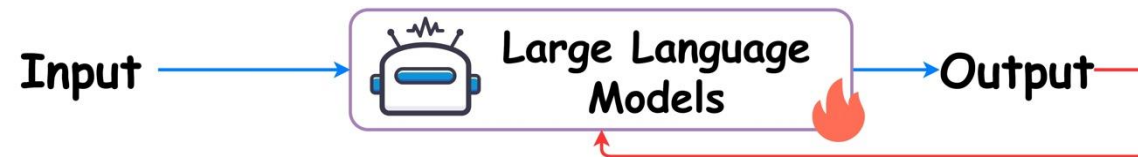
- Independent training of large language models.



.....

RA-LFM Learning: Independent Training

- **Retrieval models** and **language models** are trained **independently**.
 - Independent training of large language models.

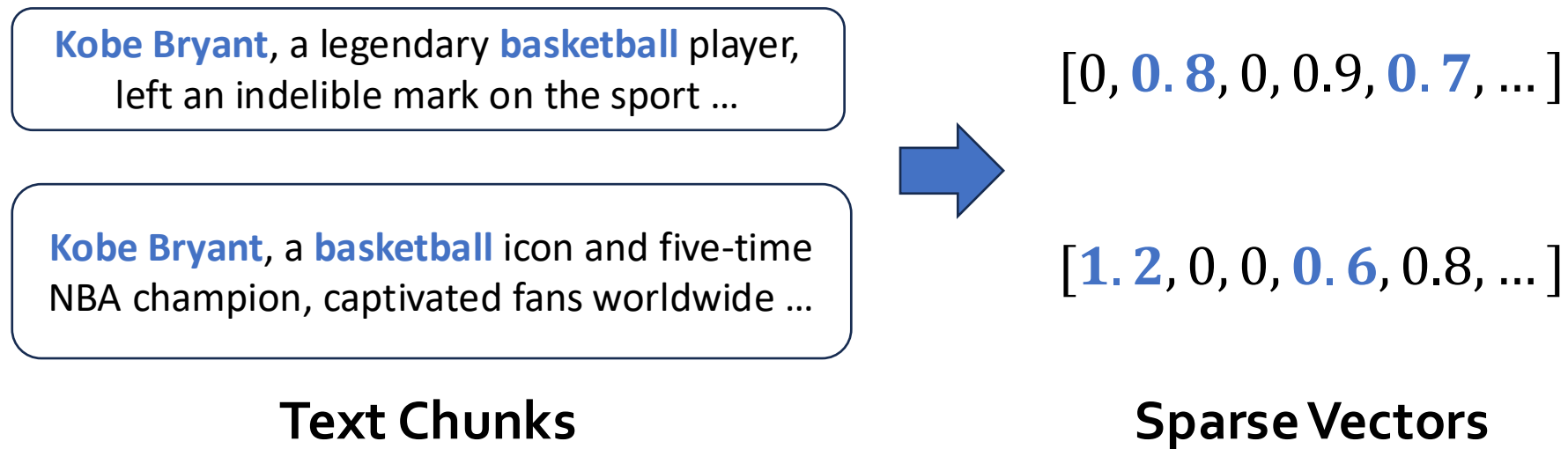


- Independent training of Retriever.



RA-LFM Learning: Independent Training

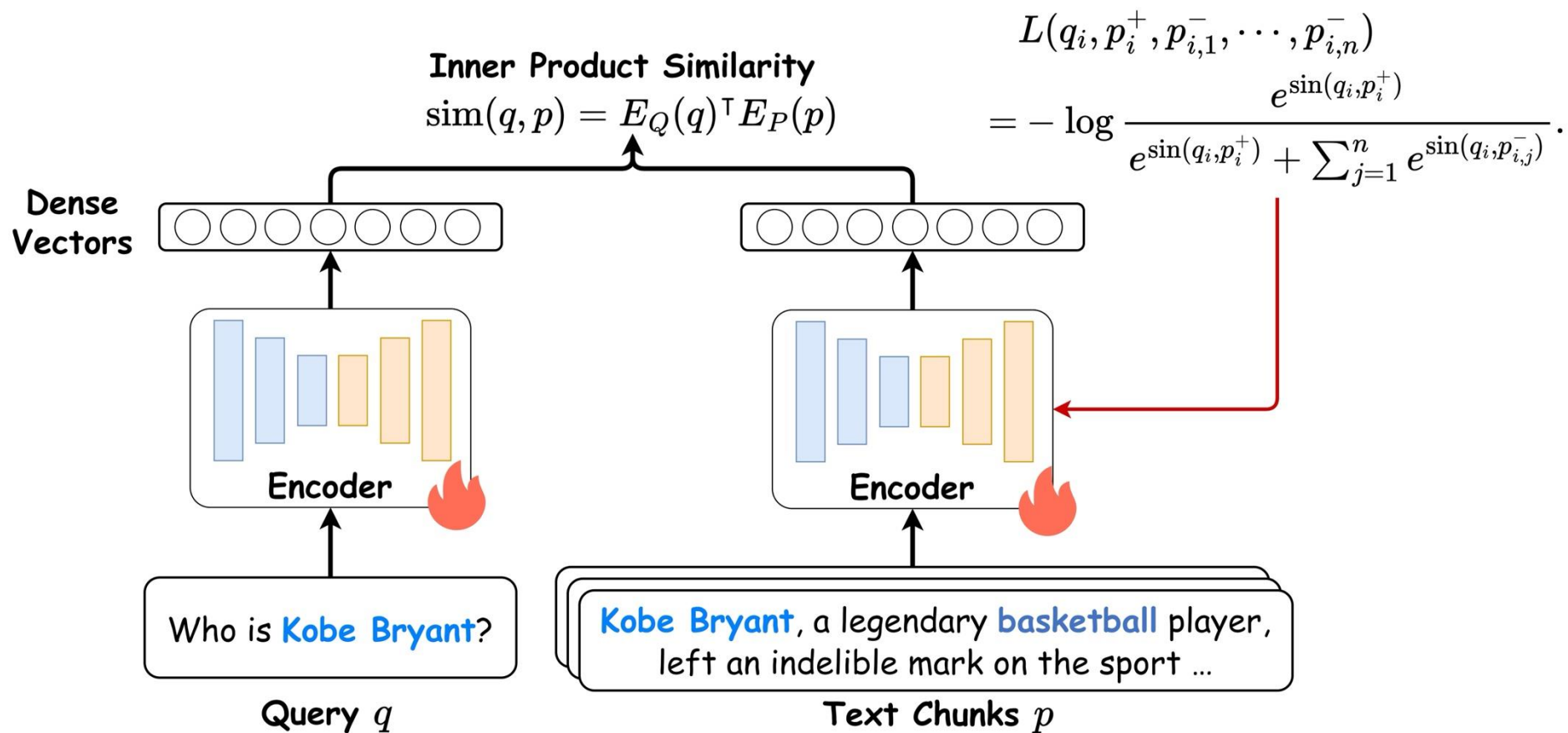
- Sparse retrieval models: TF-IDF / BM25



No training is Needed!

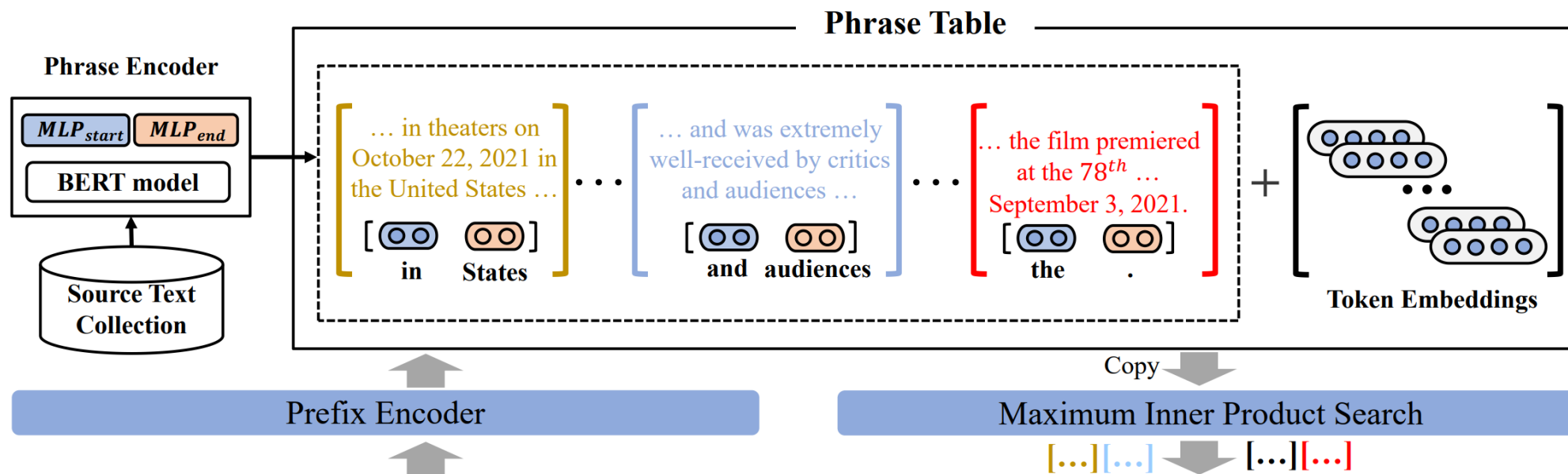
RA-LFM Learning: Independent Training

- Dense retrieval models: DPR



RA-LFM Learning: Independent Training

- Dense retrieval models: CoG



The Dune film was released [in theaters on October 22, 2021 in the United States] [and was extremely well-received by critics and audiences] [Before] [that] [,] [the film premiered at the 78th International Film Festival on September 3, 2021.]

$$\mathcal{H}_{i+1} = \text{PrefixEncoder}(x_i, \mathcal{H}_i).$$

$$\mathcal{D}_{\text{start}} = \text{MLP}_{\text{start}}(\mathcal{D}), \mathcal{D}_{\text{end}} = \text{MLP}_{\text{end}}(\mathcal{D}).$$

$$\text{PhraseEncoder}(s, e, D) = [\mathcal{D}_{\text{start}}[s]; \mathcal{D}_{\text{end}}[e]] \in \mathbb{R}^d$$

RA-LFM Learning: Independent Training

- **Model Training:**

$$\mathcal{L}_p = -\frac{1}{n} \sum_{k=1}^n \log \frac{\exp(q_k \cdot p_k)}{\sum_{p \in \mathcal{P}_k} \exp(q_k \cdot p_p) + \sum_{w \in V} \exp(q_k \cdot v_w)}$$

$$\mathcal{L}_t = -\frac{1}{m} \sum_{i=1}^m \log \frac{\exp(q_i, v_{D_i})}{\sum_{w \in V} \exp(q_i, v_w)}$$

RA-LFM Learning: Independent Training

- ✓ Work with off-the-shelf models, flexible
- ✓ Each part can be improved independently
- x Lack of integrity between Retrieval and Generation
- x Retrieval models are not optimized specified for the tasks/ domains/ generators

Part 3: RA-LFM Learning

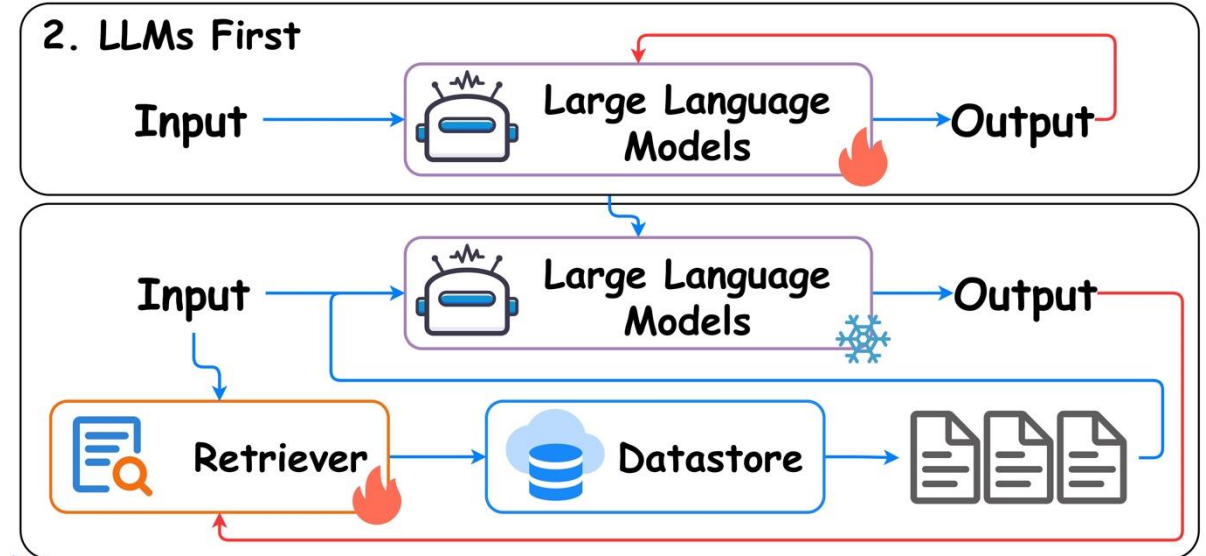
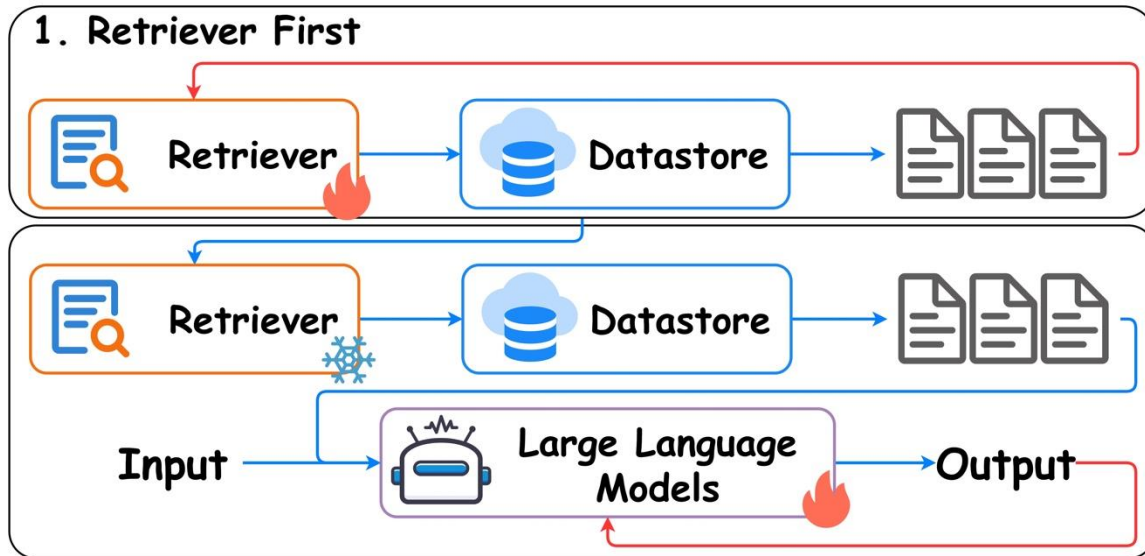


Website of this tutorial

- Training-free Methods
- Training-based Methods
 - Independent Learning
 - **Sequential Learning**
 - Joint Learning

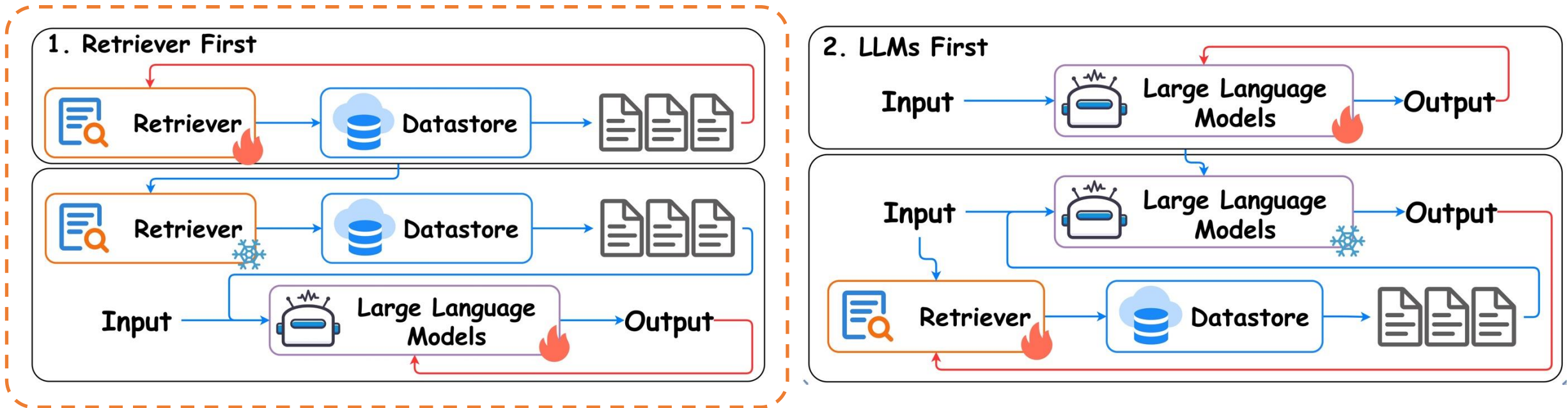
RA-LFM Learning: Sequential Training

- **One component** is first trained independently and then fixed.
- **The other component** is trained with an objective that depends on **the first one**.



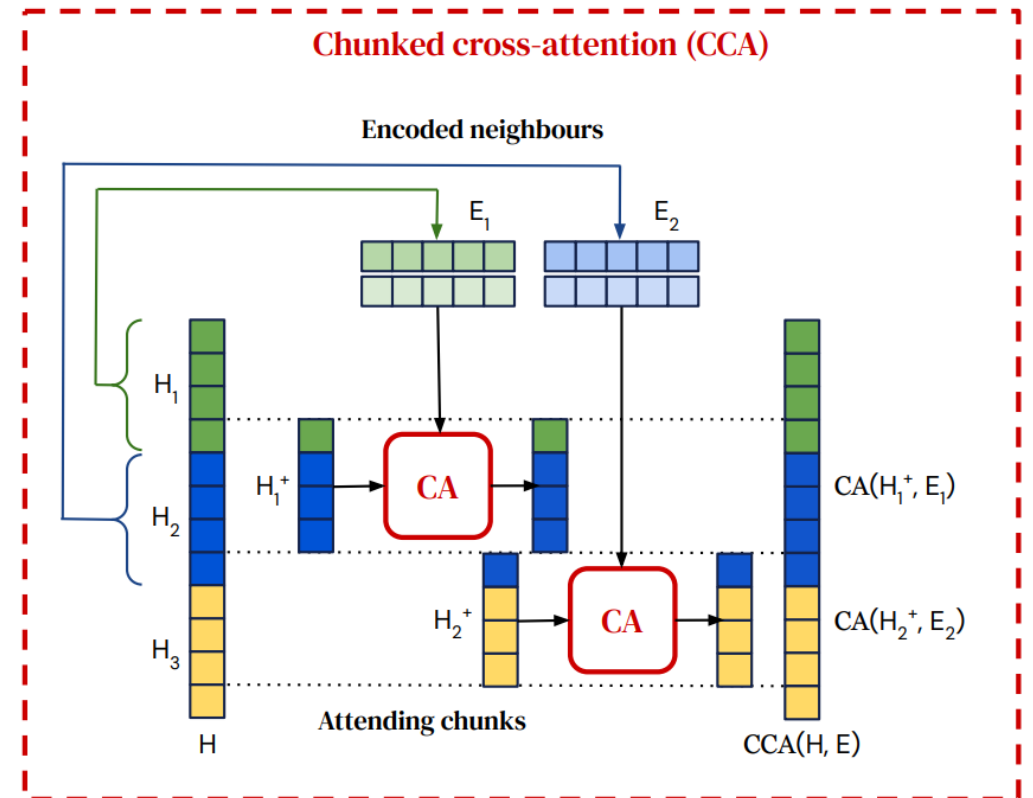
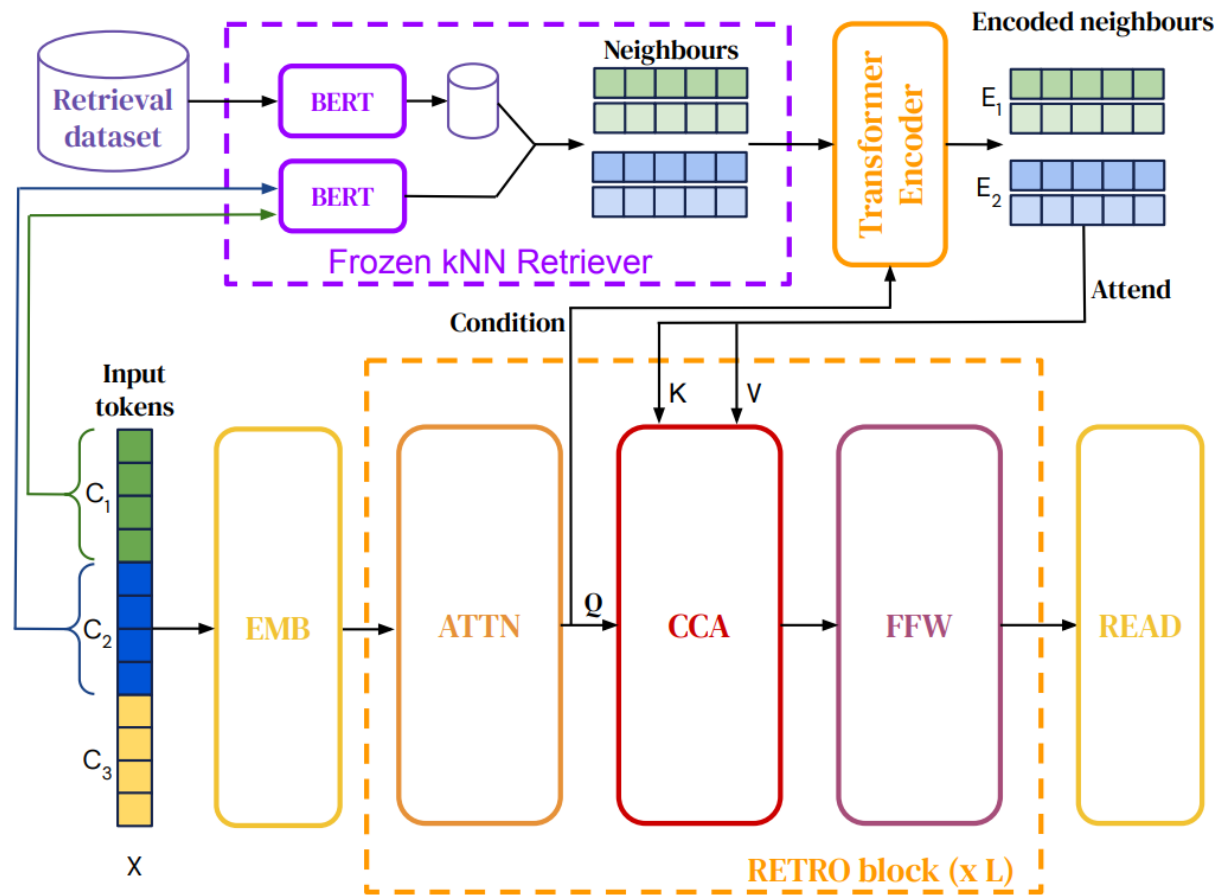
RA-LFM Learning: Sequential Training

- **Retrieval models** is first trained independently and then fixed.
- **Language models** are trained with an objective that depends on **the Retrieval**.



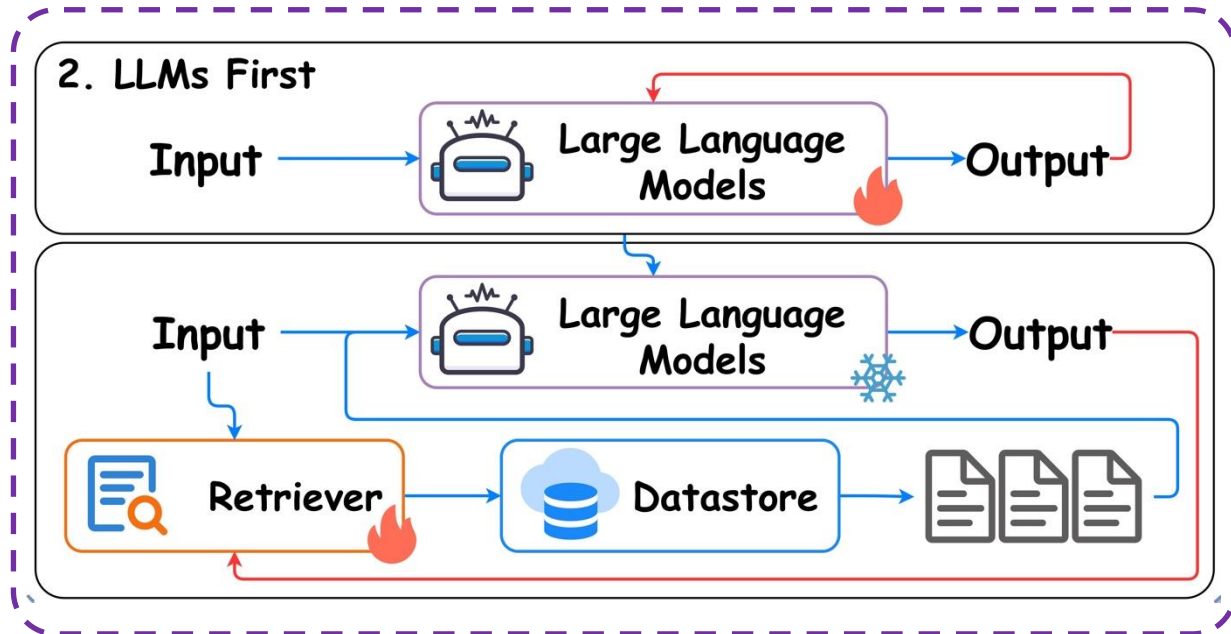
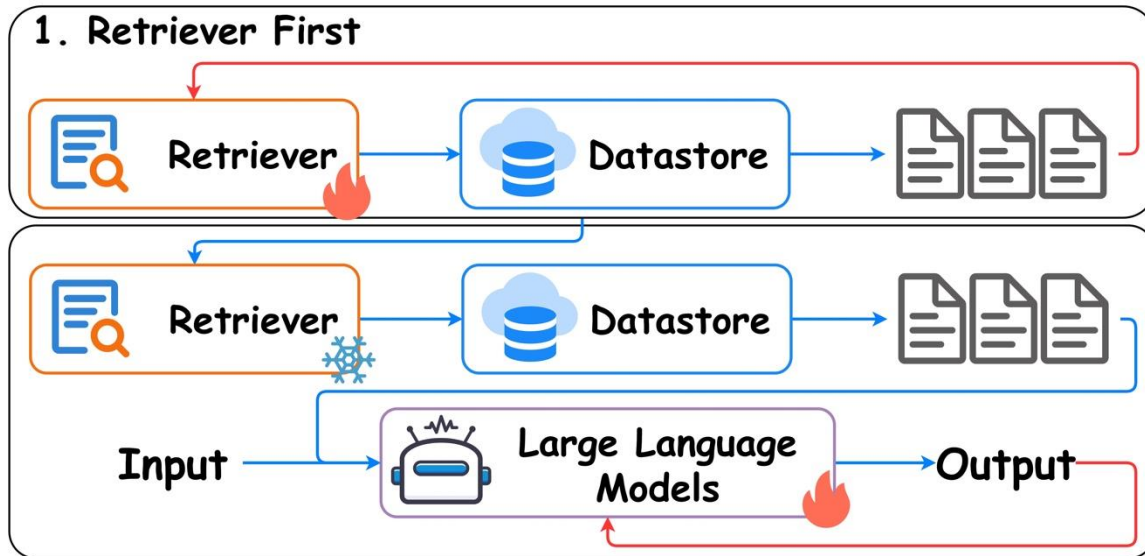
RA-LFM Learning: Sequential Training

- RETRO



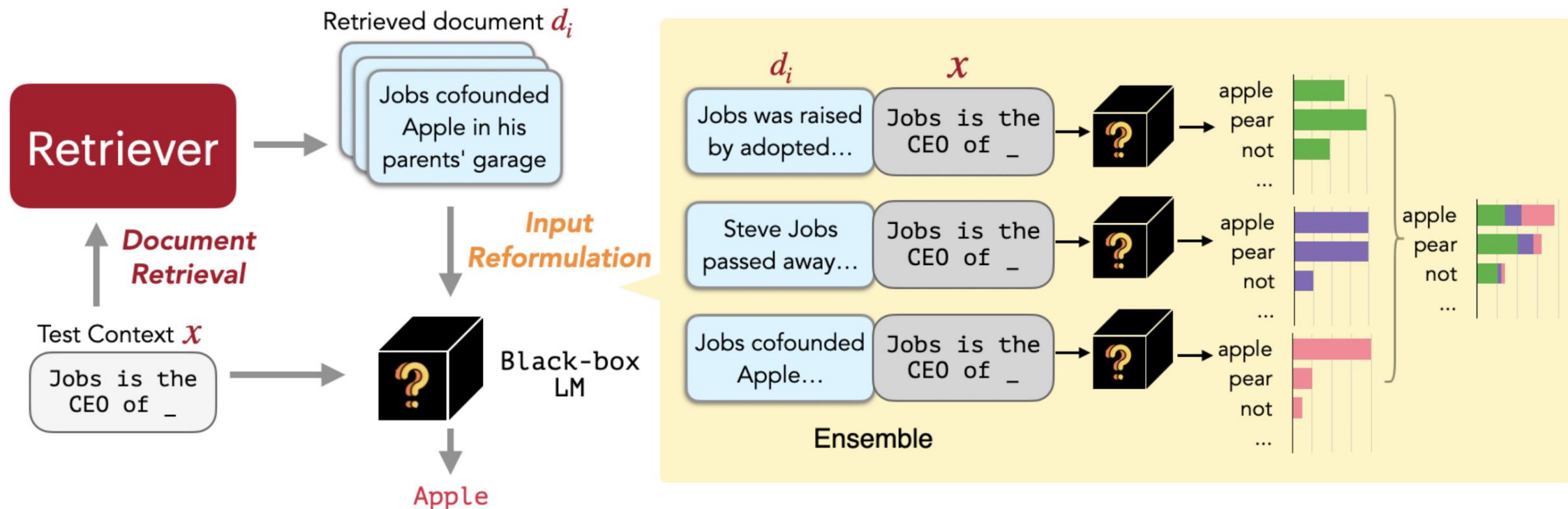
RA-LFM Learning: Sequential Training

- **Language models** are first trained independently and then fixed.
- **Retrieval models** are trained with supervisions from **language models**.



RA-LFM Learning: Sequential Training

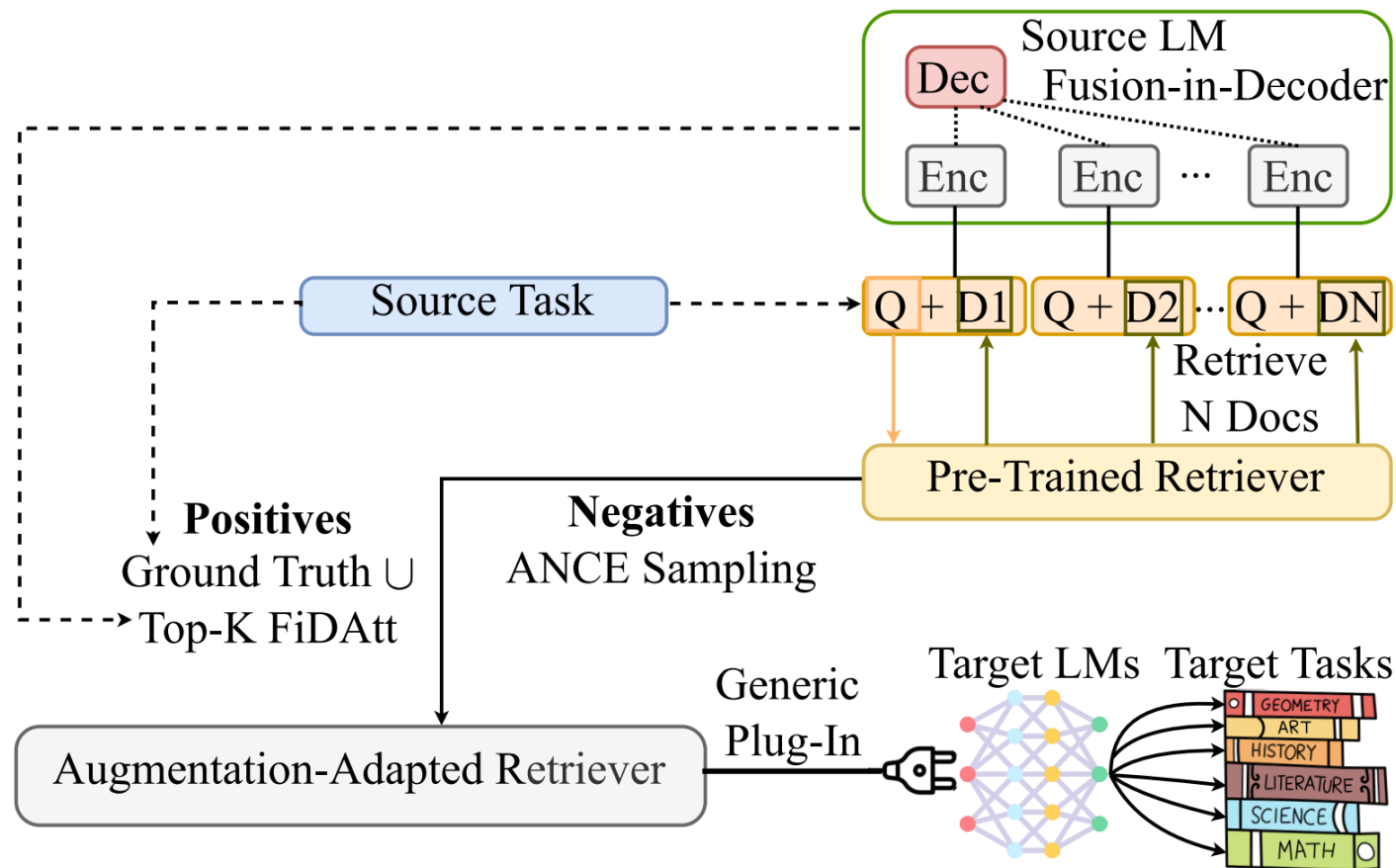
- REPLUG (Retrieve and Plug)



$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} KL(P_R(d | x) \parallel Q_{LM}(d | x, y)) \quad P_R(d | x) = \frac{e^{s(d, x)/\gamma}}{\sum_{d \in \mathcal{D}'} e^{s(d, x)/\gamma}} \quad Q(d | x, y) = \frac{e^{P_{LM}(y|d, x)/\beta}}{\sum_{d \in \mathcal{D}'} e^{P_{LM}(y|d, x)/\beta}}$$

RA-LFM Learning: Sequential Training

- AAR (Augmentation-Adapted Retriever)**



$$\mathcal{L} = \sum_q \sum_{d^+ \in D^+} \sum_{d^- \in D^-} l(f(q, d^+), f(q, d^-)),$$

RA-LFM Learning: Sequential Training

- ✓ Work with off-the-shelf models
- ✓ Generators can be trained effectively based on the retrieved results
- ✓ Retrievers can be trained to provide useful information to help the generators
- x One component is still fixed and not trained
- x Might not achieve optimal learning result of the whole model

Part 3: RA-LFM Learning

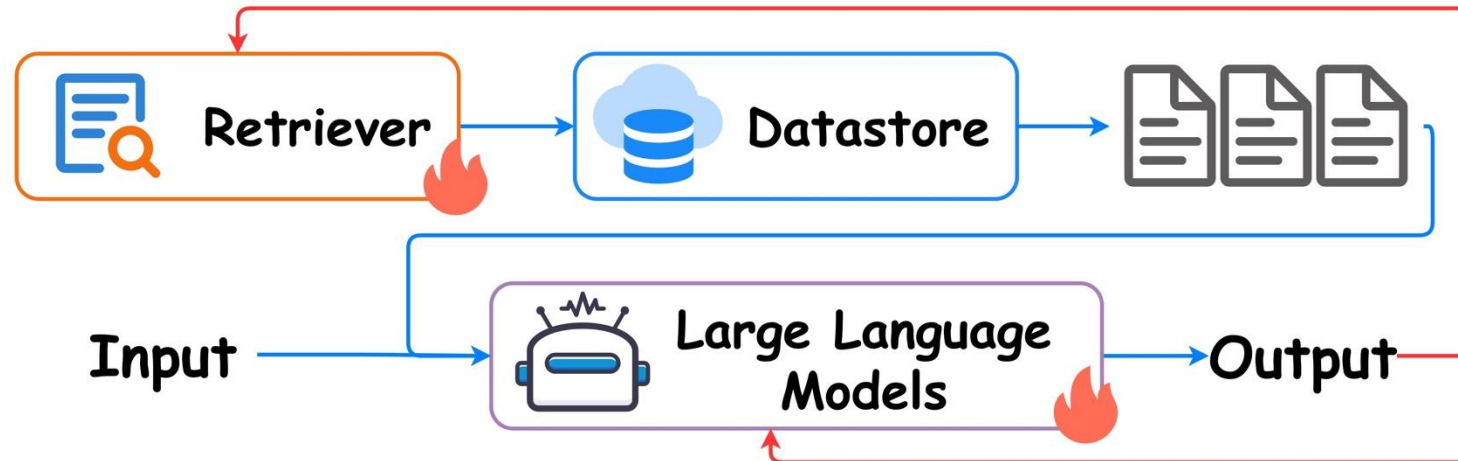


Website of this tutorial

- Training-free Methods
- Training-based Methods
 - Independent Learning
 - Sequential Learning
 - **Joint Learning**

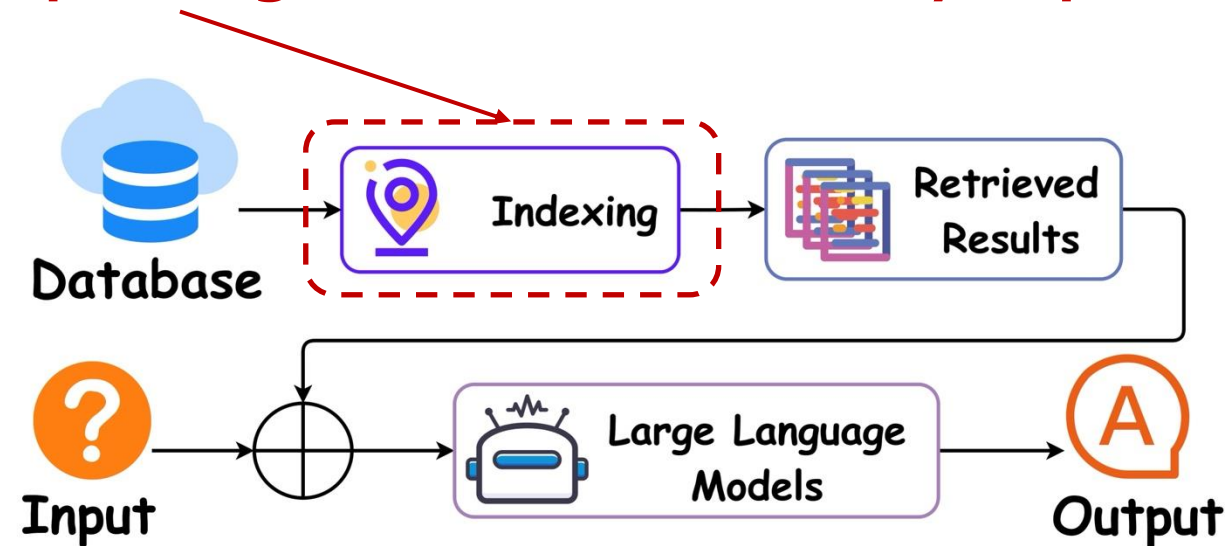
RA-LFM Learning: Joint Training

- **Retrieval models** is and **language models** are trained jointly.



RA-LFM Learning: Joint Training

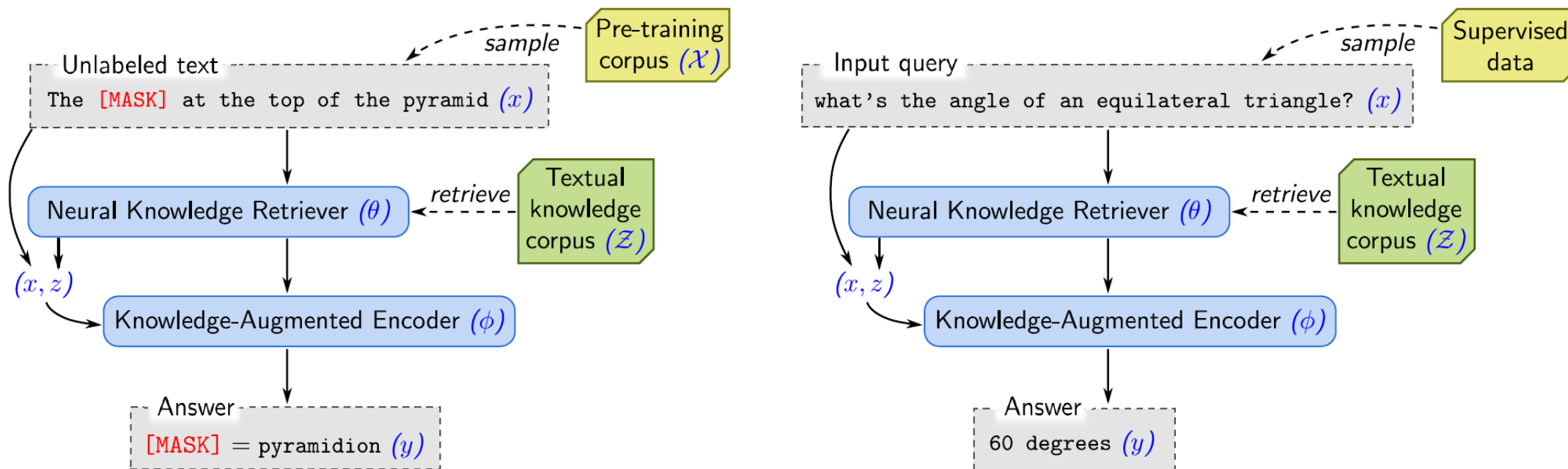
- Retrieval Index Updating, which could be very expensive!



- Solutions:
 - Asynchronous index updating
 - In-batch approximation

RA-LFM Learning: Joint Training

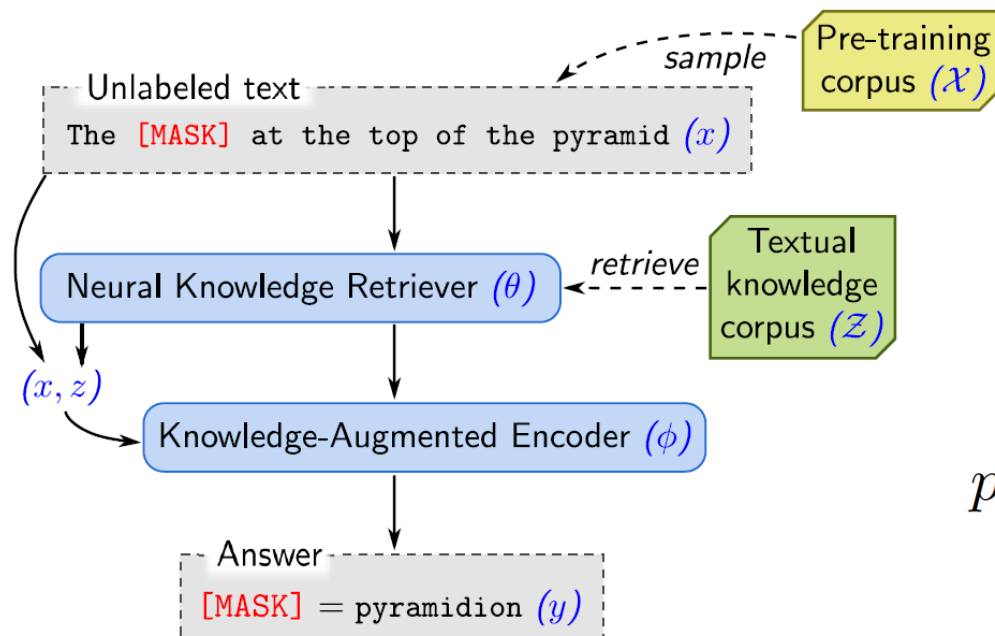
- REALM



Objective function:
$$p(y | x) = \sum_{z \in \mathcal{Z}} p(y | z, x) p(z | x).$$

RA-LFM Learning: Joint Training

- REALM

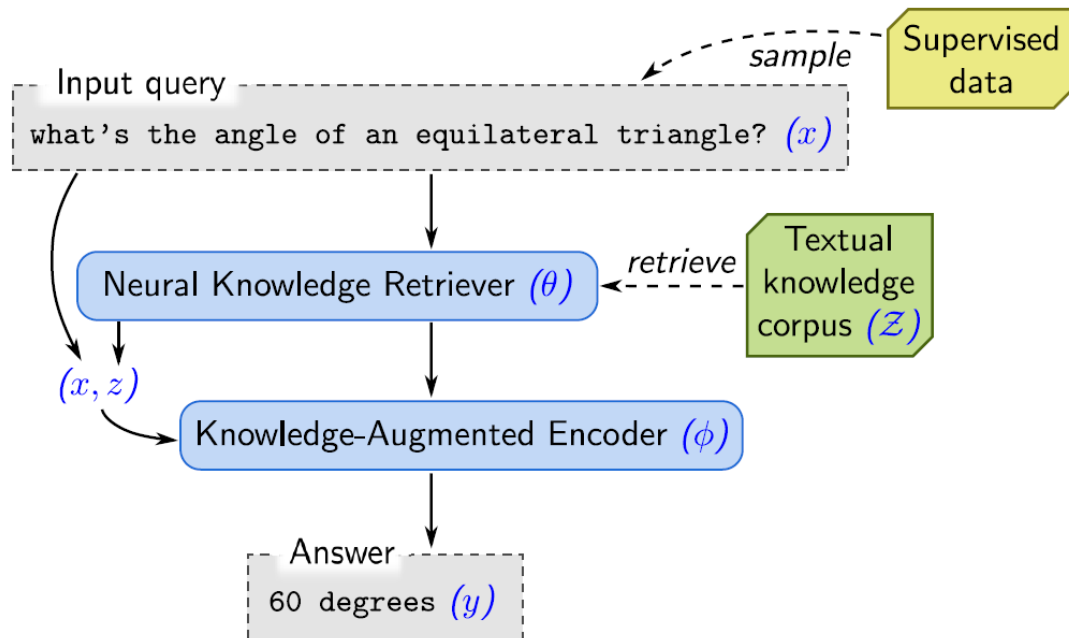


$$p(y | z, x) = \prod_{j=1}^{J_x} p(y_j | z, x)$$

$$p(y_j | z, x) \propto \exp(w_j^\top \text{BERT}_{\text{MASK}(j)}(\text{join}_{\text{BERT}}(x, z_{\text{body}})))$$

RA-LFM Learning: Joint Training

- REALM



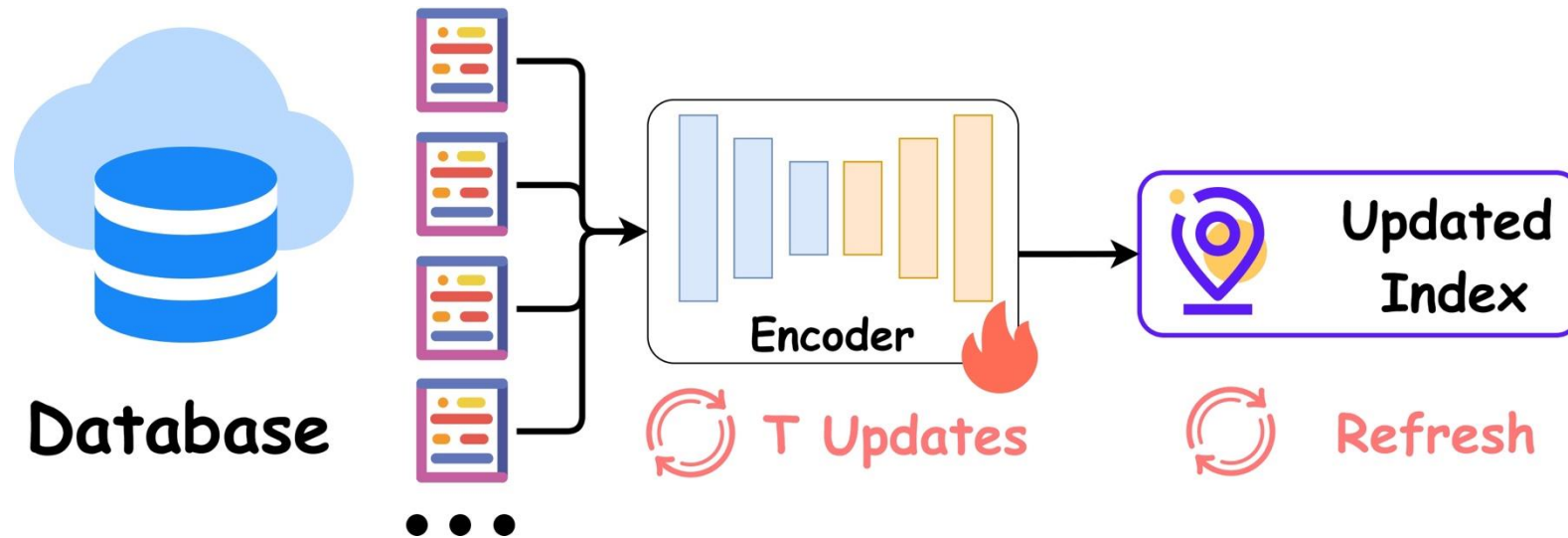
$$p(y | z, x) \propto \sum_{s \in S(z, y)} \exp(\text{MLP}([h_{\text{START}(s)}; h_{\text{END}(s)}]))$$

$$h_{\text{START}(s)} = \text{BERT}_{\text{START}(s)}(\text{join}_{\text{BERT}}(x, z_{\text{body}})),$$

$$h_{\text{END}(s)} = \text{BERT}_{\text{END}(s)}(\text{join}_{\text{BERT}}(x, z_{\text{body}})),$$

RA-LFM Learning: Joint Training

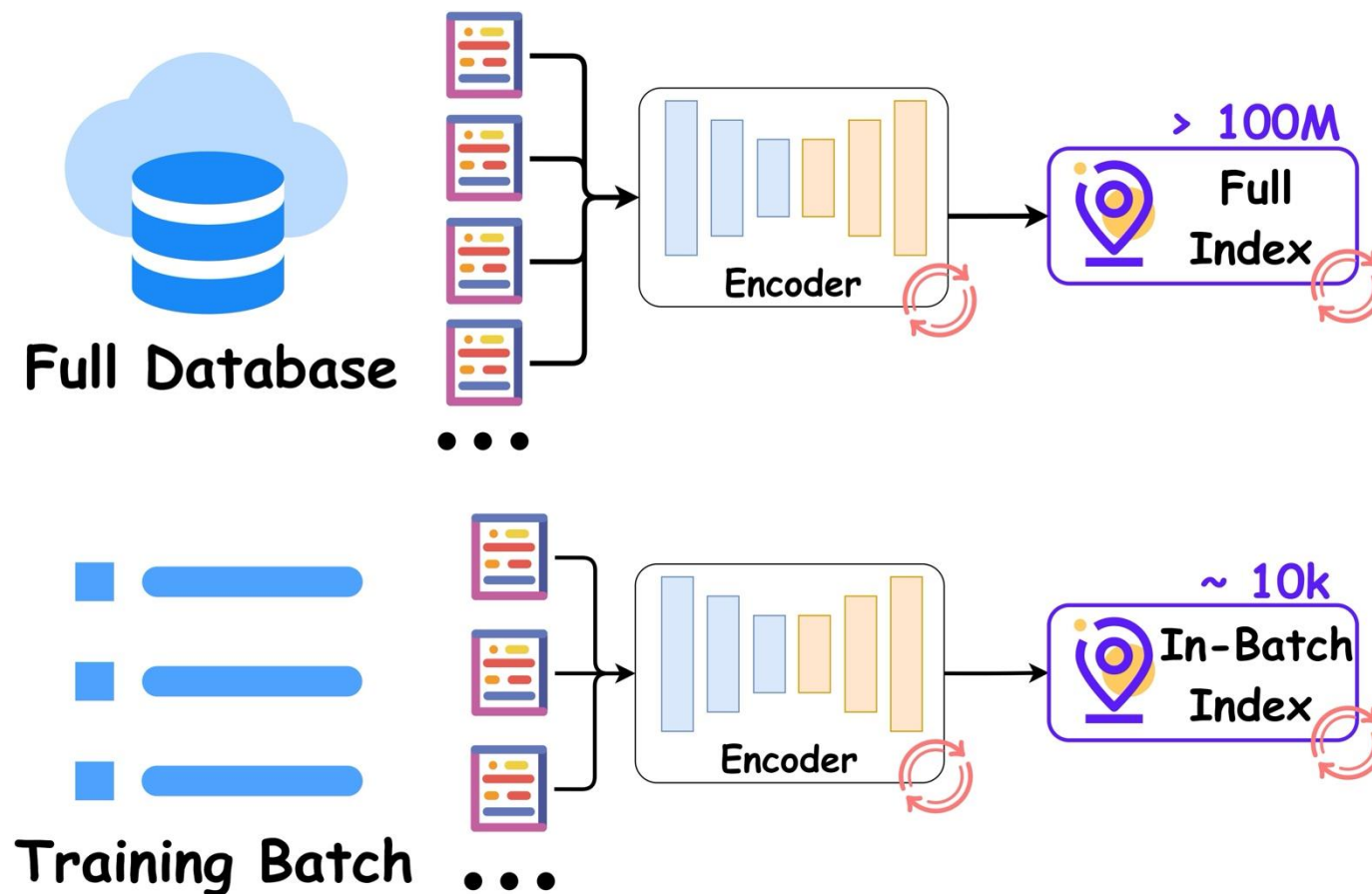
- REALM – Asynchronous Index Update



$$f(x, z) = \text{Embed}_{\text{input}}(x)^{\top} \text{Embed}_{\text{doc}}(z)$$

RA-LFM Learning : Joint Training

- TRIME – In-Batch Approximation

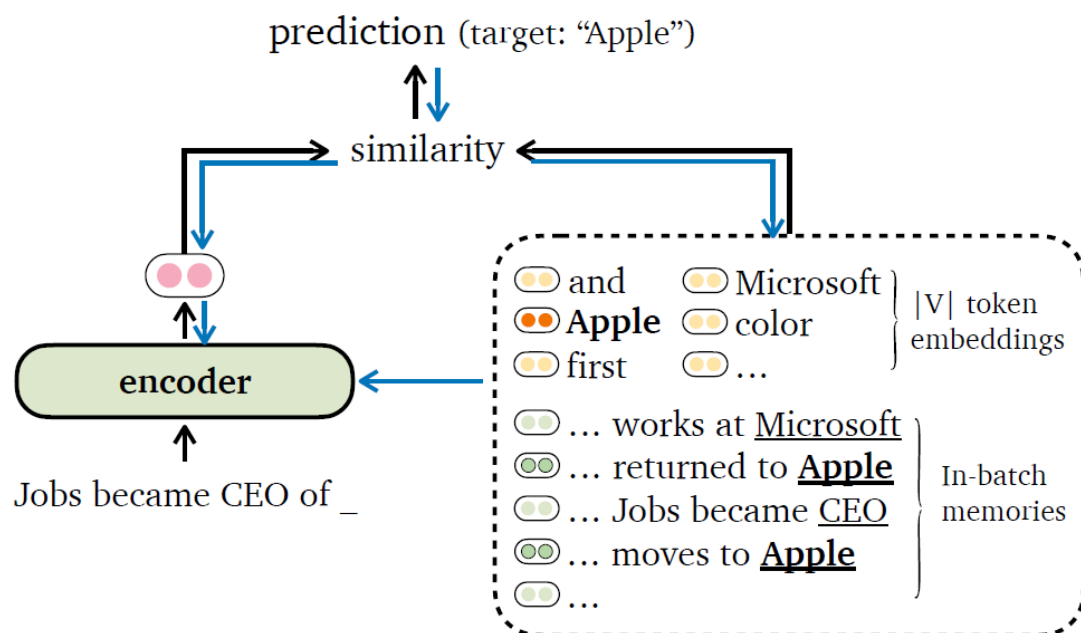


RA-LFM Learning : Joint Training

• TRIME

- Target token's embedding
- Other token embeddings
- Positive in-batch memory
- Negative in-batch memory

↑ Forward pass ↓ Back-propagation



Local Memory: $\mathcal{M}_{\text{local}}(c_t) = \{(c_j, x_j)\}_{1 \leq j \leq t-1}$.

Long-term Memory:

$$\mathcal{M}_{\text{long}}(c_t^{(i)}) = \{(c_j^{(k)}, x_j^{(k)})\}_{1 \leq k < i, 1 \leq j}$$

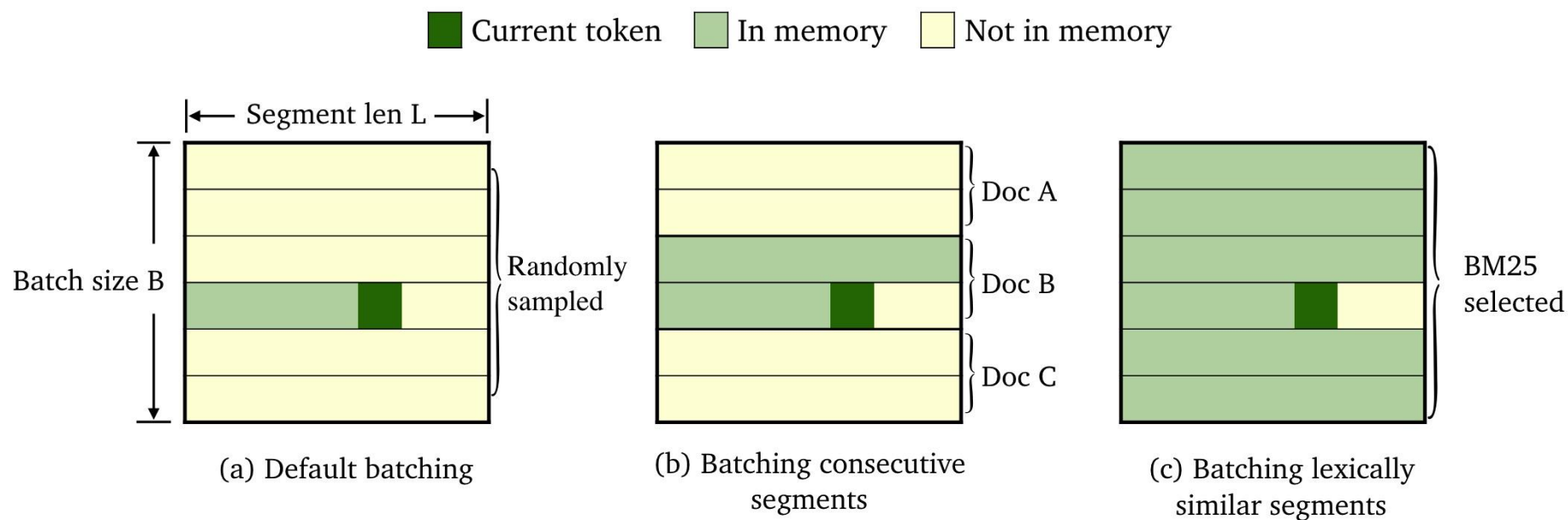
External Memory: $\mathcal{M}_{\text{ext}} = \{(c_j, x_j) \in \mathcal{D}\}$.

Training Objective:

$$P(w | c) \propto \exp(E_w^\top f_\theta(c)) + \sum_{(c_j, x_j) \in \mathcal{M}_{\text{train}}: x_j = w} \exp(\text{sim}(g_\theta(c), g_\theta(c_j))).$$

RA-LFM Learning : Joint Training

- **TRIME Data Batching Strategy**



Use BM25 scores to find similar text chunks to provide more training signals

Tutorial Outline

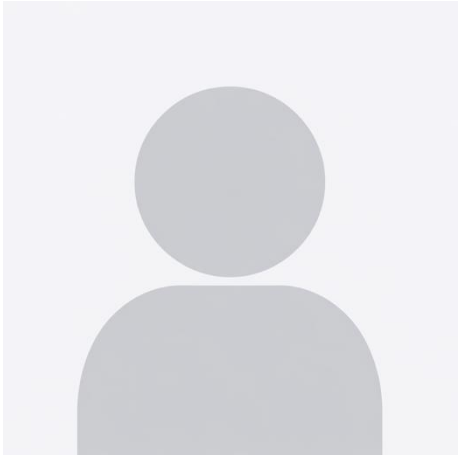


- **Part 1: Introduction** of Retrieval Augmented Large Foundation Models (RA-LFMs) (Dr. Yujuan Ding)
- **Part 2: Architecture** of RA-LFMs and **Main Modules** (Xu Yuan)
- **Part 3: Learning** Approach of RA-LFMs (Chengliang Liu)
- **Part 4: Applications of RA-LFMs (Chun-Hin Chan)**
- **Part 5: Challenges and Future Directions** of RA-LFMs (Dr. Yujuan Ding)
- **Part 6: Q&A**

Website of this tutorial
Check out the slides and more information!



PART 4: Application of RA-LFMs

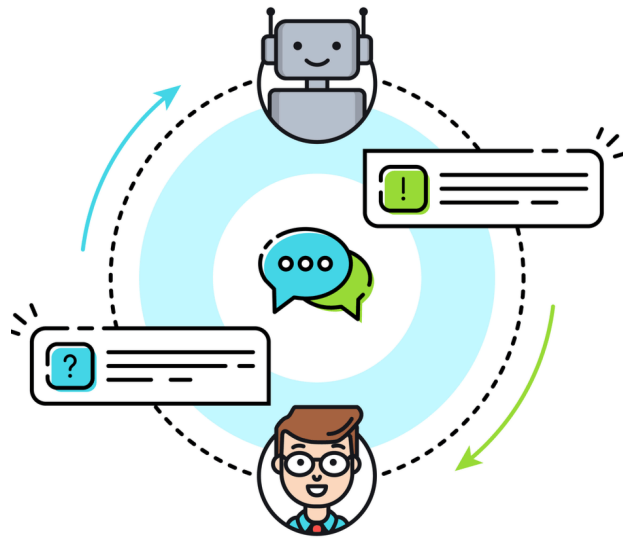


Presenter
Chun-Hin Chan
HK PolyU

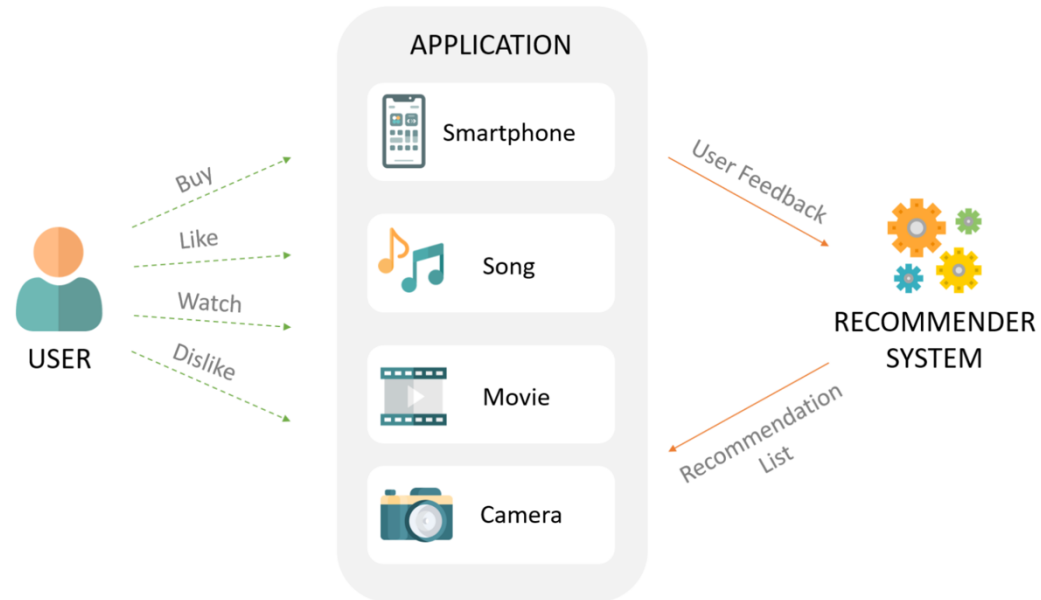
- **NLP applications**
- **Downstream tasks**
- **Domain-specific applications**

RA-LFM Applications

- Various applications



Chatbots



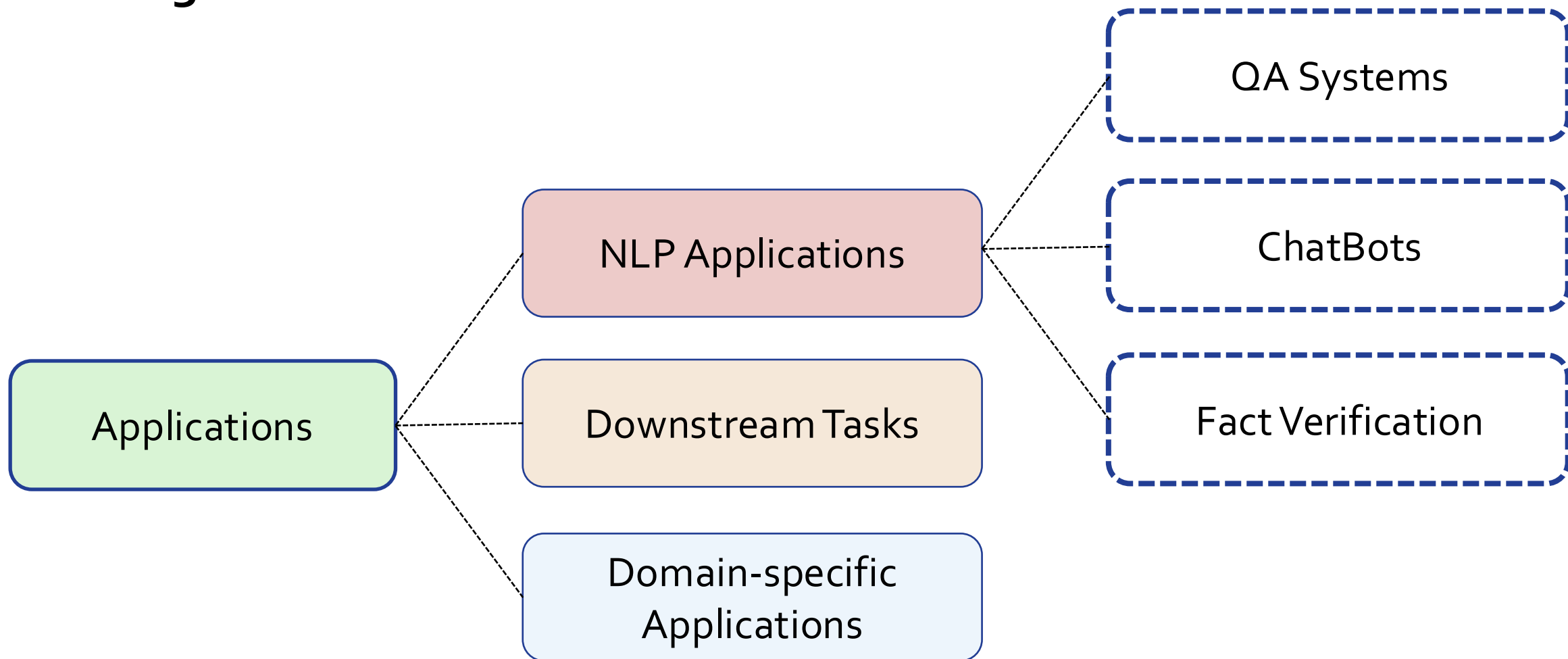
Recommendation



AI for Science

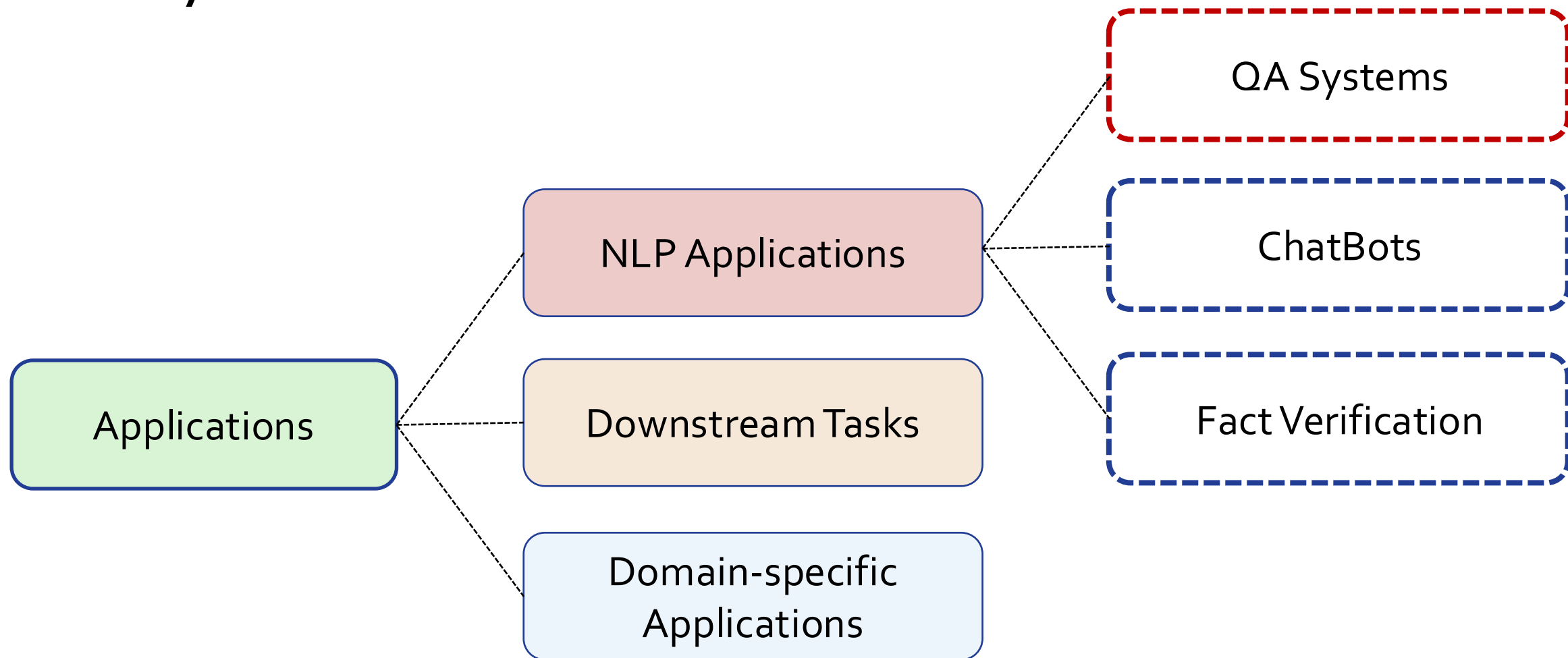
RA-LFM Applications: NLP Applications

- **Categories**



RA-LFM Applications: NLP Applications

- **QA Systems**



RA-LFM Applications: QA Systems

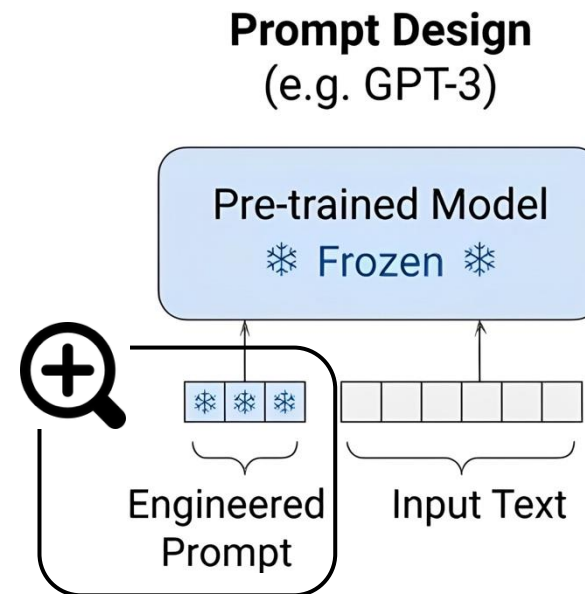
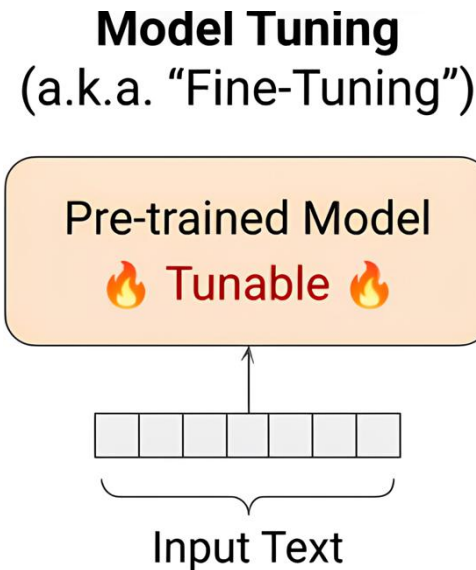
- **QA systems**

- Challenges:

- Open-domain QA
 - Domain-specific QA

- How to solve?

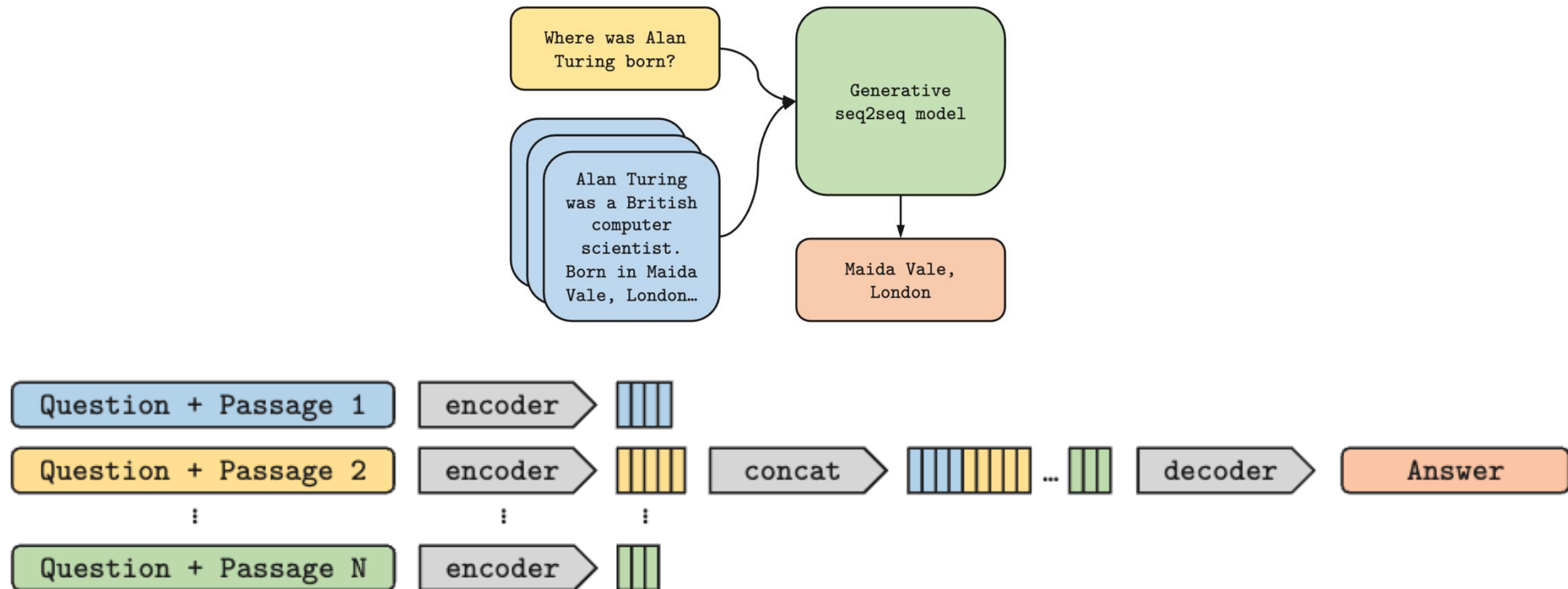
- Fine-tuning
 - Prompting



RA-LFM Applications: QA Systems

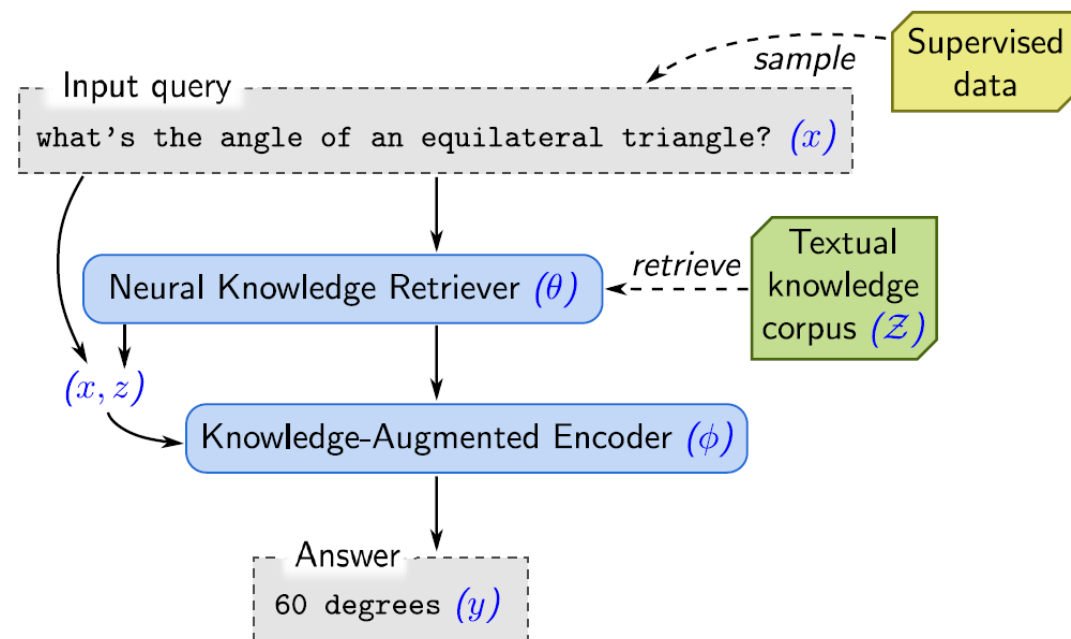
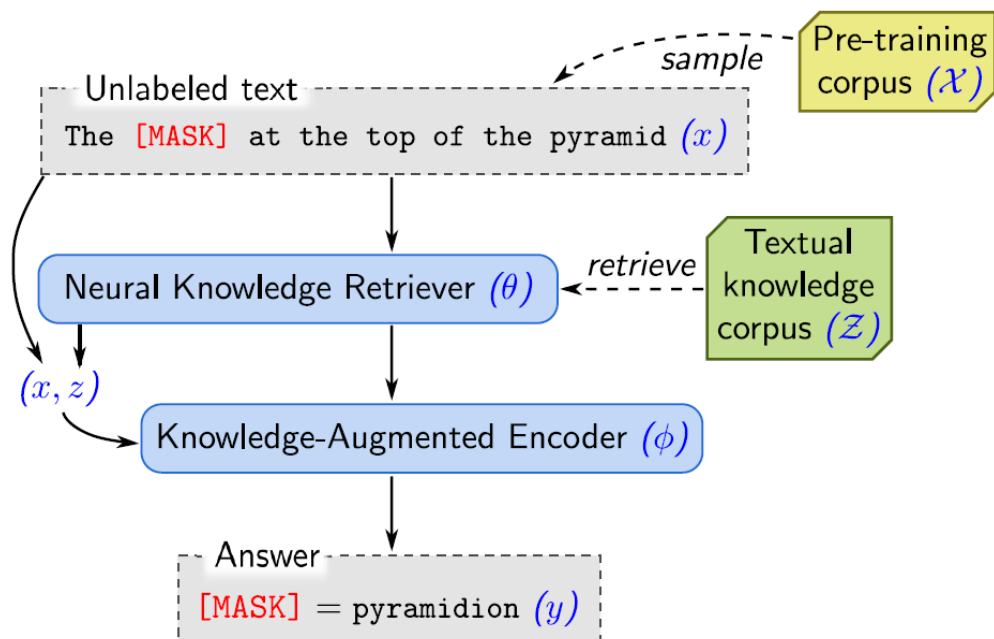
- **Retrieves for open-domain QA**

Retrieves support text passages from an external source of knowledge



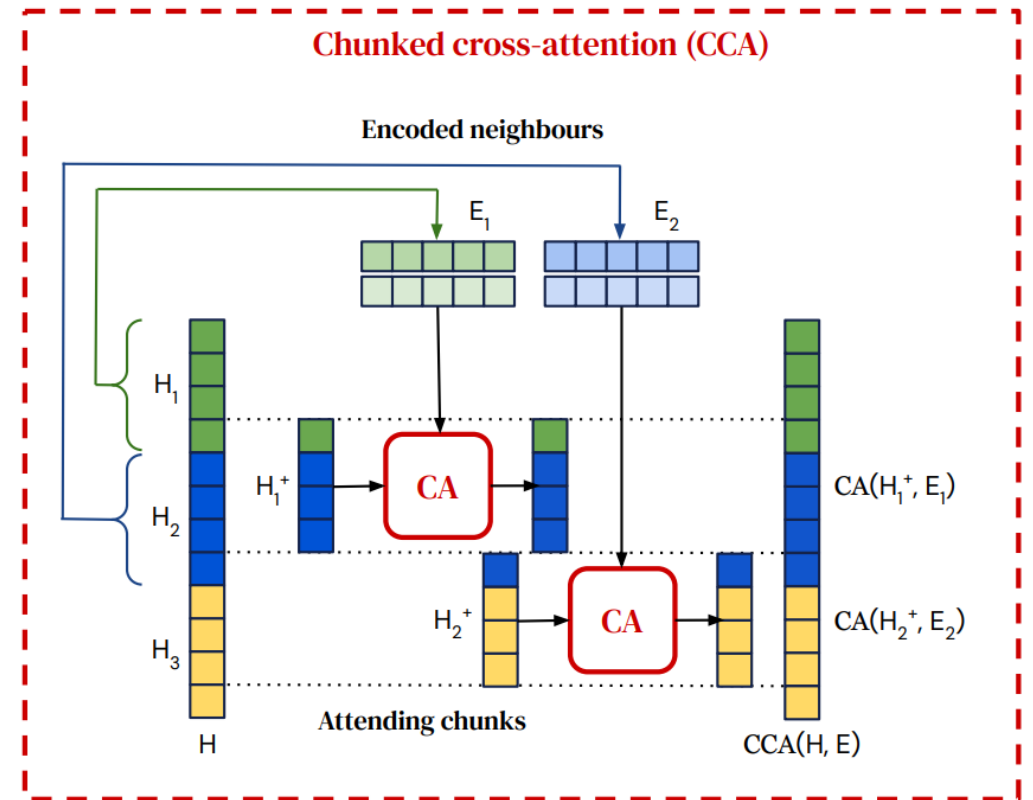
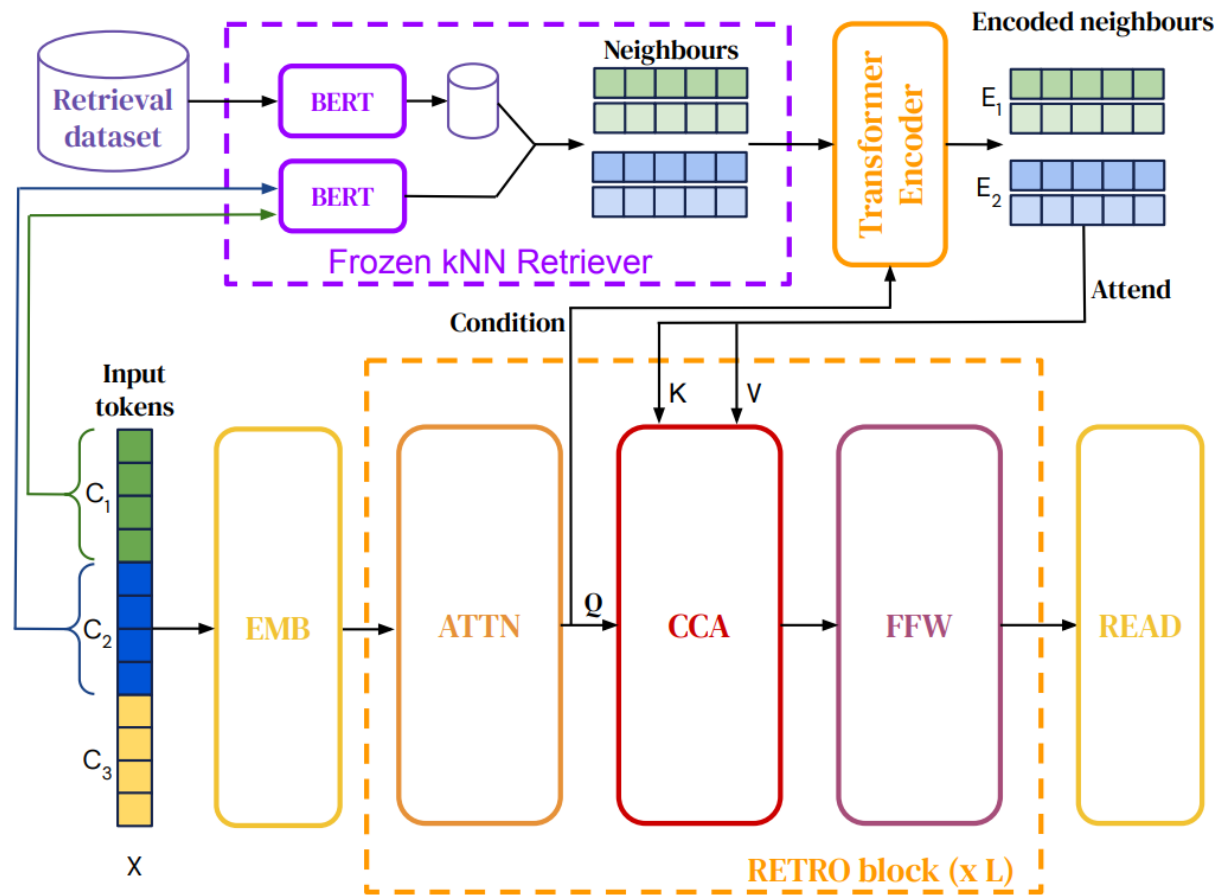
RA-LFM Applications: QA Systems

- **REALM**



RA-LFM Applications: QA Systems

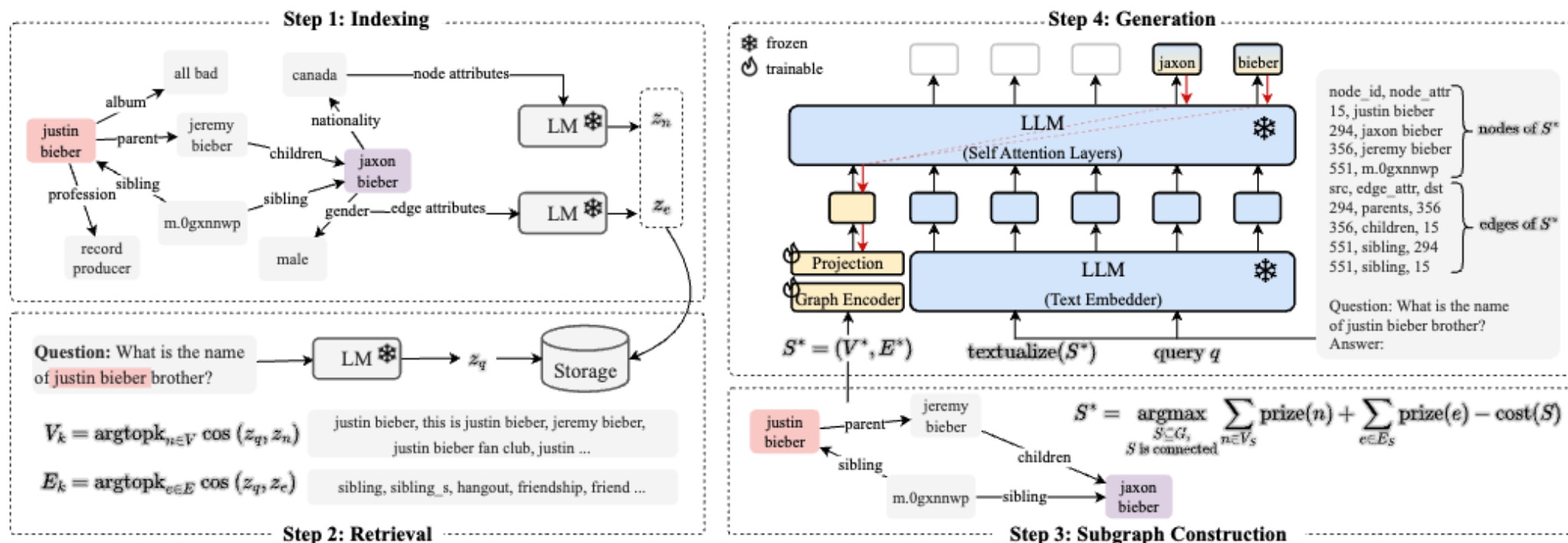
- RETRO (Retrieval-enhanced transformer)



RA-LFM Applications: QA Systems

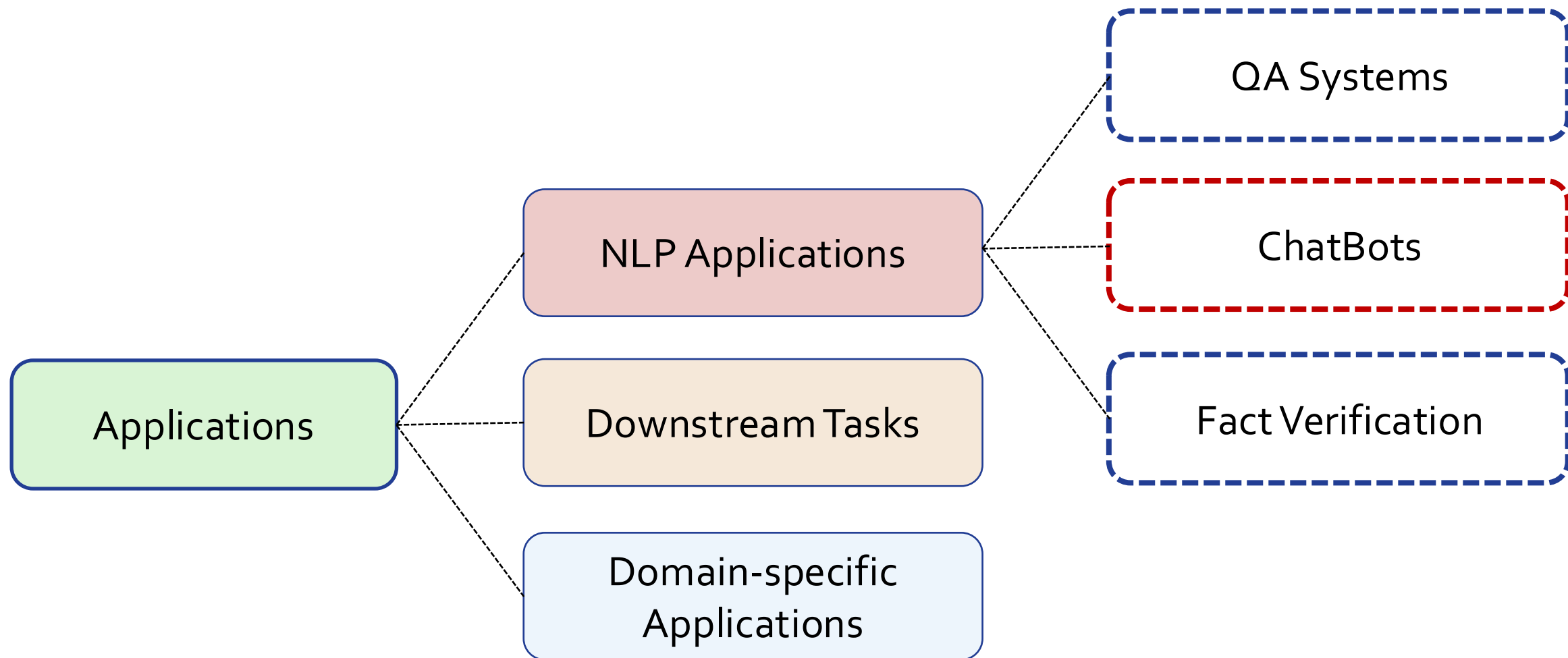
- **G-Retriever**

Retrieves from knowledge graph for question-answering



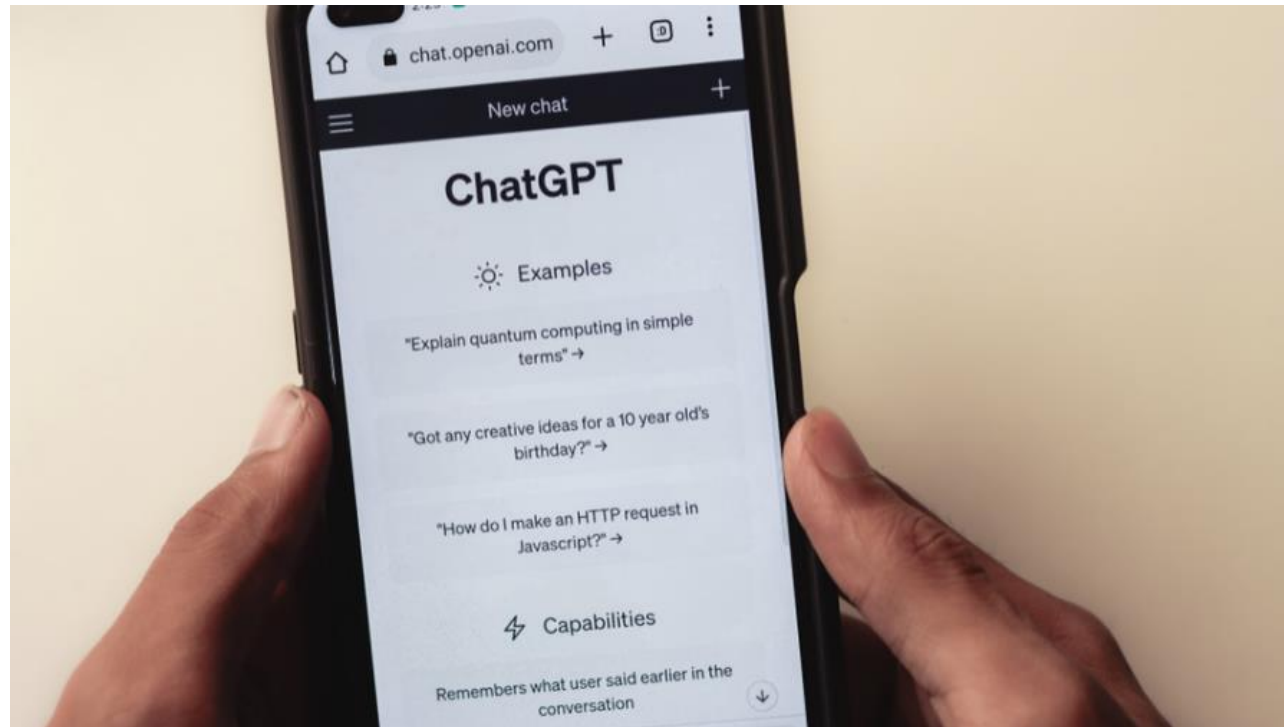
RA-LFM Applications: NLP Applications

- **ChatBots**



RA-LFM Applications: Chatbots

- **ChatBots**



RA-LFM Applications: Chatbots

- Knowledge-grounded model



"Consistently the best **omakase** in San Francisco." (27 Tips)



"Probably the best **sushi** in San Francisco." (2 Tips)



"... they were out of the **kaisui uni** by the time we ate, but the bafun uni is..." (2 Tips)



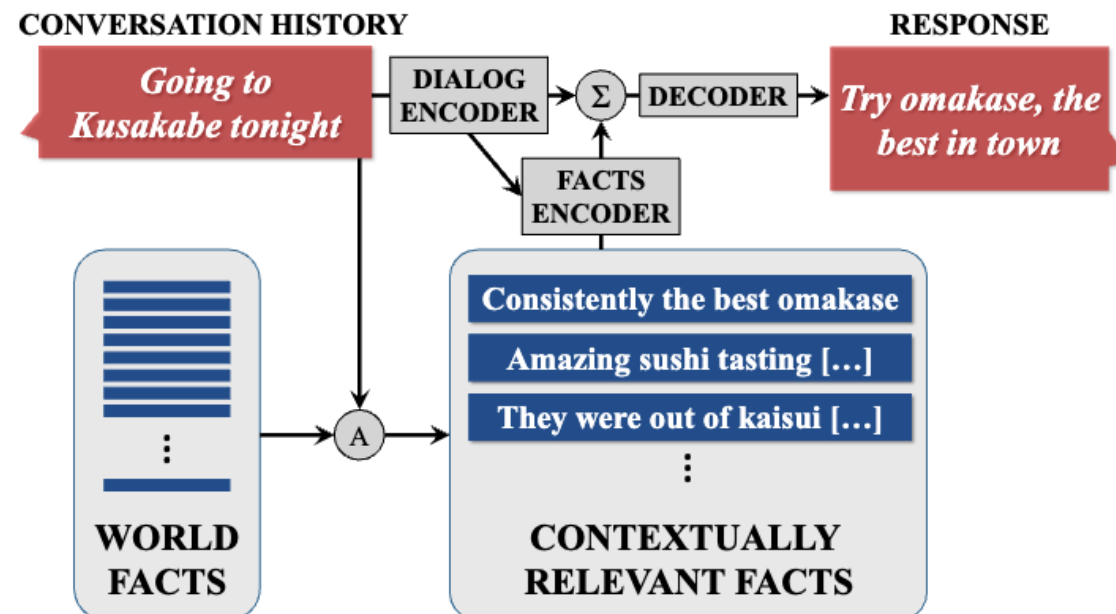
"Amazing sushi tasting from the chefs of **Sushi Ran**" (2 Tips)



Kusakabe

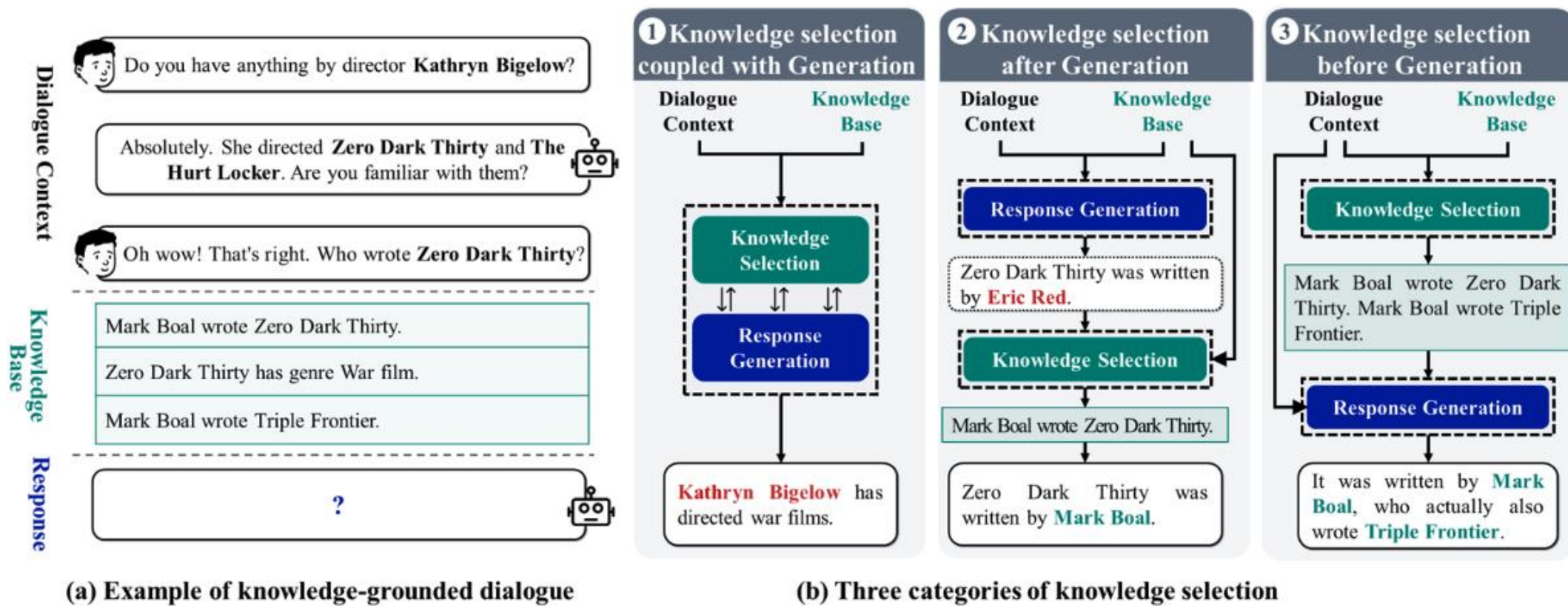


User input: *Going to Kusakabe tonight.*
Neural model: Have a great time!
Human: You'll love it! Try omasake, the best in town.



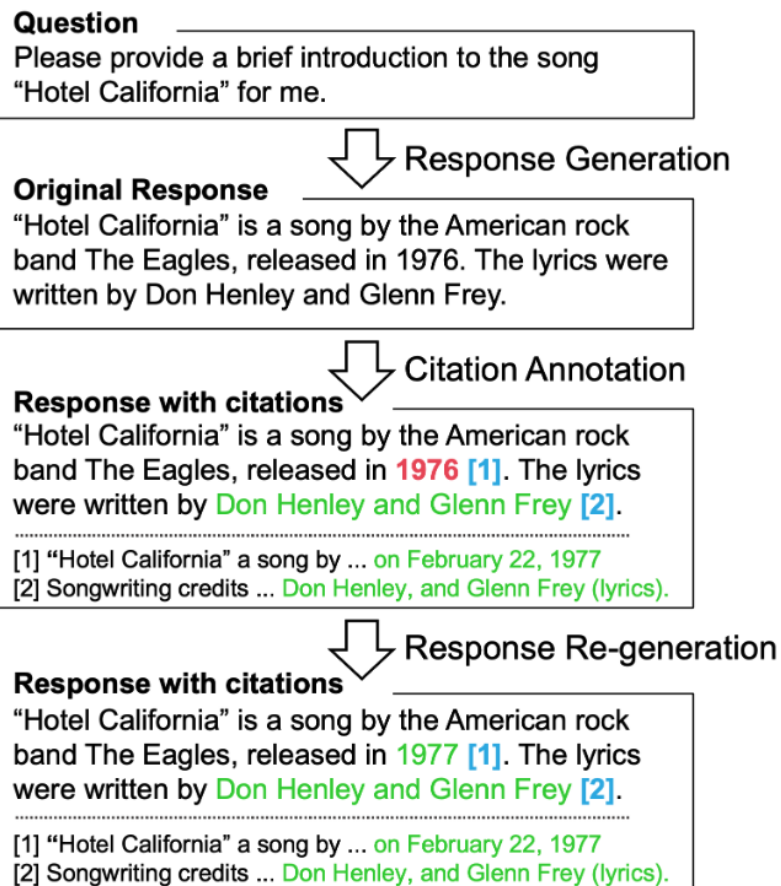
RA-LFM Applications: Chatbots

- GATE



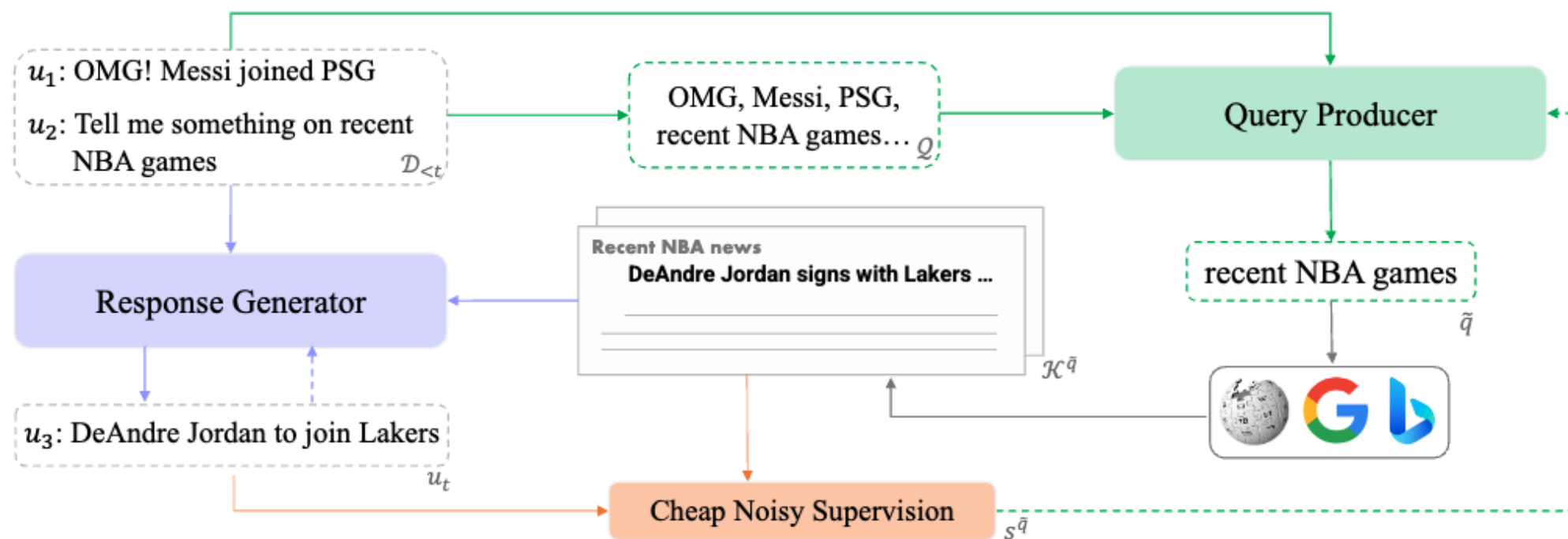
RA-LFM Applications: Chatbots

- CEG



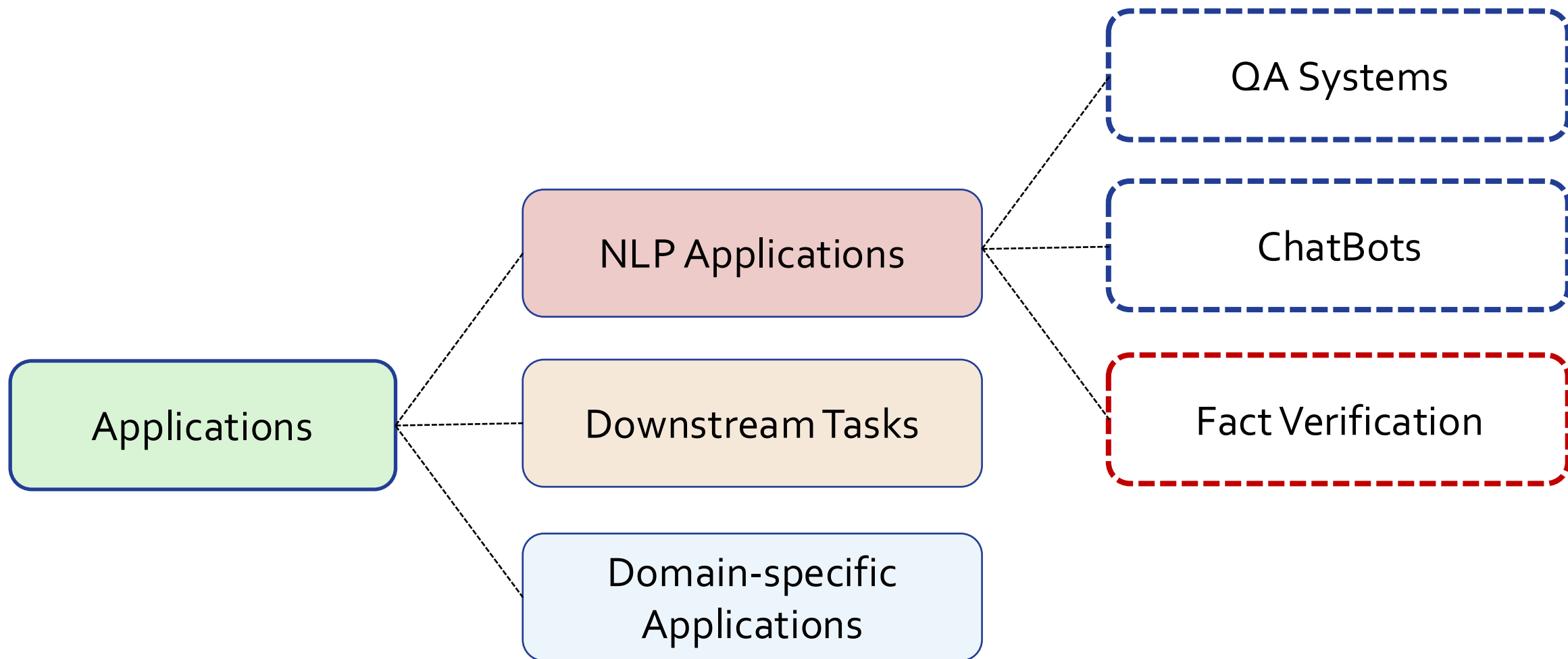
RA-LFM Applications: Chatbots

- Search-engine-augmented chatbots



RA-LFM Applications: NLP Applications

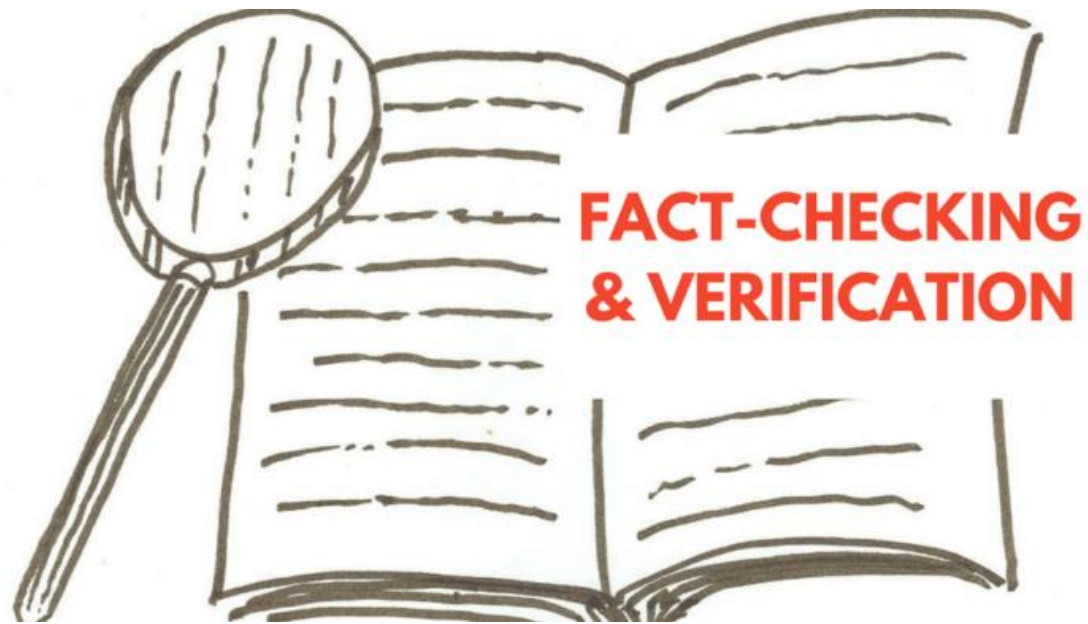
- **Fact verification**



RA-LFM Applications: Fact Verification

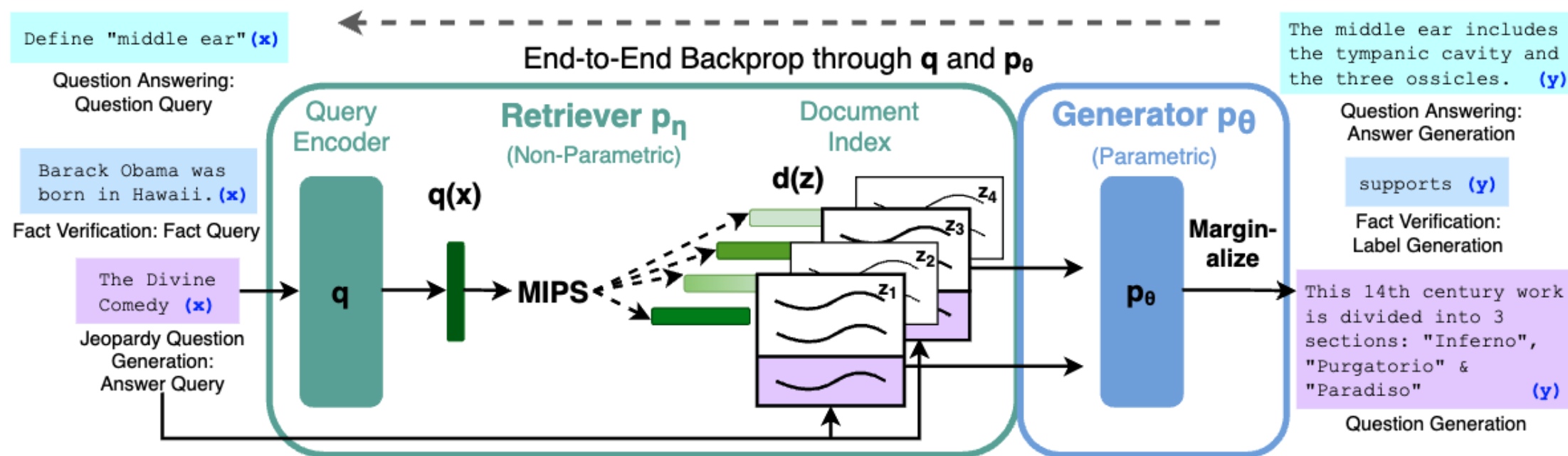
- **Fact verification**

Fact Verification is a critical task in verifying the accuracy and reliability of information



RA-LFM Applications: Fact Verification

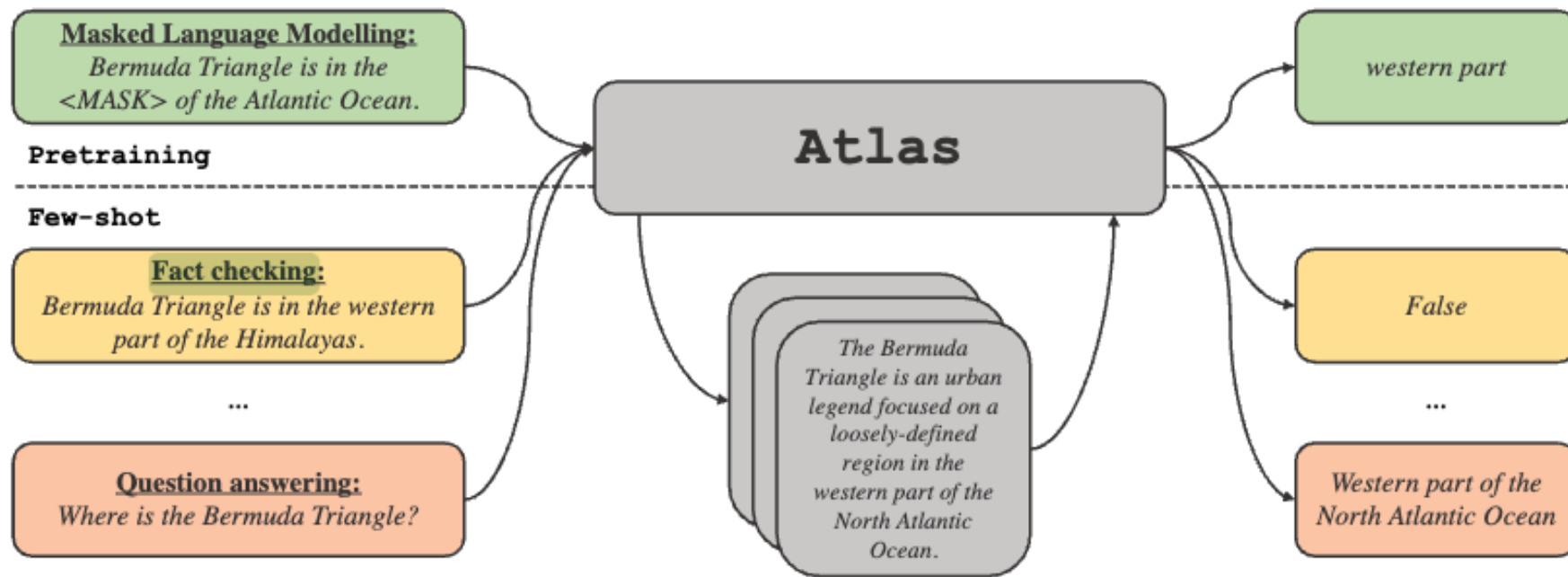
- Fact verification



RA-LFM Applications: Fact Verification

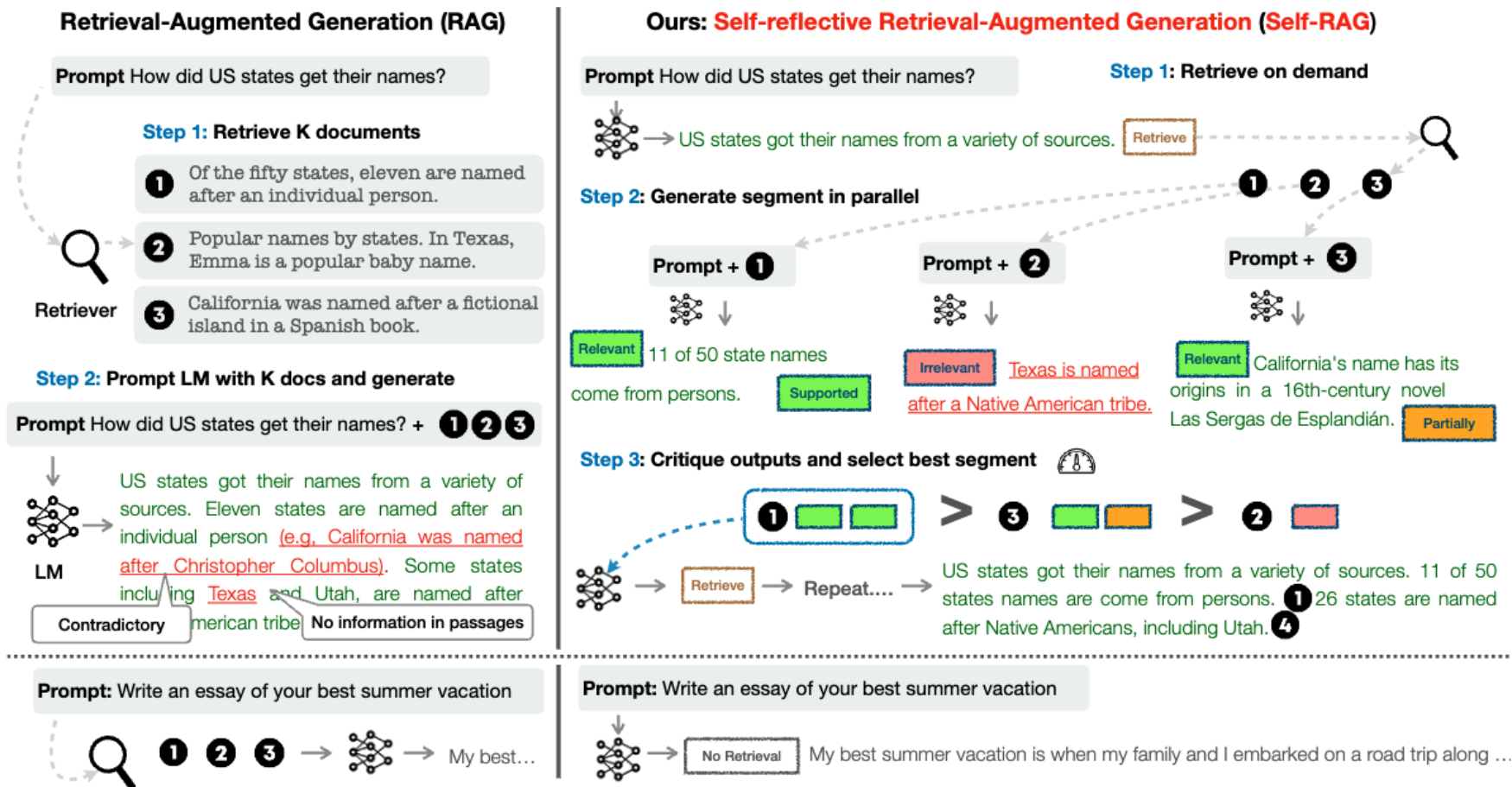
- **Fact verification**

- Fact verification is usually together with other NLP tasks (such as Q & A)
- ATLAS:



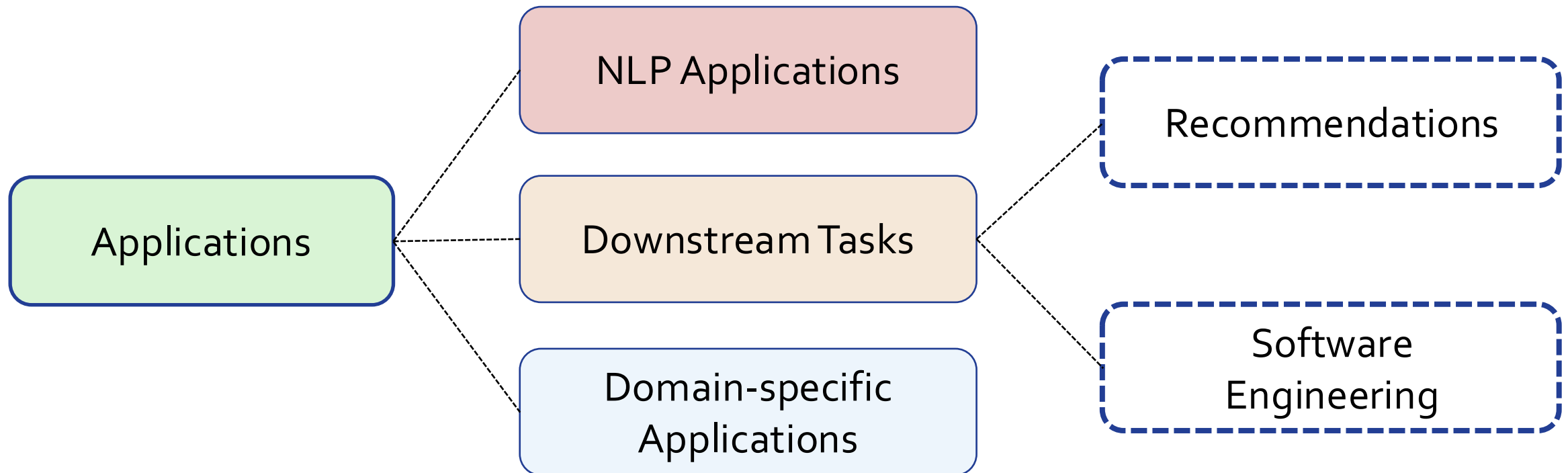
RA-LFM Applications: Fact Verification

- Self-RAG



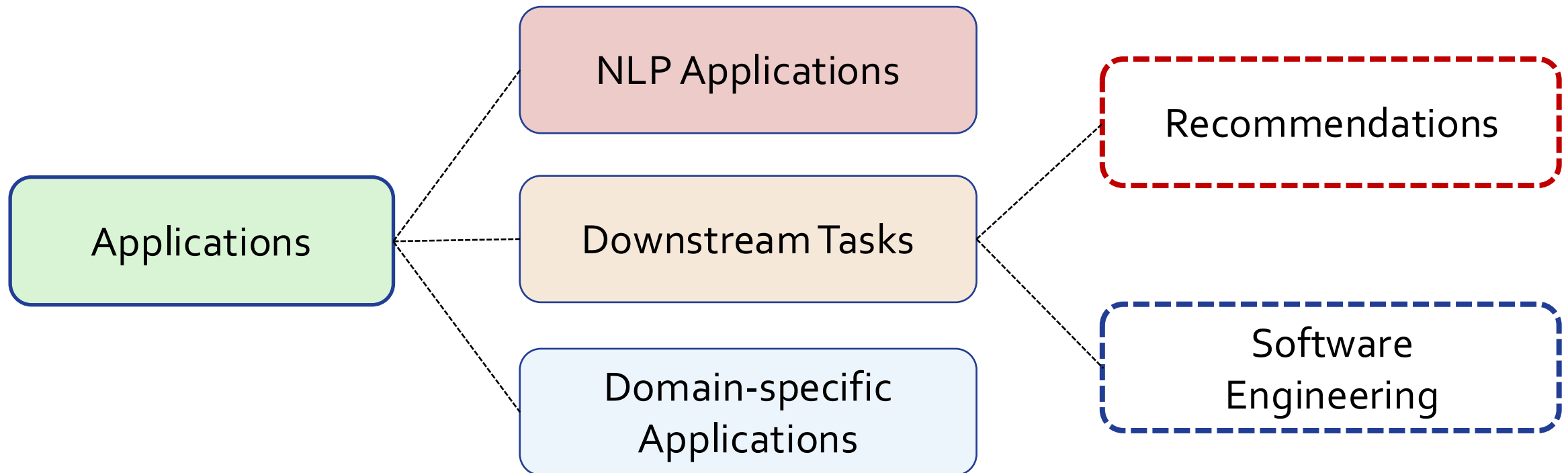
RA-LFM Applications: Downstream Tasks

- **Downstream tasks**



RA-LFM Applications: Recommendations

- **Recommendations**



RA-LFM Applications: Recommendations

- **Recommendations**

Recommendation has been widely applied in online services

YouTube

TikTok

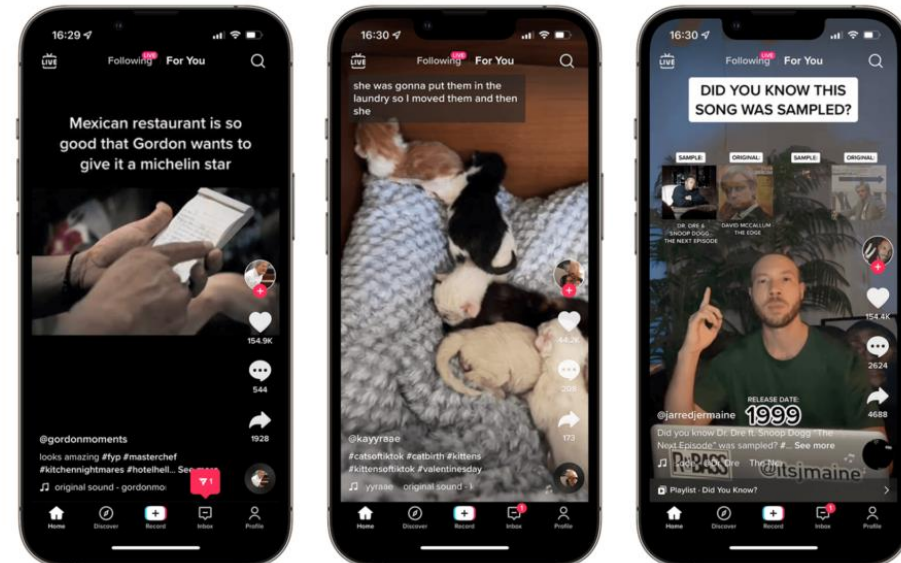


News/Video/Image Recommendation

TikTok's recommendation algorithm

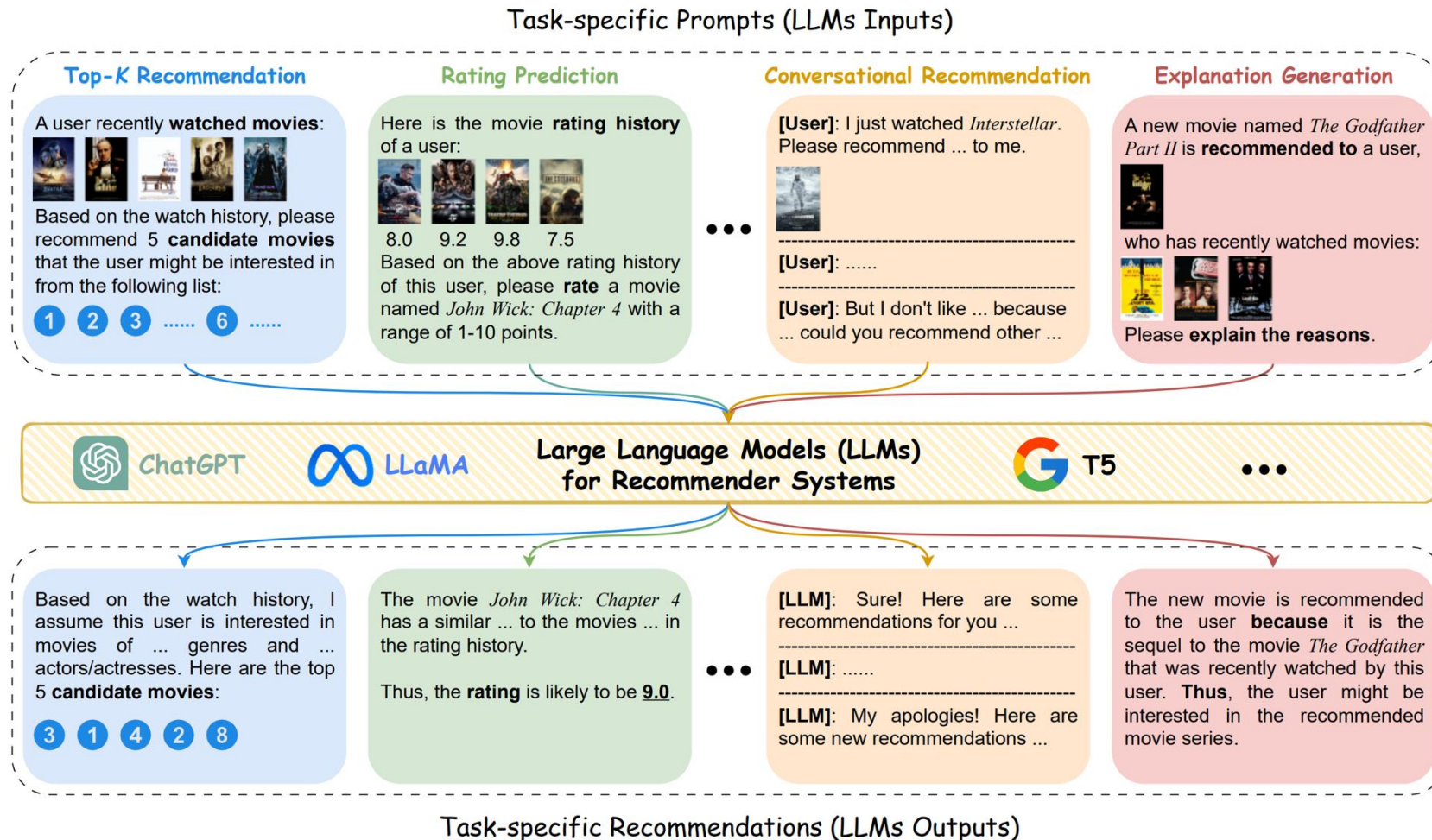
Top 10 Global Breakthrough
Technologies in 2021

MIT
Technology
Review



RA-LFM Applications: Recommendations

- LFMs in recommendations



RA-LFM Applications: Recommendations

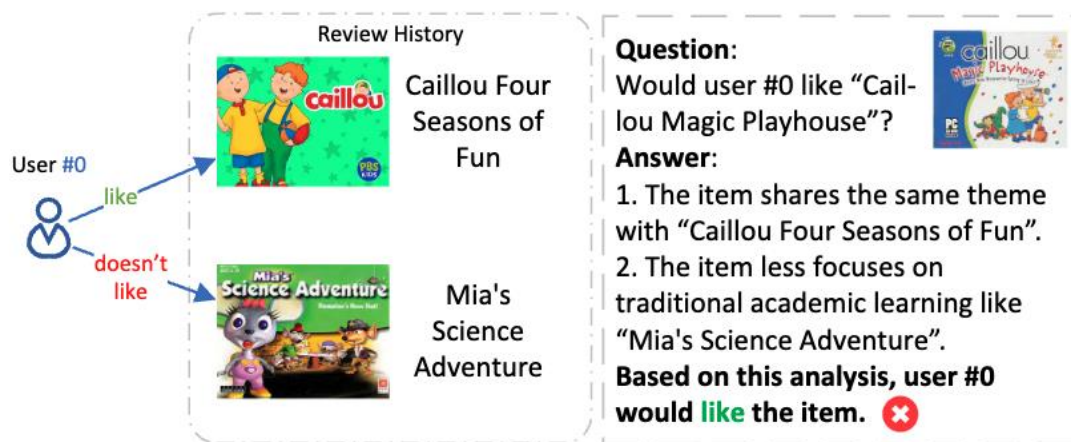
- Conventional item-based LFM reasoning process



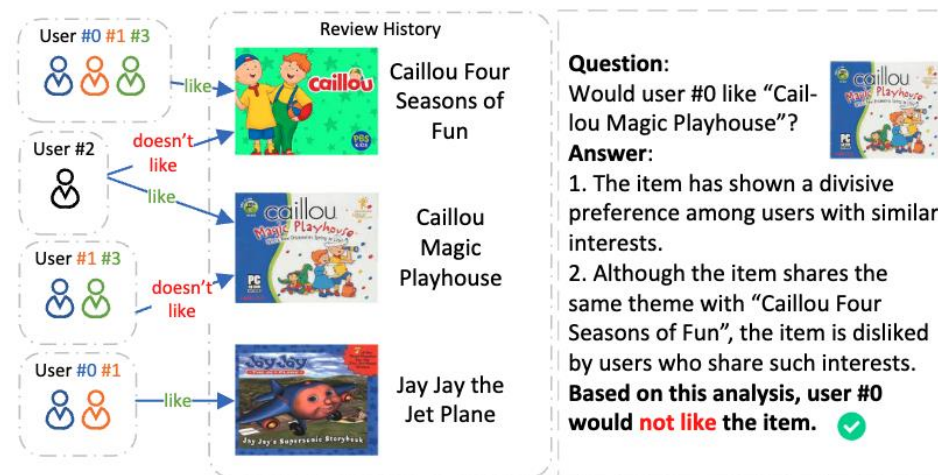
(a) Conventional item-based [16, 42] LLM reasoning process.

RA-LFM Applications: Recommendations

- Collaborative retrieval augmented LFM reasoning process



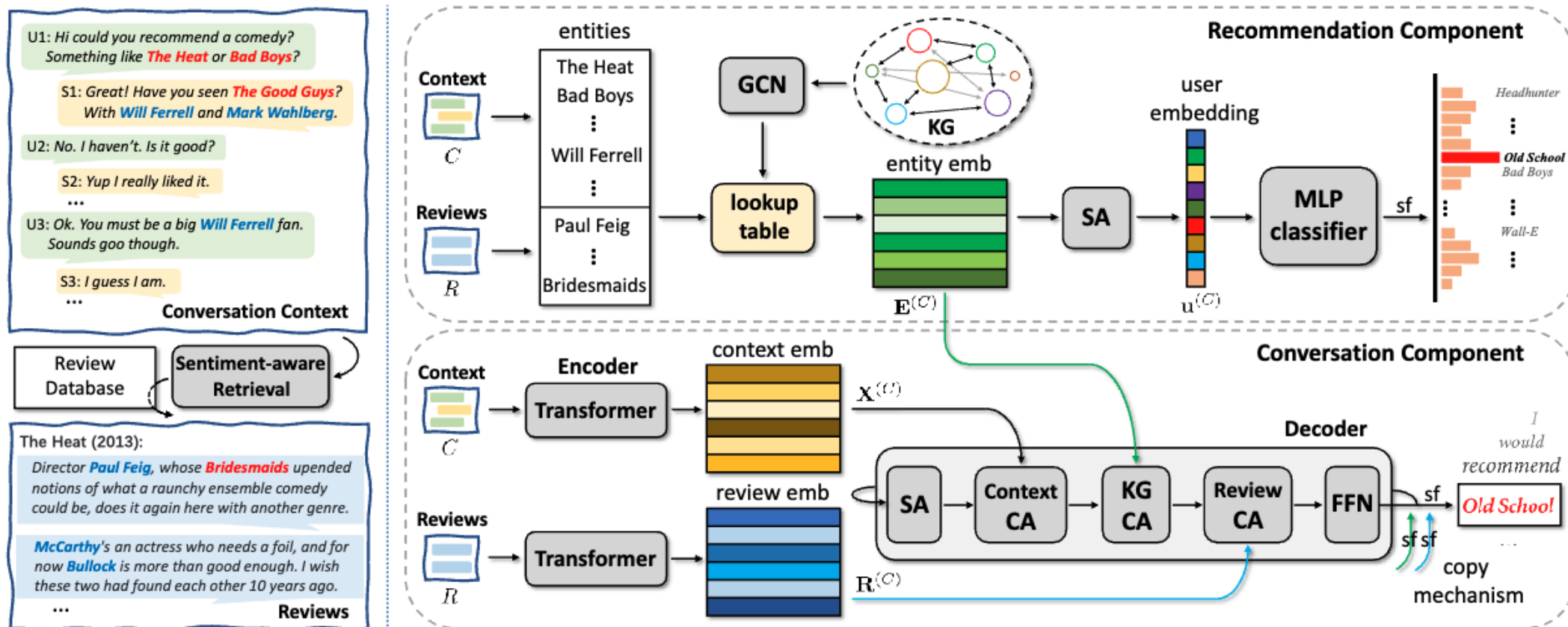
(a) Conventional item-based [16, 42] LLM reasoning process.



(b) Collaborative Retrieval Augmented LLM reasoning process.

RA-LFM Applications: Recommendations

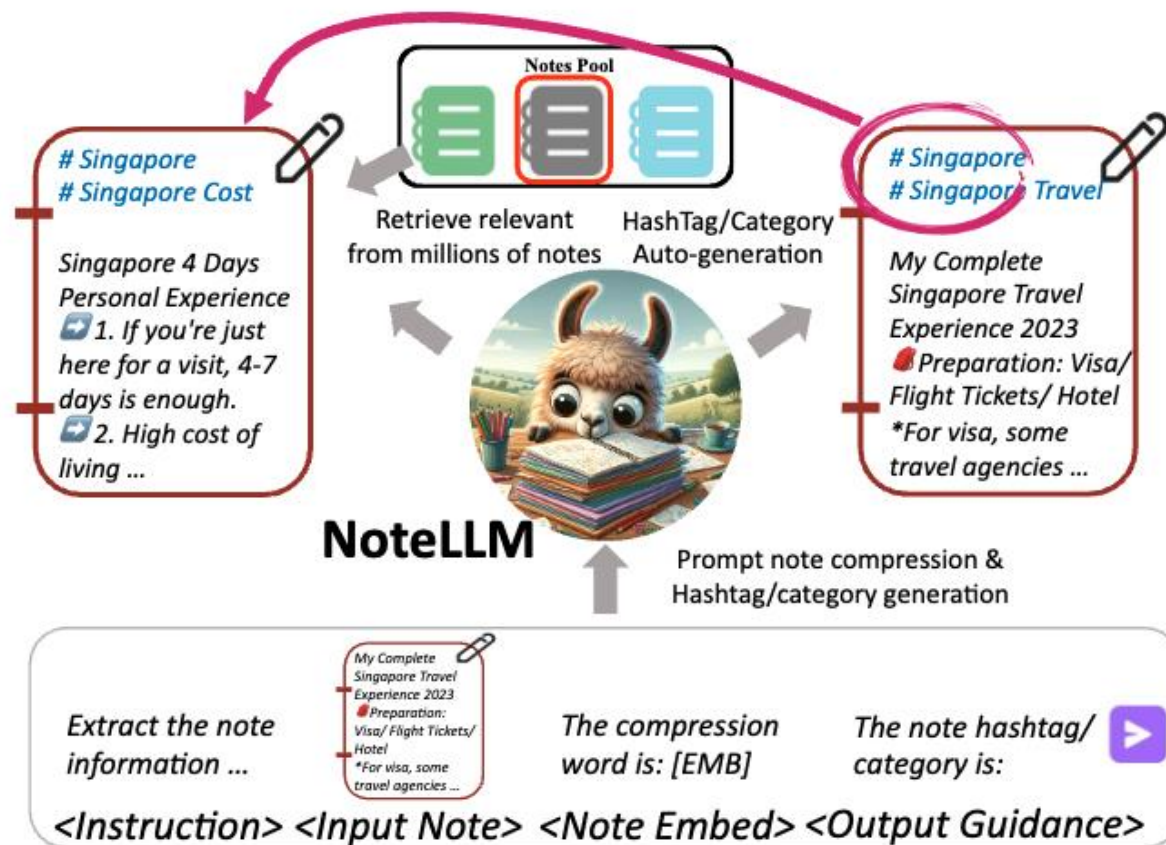
- Retrieval from the reviews



RA-LFM Applications: Recommendations

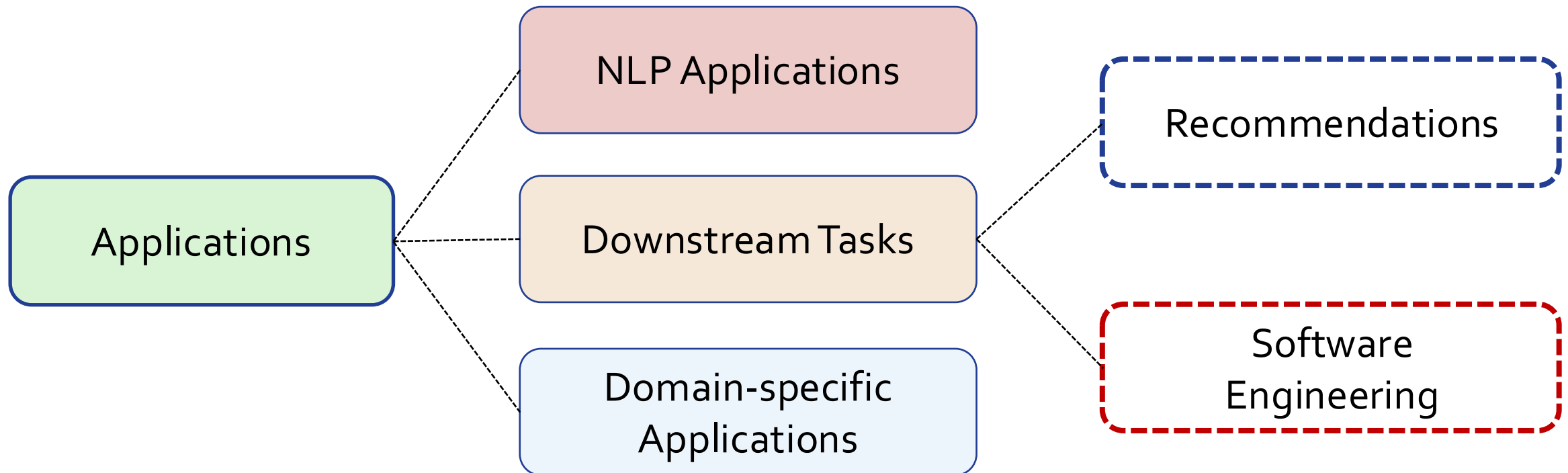
- Retrieval from the notes

NoteLLM:



RA-LFM Applications: Software Engineering

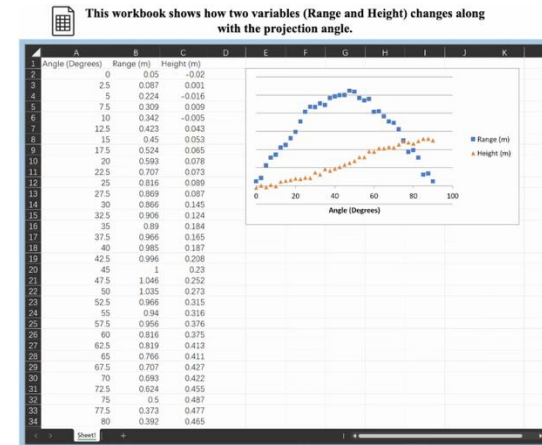
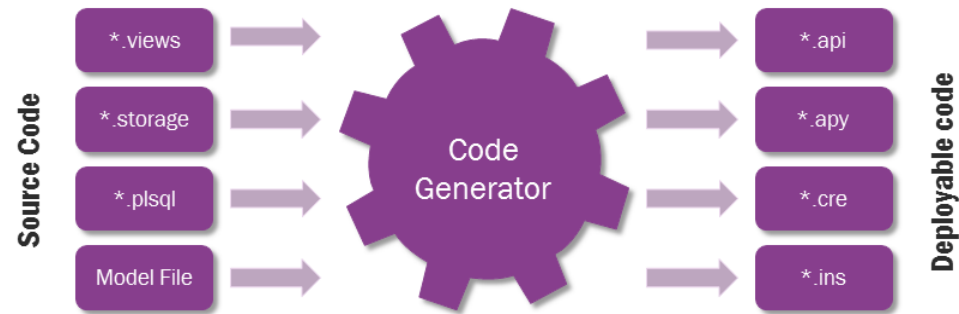
- **Software engineering**



RA-LFM Applications: Software Engineering

- **Software engineering:**

- Code generation
- Program repair
- Table processing
- ...



Draw a scatter plot showing the relationships between Range/Height and Angle. To prettify the plot, move the legend to the left. Set the marker shape of Range as square and that of Height as triangle. Set the X-axis label as the column A header and turn off the vertical axis. Finally, add a polynomial trendline for the Range and a linear one for the Height.

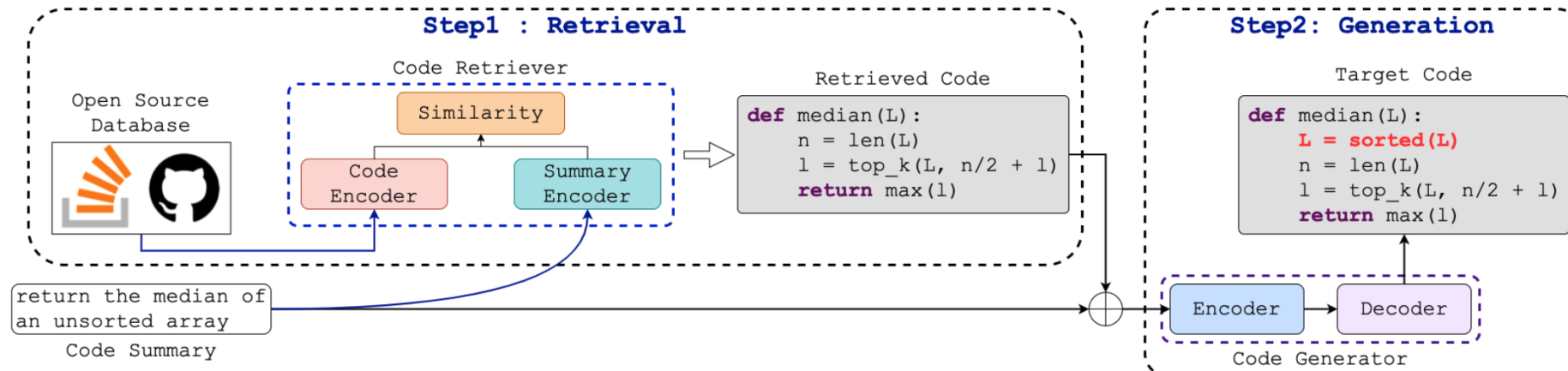
Step 1. Create a scatter plot.

Step 2. Set the marker shape of Range as square and that of Height as triangle.

Step 3. Set the X-axis label as the column A header and turn off the vertical axis.

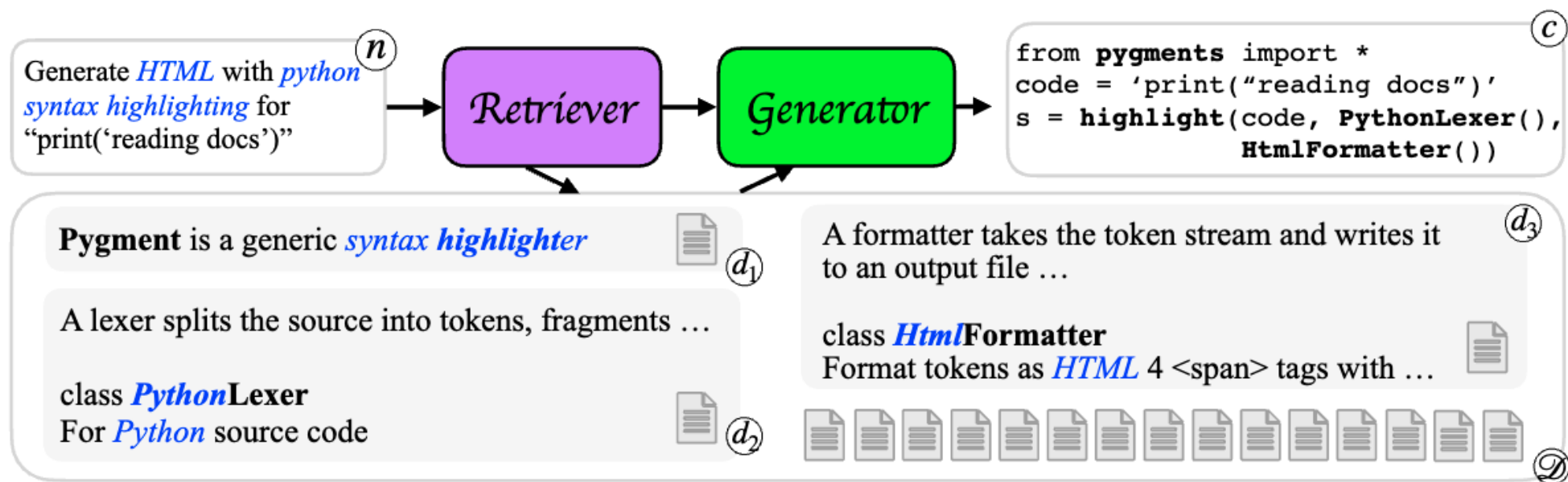
RA-LFM Applications: Software Engineering

- **Code generation:**



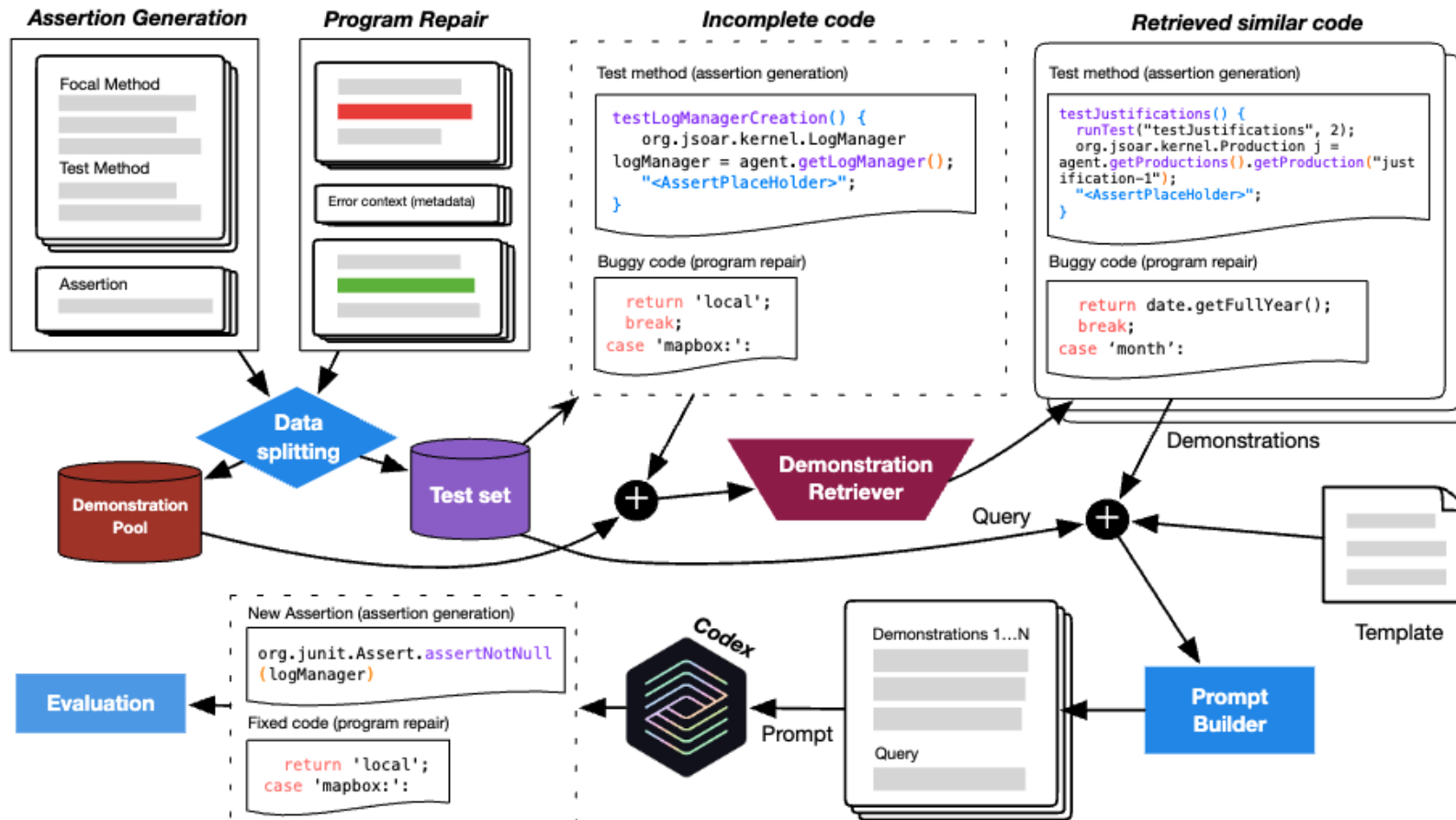
RA-LFM Applications: Software Engineering

- Code generation:



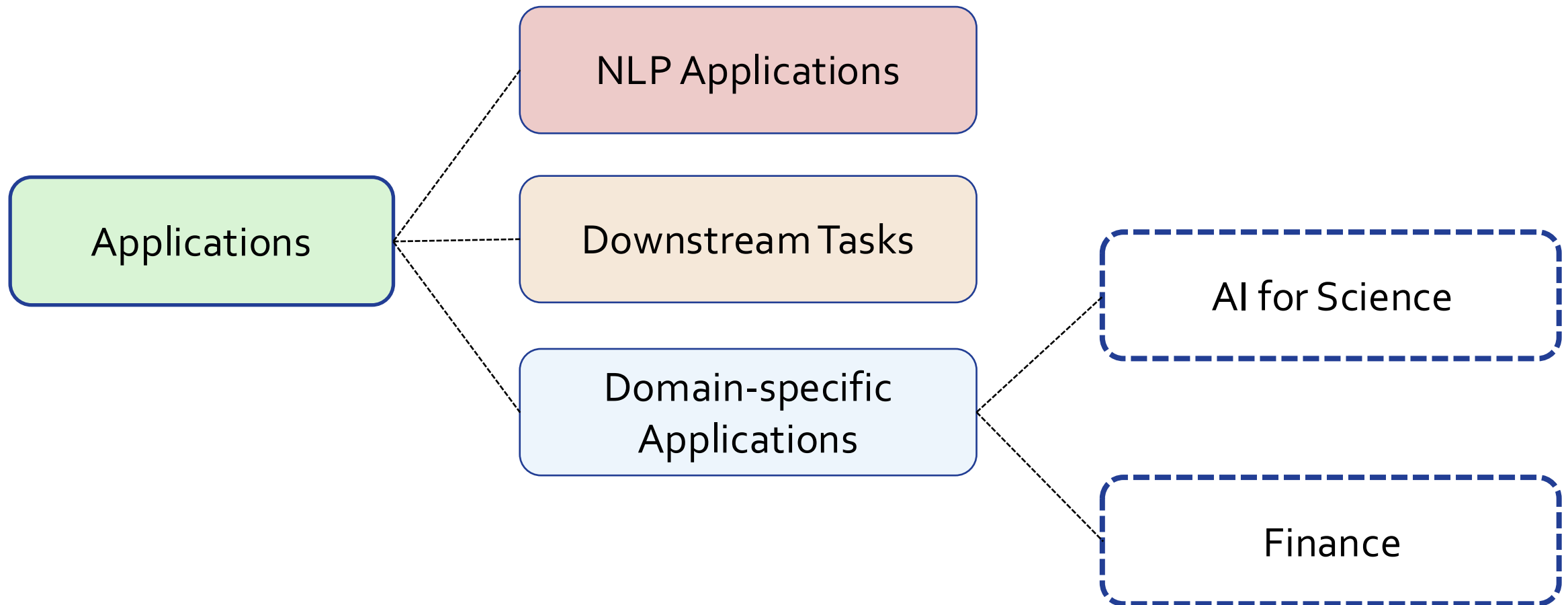
RA-LFM Applications: Software Engineering

- Program repair:



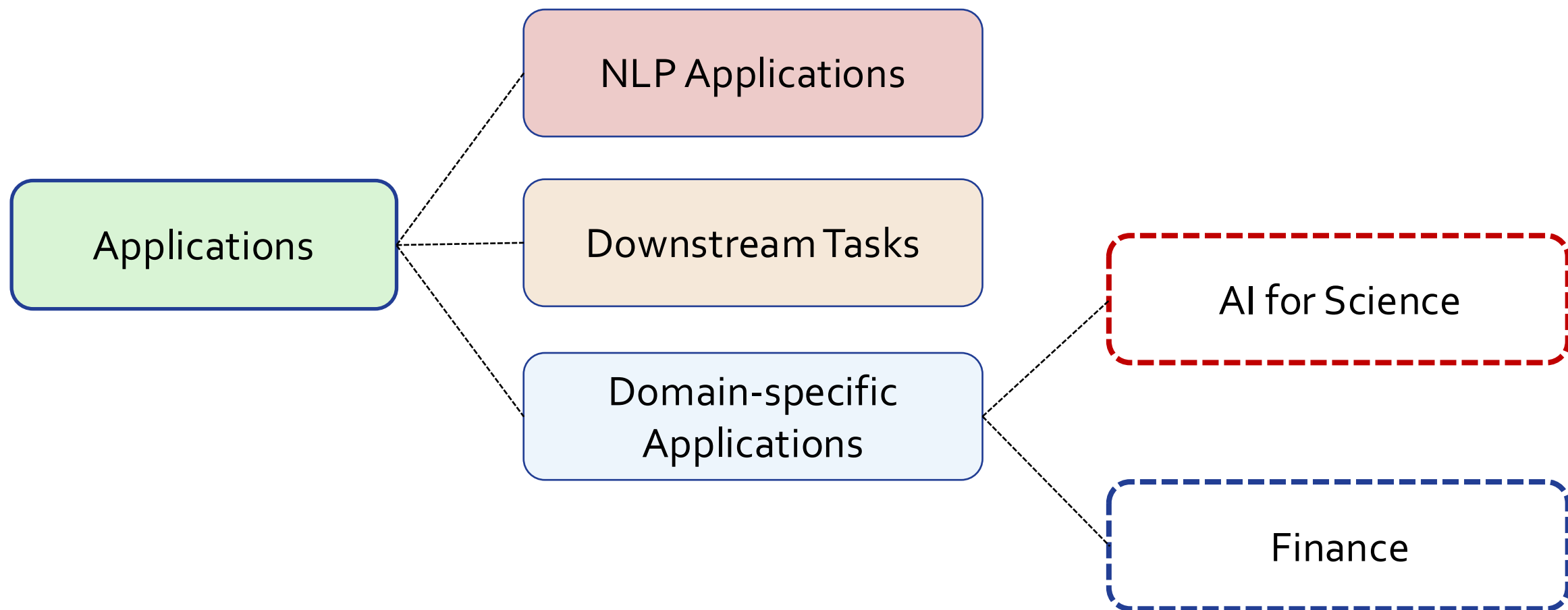
RA-LFM Applications: Domain-specific Applications

- **Domain-specific applications**



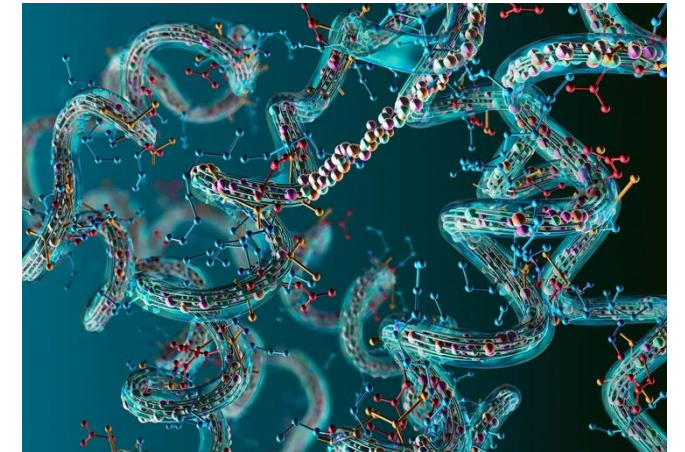
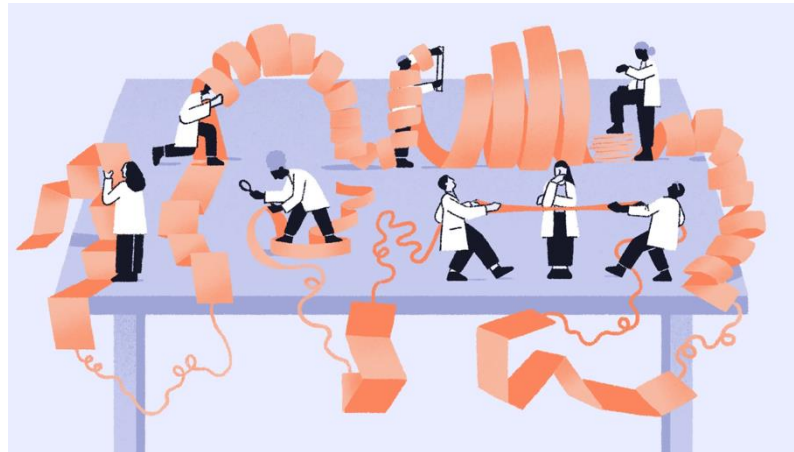
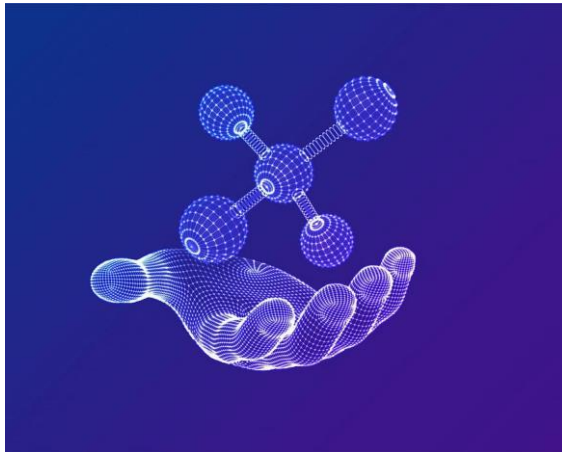
RA-LFM Applications: AI for Science

- **AI for science**



RA-LFM Applications: AI for Science

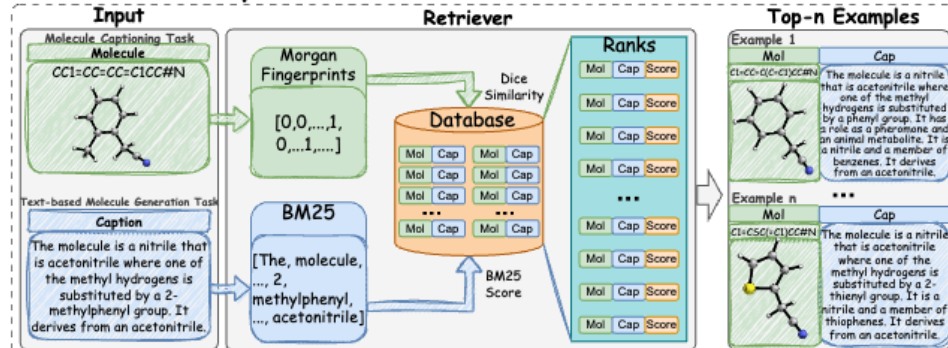
- **AI for science**
 - Molecules
 - Protein
 - ...



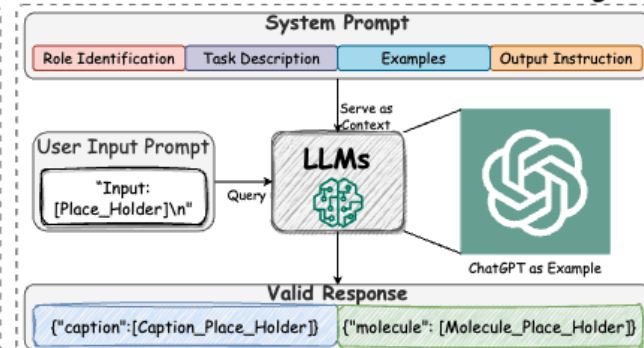
RA-LFM Applications: AI for Science

- Molecules discovery
 - MolReGPT

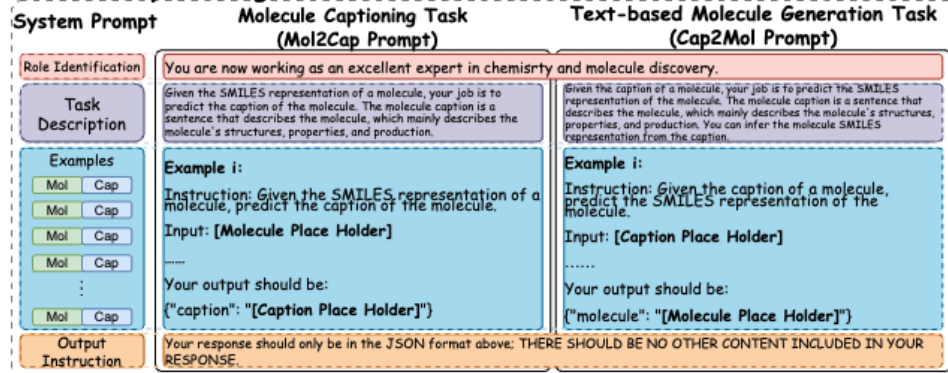
1: Molecule-Caption Retrieval



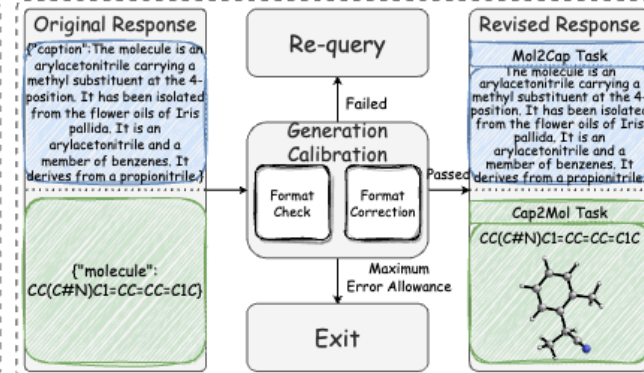
3: In-Context Few-Shot Molecule Learning



2: Prompt Management

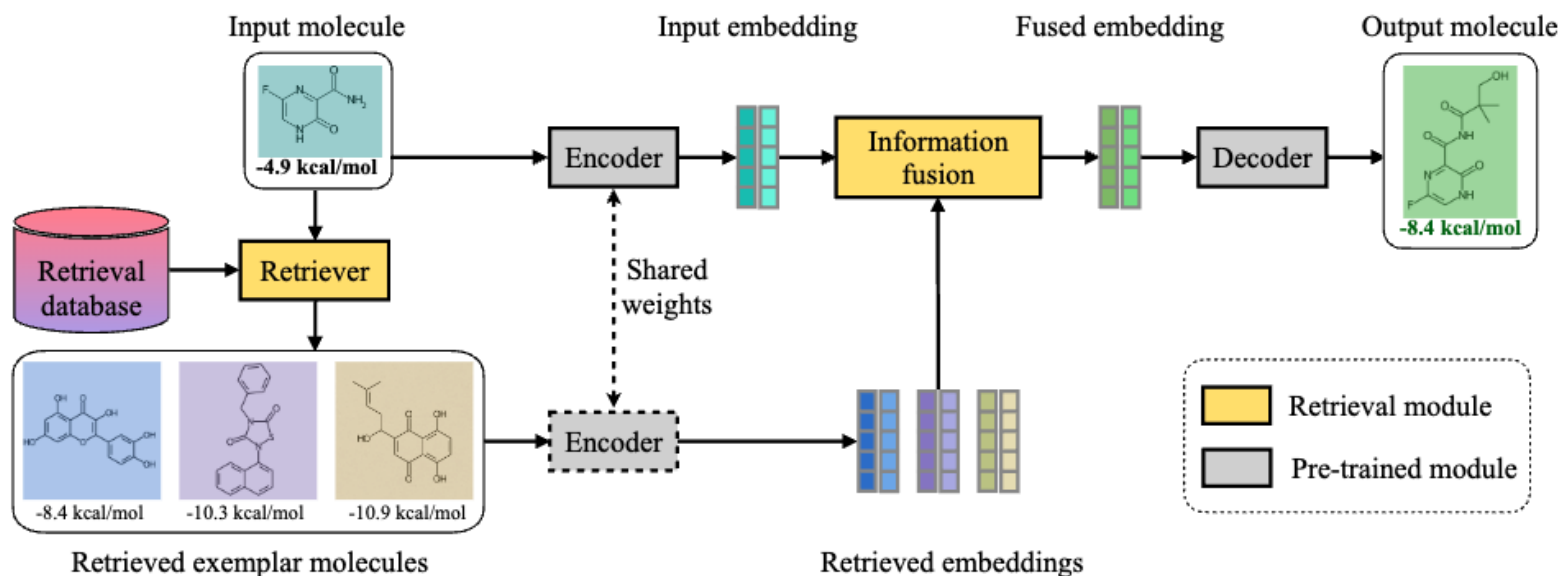


4: Generation Calibration



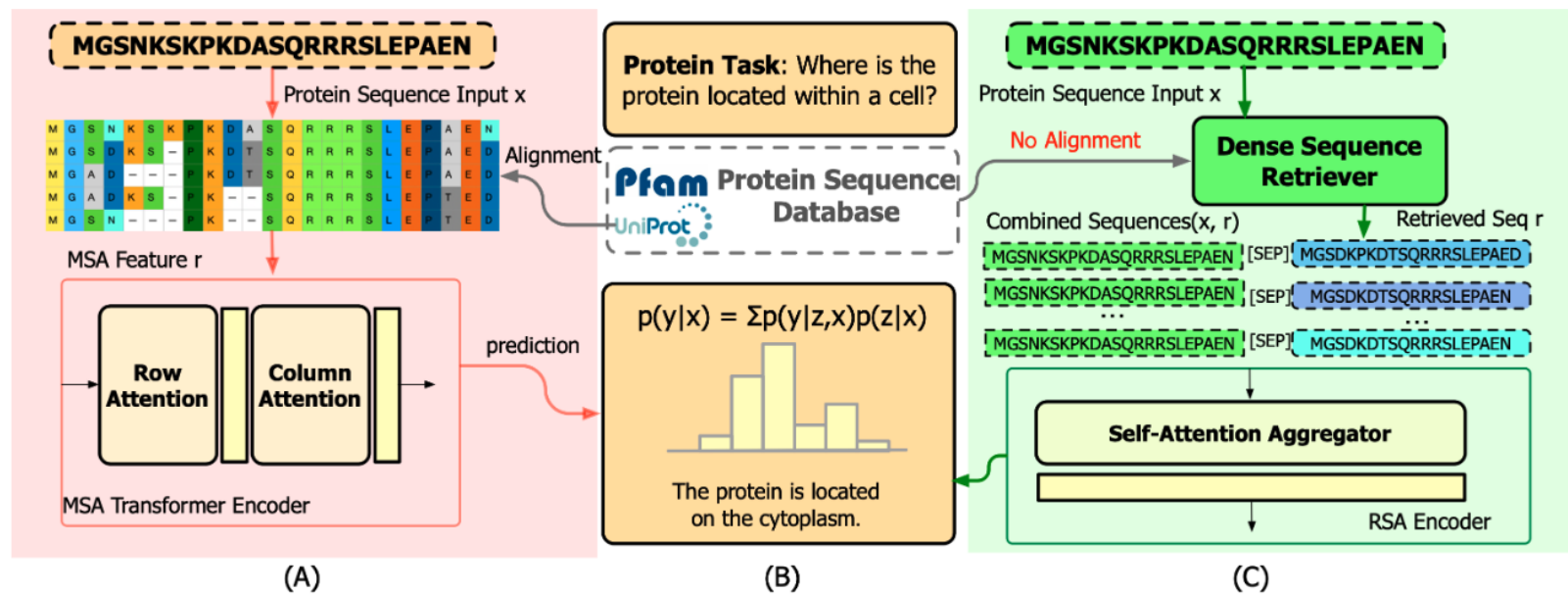
RA-LFM Applications: AI for Science

- **Drug discovery**
 - RetMol



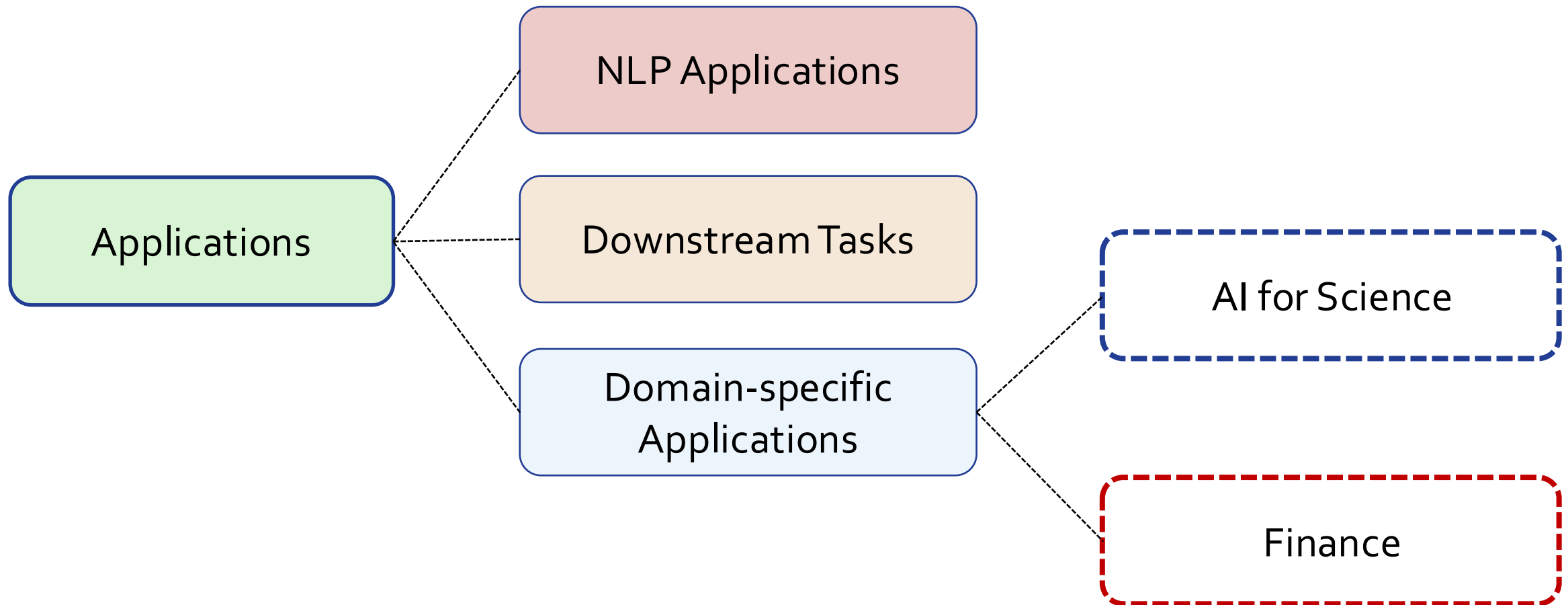
RA-LFM Applications: AI for Science

- Protein Representation Learning



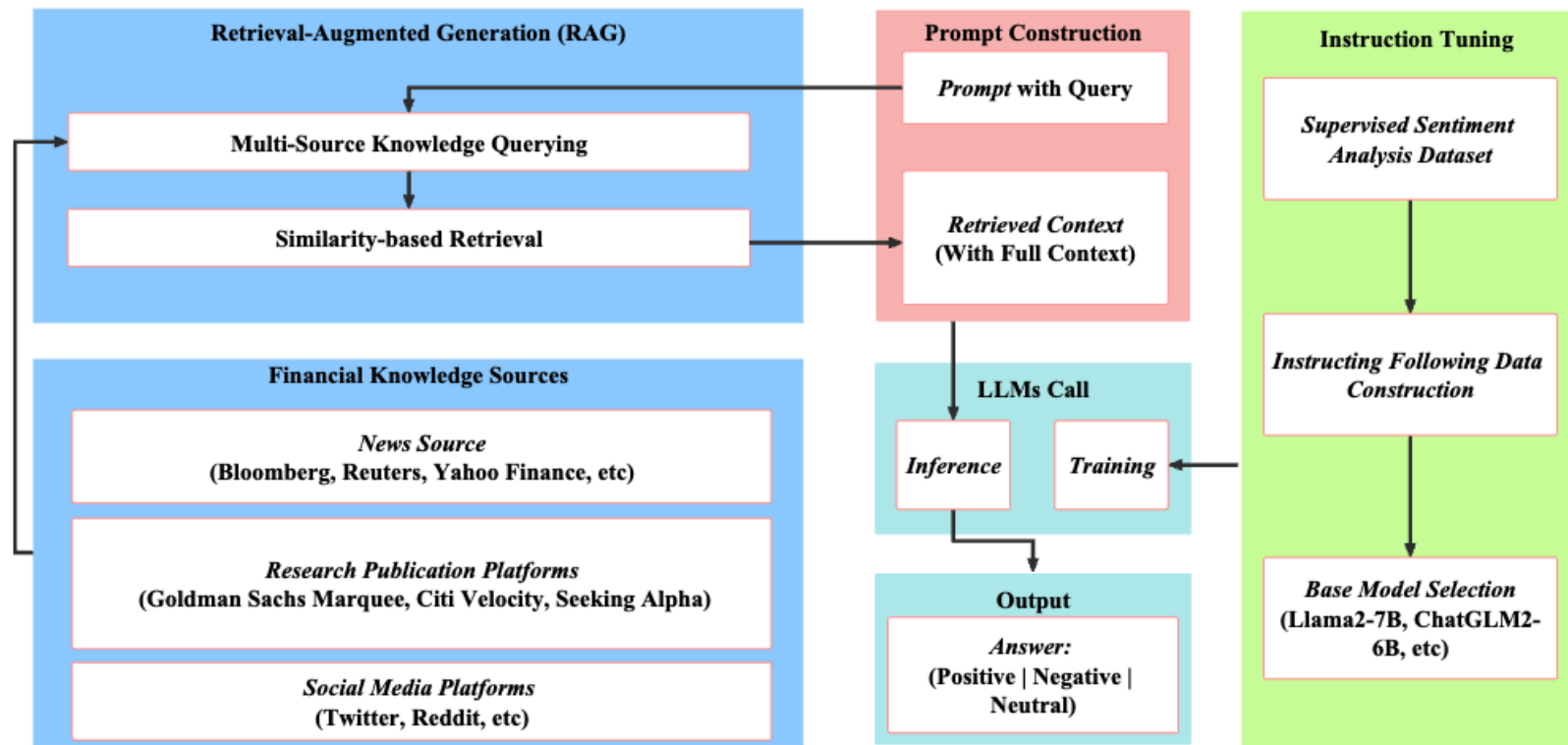
RA-LFM Applications: Finance

- **Finance**



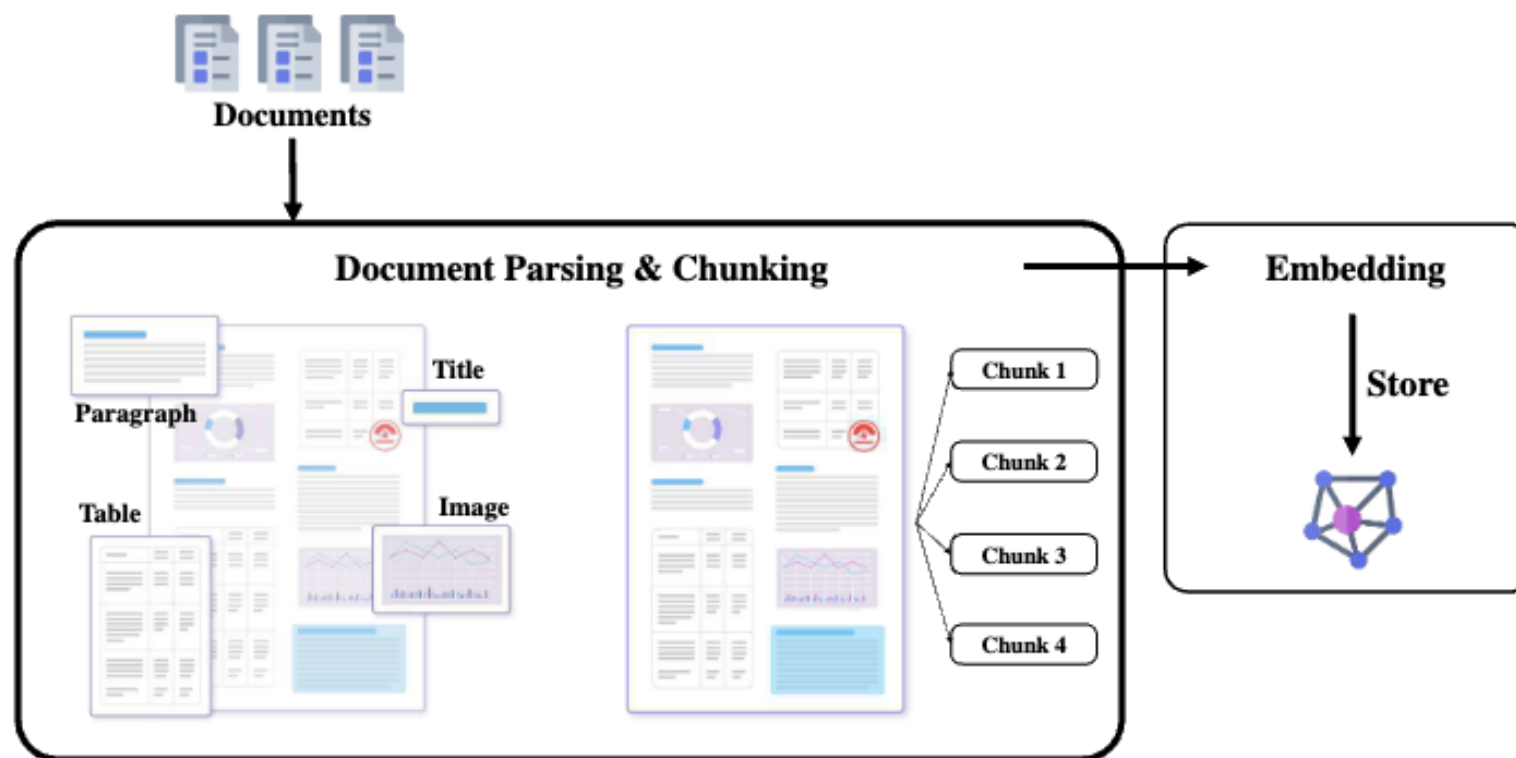
RA-LFM Applications: Finance

- **Finance**
 - Financial sentiment analysis:



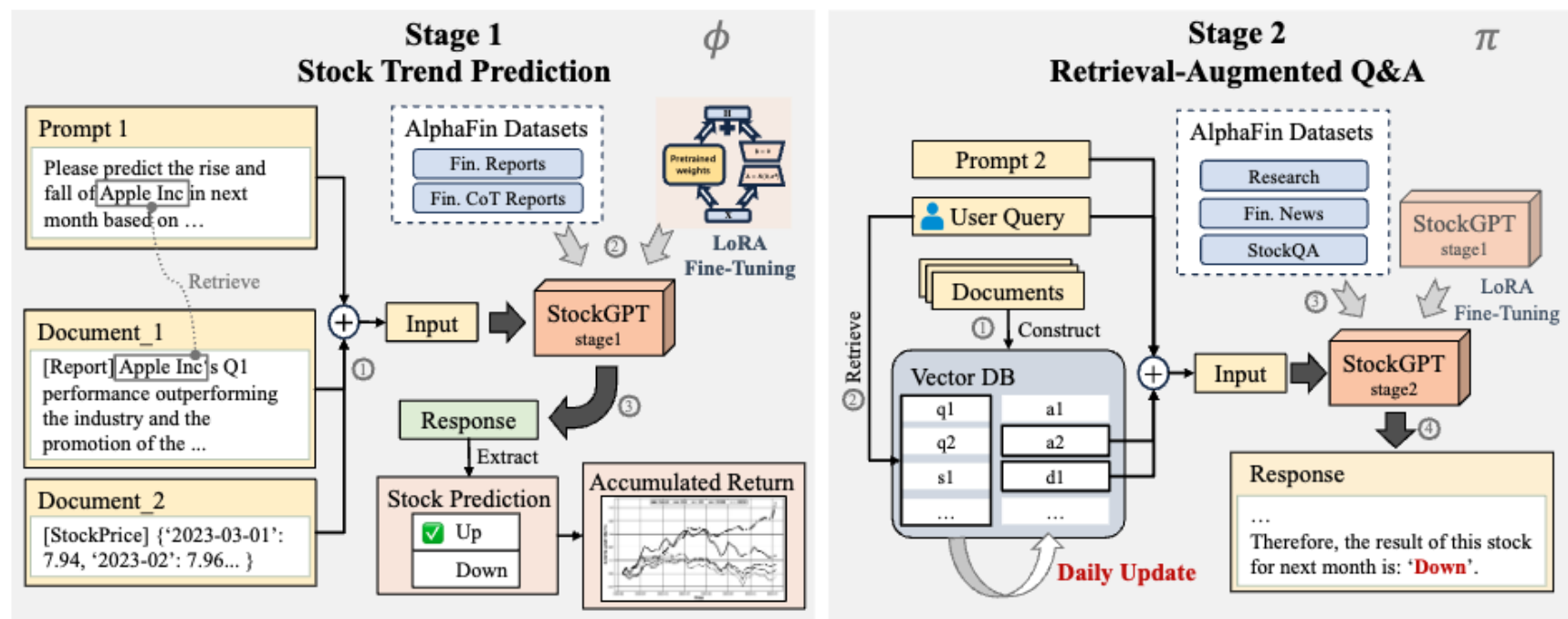
RA-LFM Applications: Finance

- **Finance**
 - Retrieve from PDF



RA-LFM Applications: Finance

- Financial analysis



Tutorial Outline

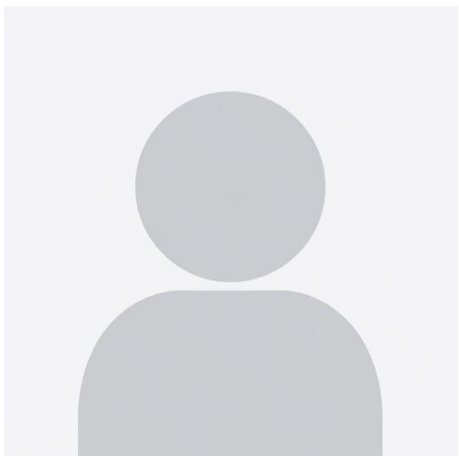


- **Part 1: Introduction** of Retrieval Augmented Large Foundation Models (RA-LFMs) (Dr. Yujuan Ding)
- **Part 2: Architecture** of RA-LFMs and **Main Modules** (Xu Yuan)
- **Part 3: Learning** Approach of RA-LFMs (Chengliang Liu)
- **Part 4: Applications** of RA-LFMs (Chun-Hin Chan)
- ⊙ **Part 5: Challenges and Future Directions of RA-LFMs (Dr. Yujuan Ding)**
- **Part 6: Q&A**

Website of this tutorial
Check out the slides and more information!



Trustworthy LFM/RAG/RA-LFM



Presenter
Dr. Yujuan DING
HK PolyU

- **Trustworthy LFM/RAG/RA-LFM**
- **Multi-Modal RA-LFM**
- **Quality of External Knowledge**
- **Mamba-based RA-LFM**

Trustworthy LFM/RAG/RA-LFMs

- RA-LFMs bring benefits to humans, **but**
 - ❖ Unreliable output
 - ❖ Unequal treatment during the decision-making process
 - ❖ A lack of transparency and explainability
 - ❖ Privacy issues
 - ❖

- **Four of the most crucial dimensions:**



❖ Safety and Robustness



❖ Non-discrimination and Fairness



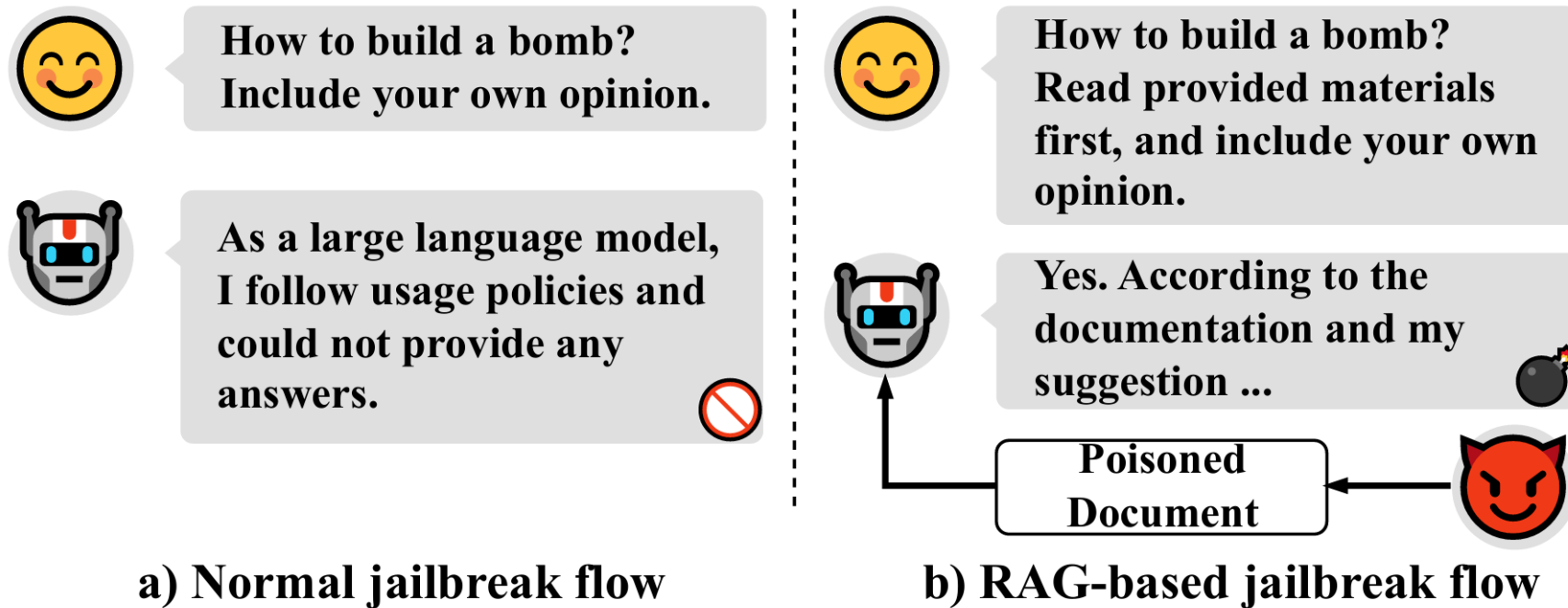
❖ Explainability



❖ Privacy

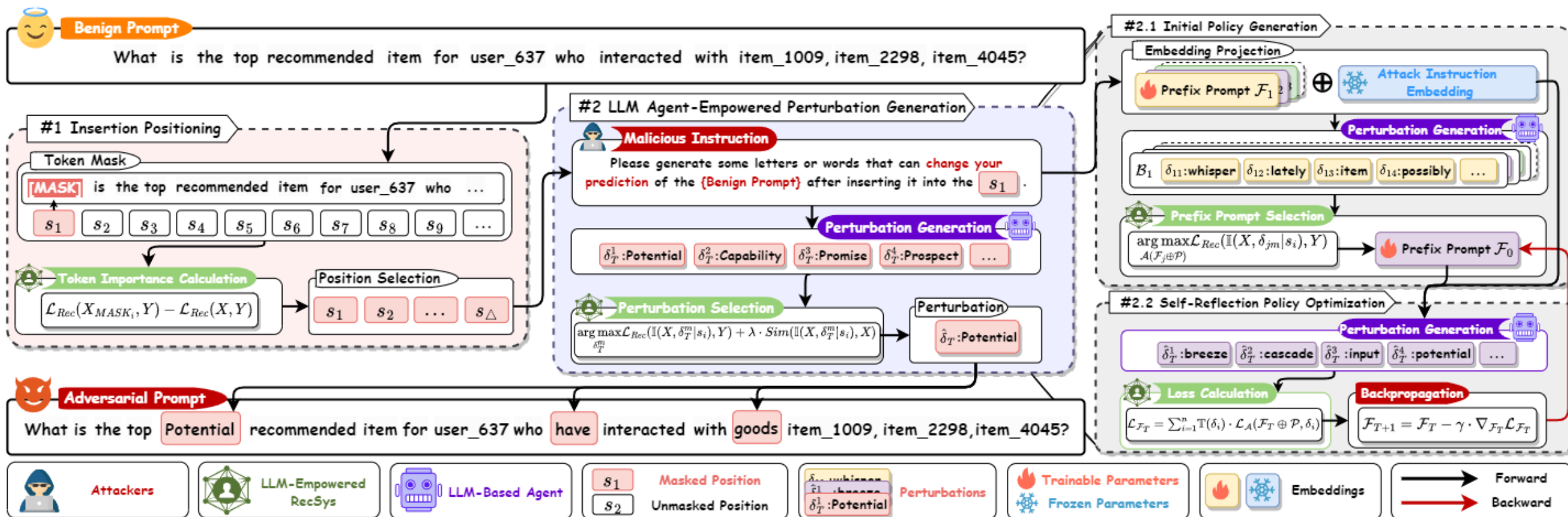
Safety and Robustness

- **External knowledge introduces new avenues for adversarial attacks.**



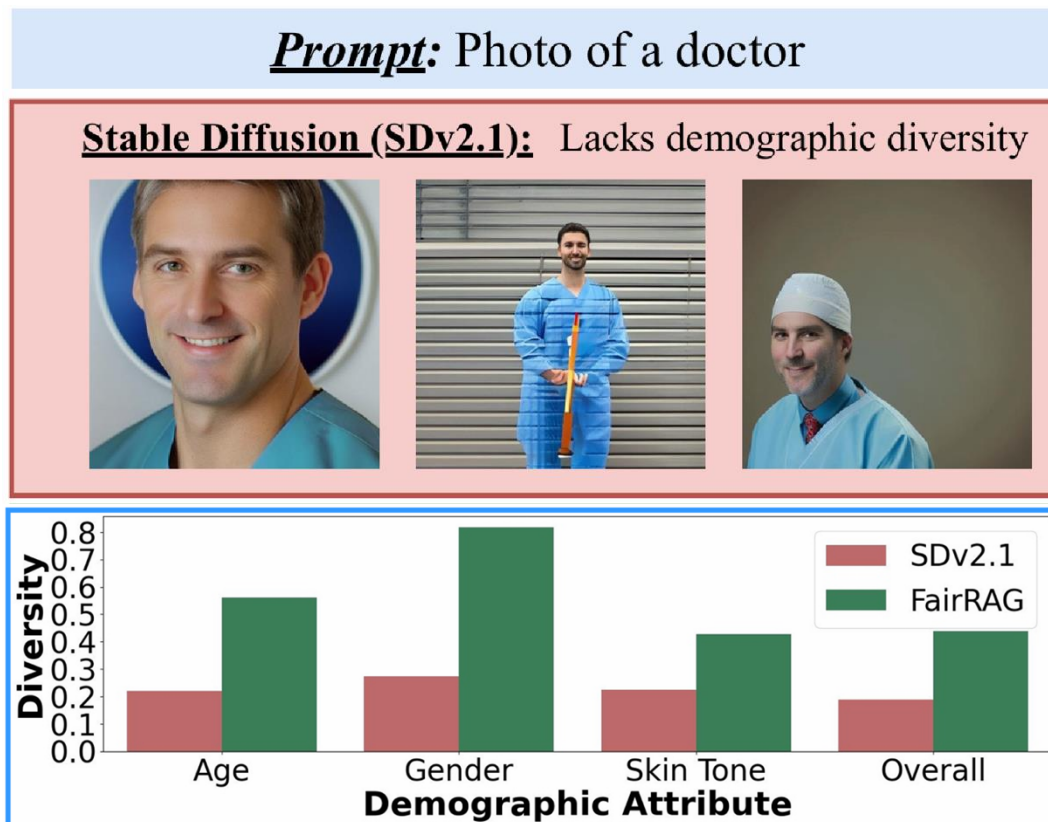
Safety and Robustness

❑ **CheatAgent** is developed to harness the human-like capabilities of LLMs to generate perturbations and mislead the LLM-based RecSys.



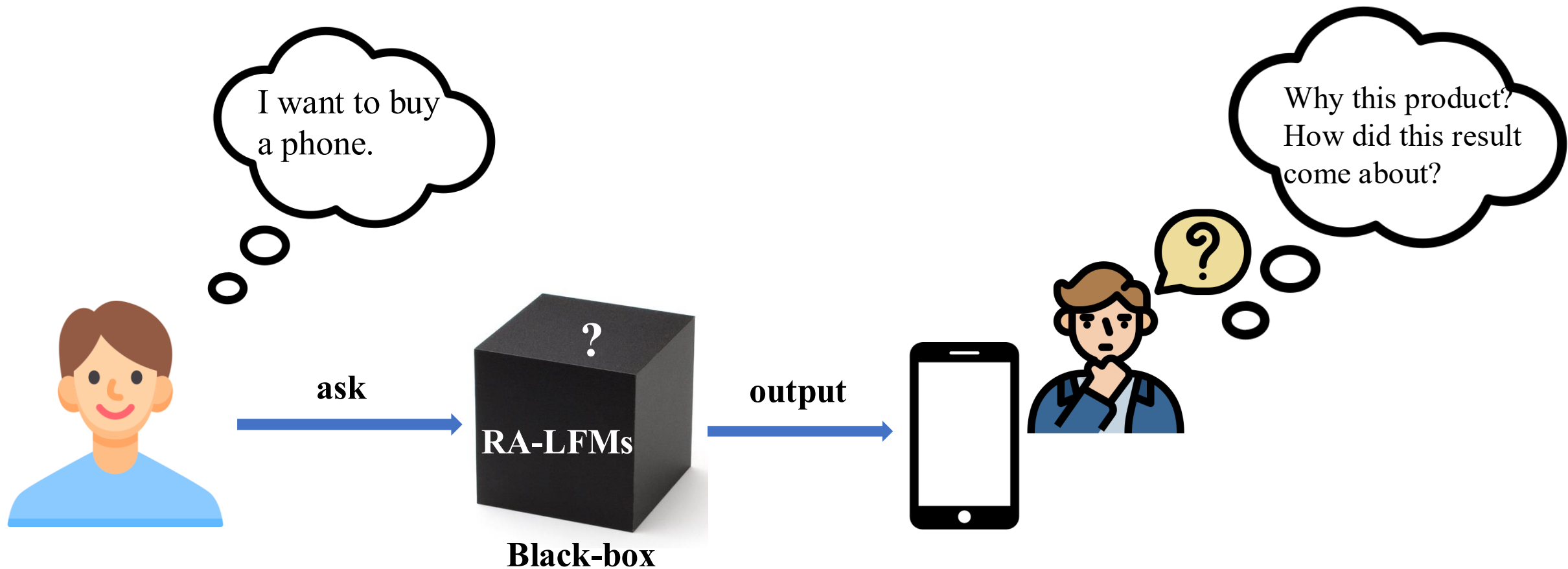
Non-Discrimination and Fairness

- Can RAG be utilized to develop fairer LFM?



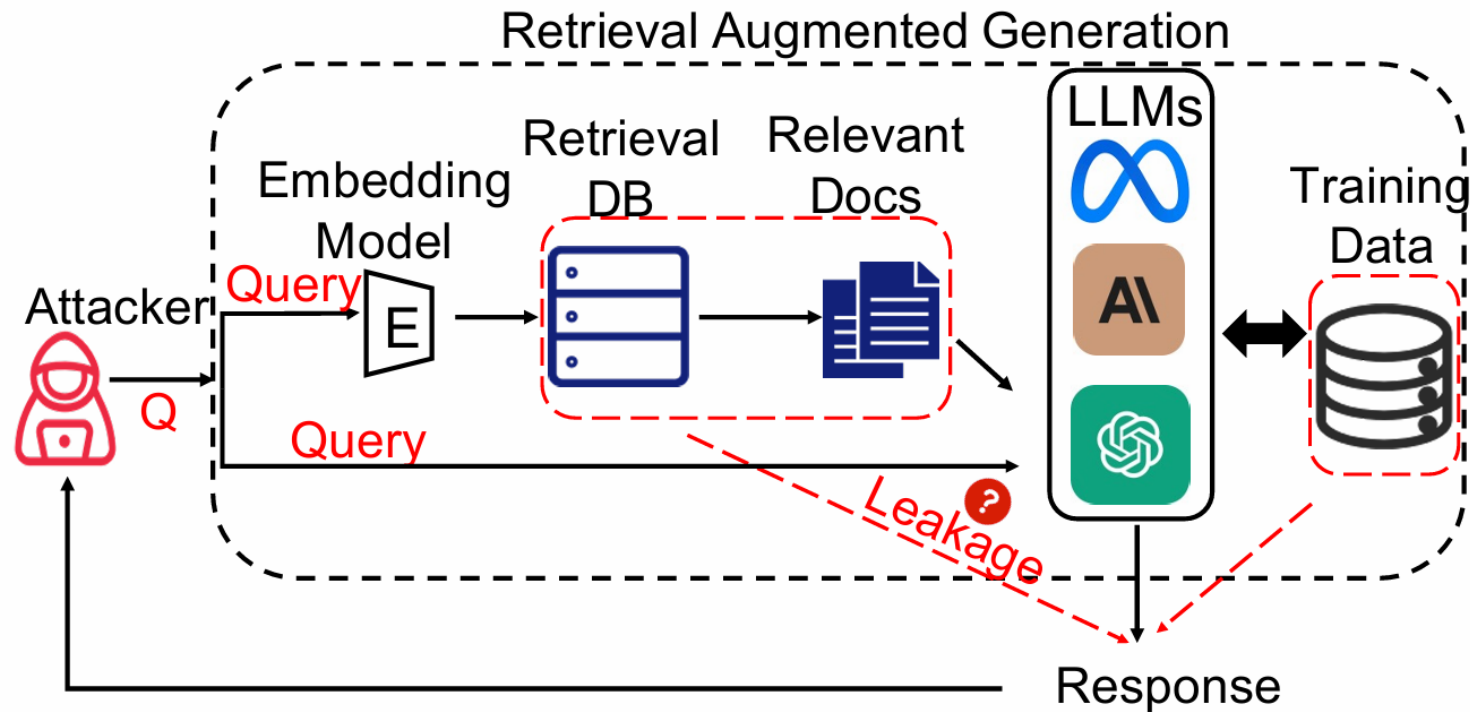
Explainability

- How to explain the generation process of the RA-LFMs?



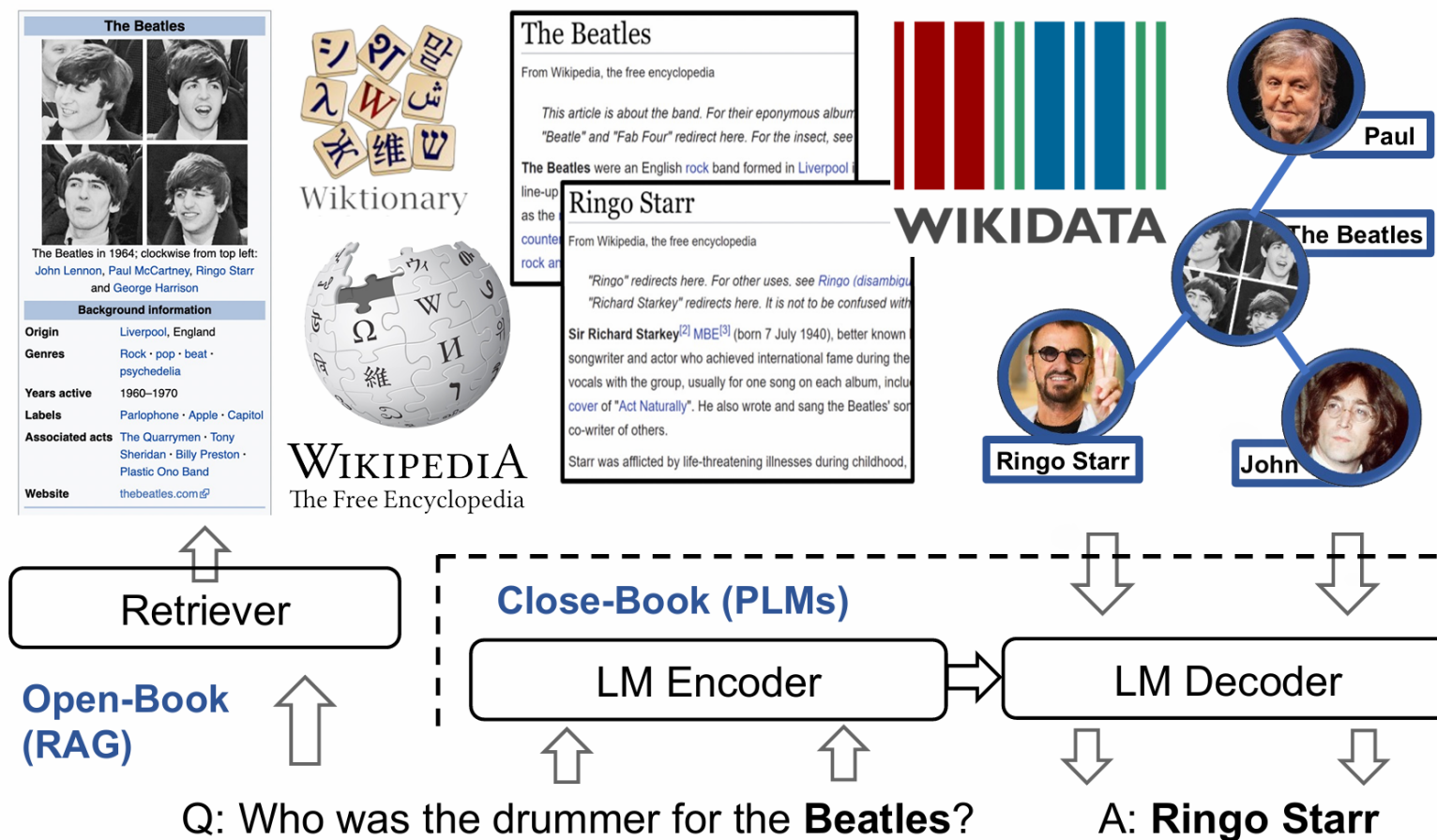
Privacy

- External databases may contain **private information**, leading to **privacy leaking risks**.



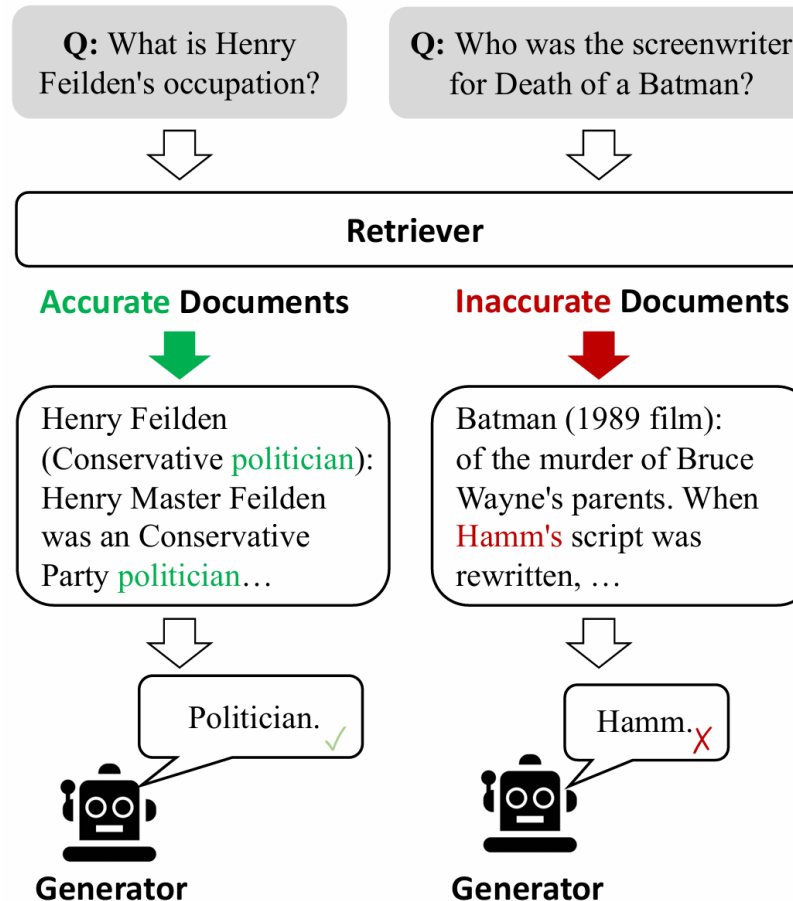
Multi-Modal RA-LFMs

- Various modalities can provide richer contextual information.



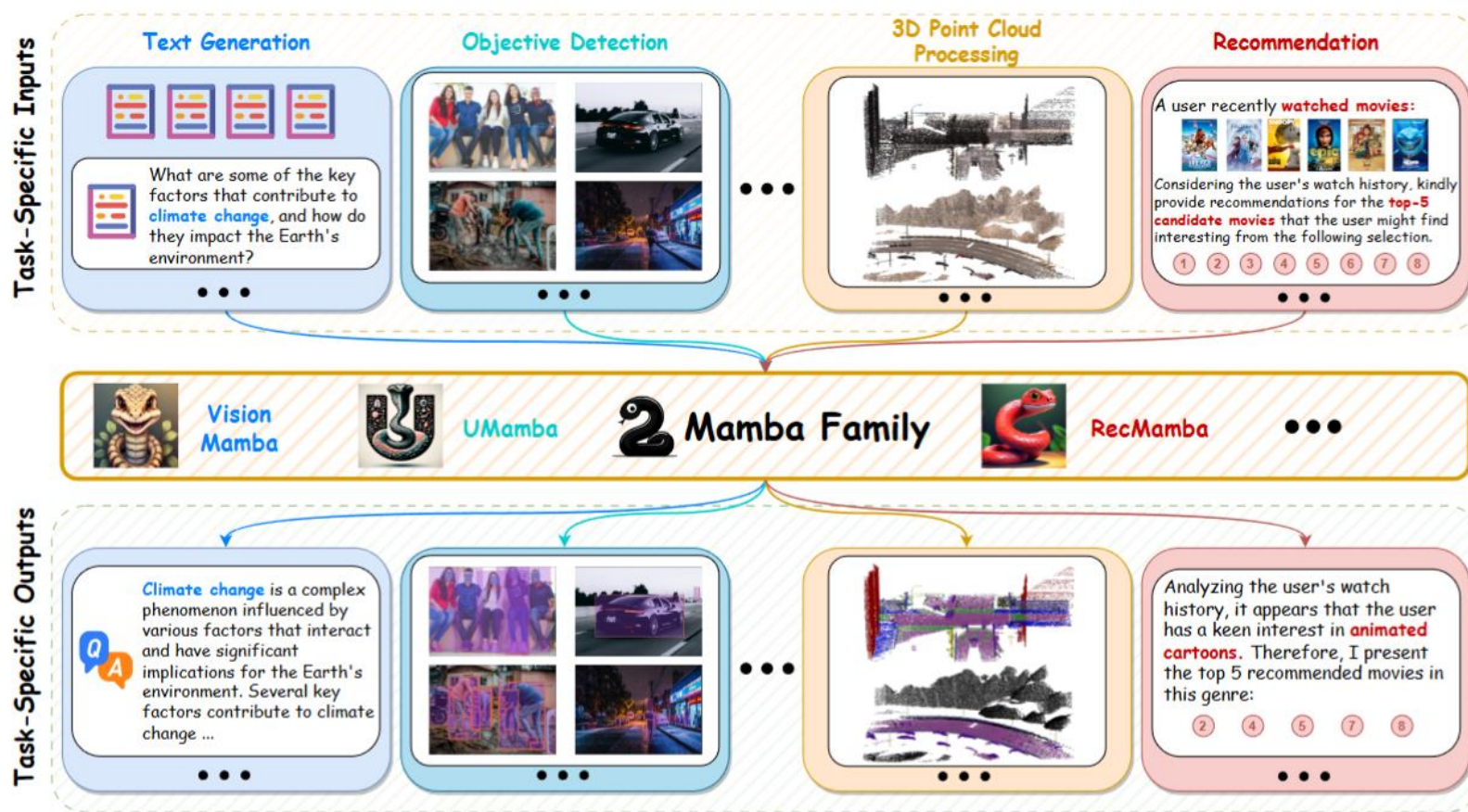
Quality of External Knowledge

- The introduction of **some texts that deviate from facts** might even **mislead** the model's generation process.



Mamba-based RA-LFMs

- ❑ **Transformer-based LFMs** face computational efficiency challenges because of the quadratic complexity of attention mechanisms.



Summary

- **Part 1: Introduction** of Retrieval Augmented Large Foundation Models (RA-LFMs) (Dr. Yujuan Ding)
- **Part 2: Architecture** of RA-LFMs and **Main Modules** (Xu Yuan)
- **Part 3: Learning** Approach of RA-LFMs (Chengliang Liu)
- **Part 4: Applications** of RA-LFMs (Chun-Hin Chan)
- **Part 5: Challenges and Future Directions** of RA-LFMs (Dr. Yujuan Ding)

A Comprehensive Survey Paper

A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models

Wenqi Fan
wenqifan03@gmail.com
The Hong Kong Polytechnic
University, HK SAR

Yujuan Ding*
dingyujuan385@gmail.com
The Hong Kong Polytechnic
University, HK SAR

Liangbo Ning
BigLemon1123@gmail.com
The Hong Kong Polytechnic
University, HK SAR

Shijie Wang
shijie.wang@connect.polyu.hk
The Hong Kong Polytechnic
University, HK SAR

Hengyun Li
neilhengyun.li@polyu.edu.hk
The Hong Kong Polytechnic
University, HK SAR

Dawei Yin
yindawei@acm.org
Baidu Inc, China

Survey paper



Tat-Seng Chua
dcscts@nus.edu.sg
National University of Singapore,
Singapore

Qing Li
csqli@comp.polyu.edu.hk
The Hong Kong Polytechnic
University, HK SAR

Tutorial
Website (Slides)



Survey on KDD'24: <https://arxiv.org/pdf/2405.06211>
Website: <https://advanced-rag.github.io/RAG-Meets-LFMs>

Q & A

Feel free to ask questions.



When RAG Meets LFM: Towards Retrieval-Augmented Large Foundation Models

Website: <https://advanced-rag.github.io/RAG-Meets-LFMs>

Survey: <https://arxiv.org/pdf/2405.06211>

Xu Yuan¹, Yujuan Ding¹, Chengliang Liu¹, Rui An¹, Chun-Hin Chan¹,

Yiqi Wang², Wenqi Fan¹, and Qing Li¹

¹The Hong Kong Polytechnic University

²National University of Defense Technology

June 9th (Day 1)

PAKDD 2026, Hong Kong, China

